

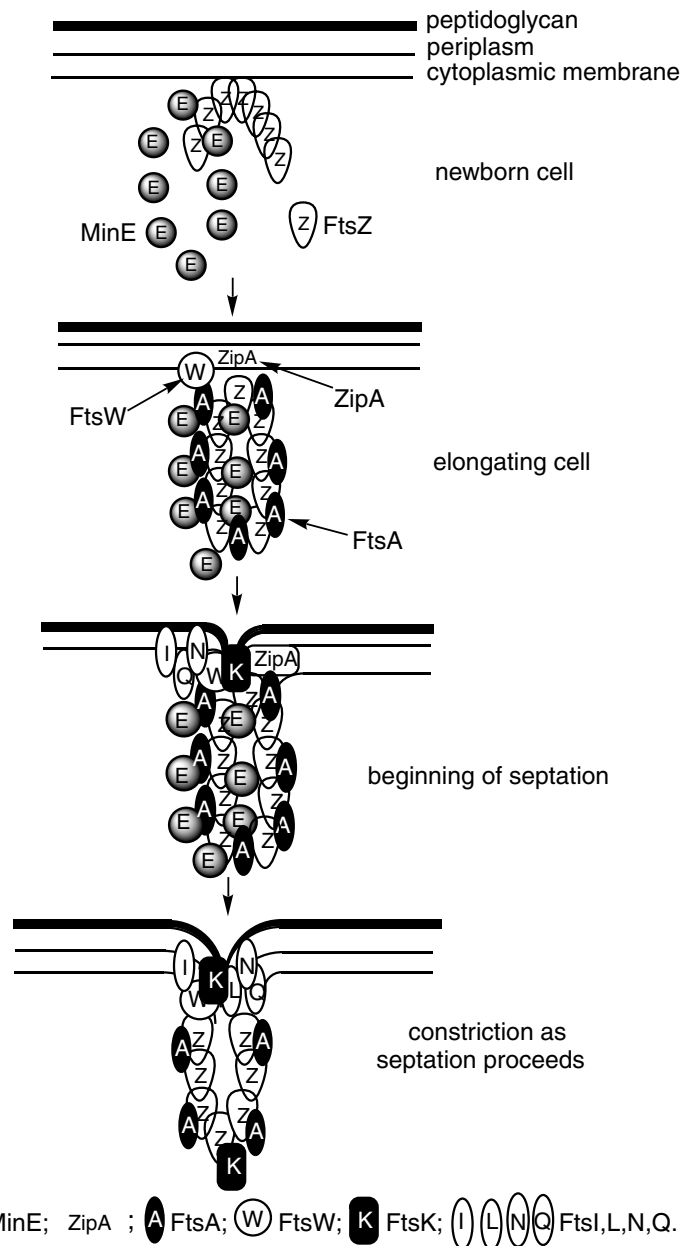


Fig. 17-1. Conceptual diagram of the role of cell division components in *E. coli*. The process begins with the assembly of FtsZ at the midpoint of the cell and targets MinE to the same location. In the initial stage of septation, ZipA, FtsA, and FtsW associate with the FtsZ ring followed shortly by FtsK, FtsI, FtsL, FtsN, and FtsQ. As septation proceeds, the ring contracts and a few of the FtsZ components begin leaving the ring. FtsK may aid in binding the FtsZ ring to the membrane. See Table 17-1 for a description of the function of the individual components.

where it forms a ring that remains associated with the leading edge of the invaginating septum until septation is completed.

The FtsZ ring system that forms the cytoskeletal framework of the division process is homologous to the tubulin of eukaryotic cells. FtsZ shows sequence homology with tubulin, and the atomic structure reveals virtually identical folds. Both FtsZ and tubulin assemble into protofilaments that can switch from a straight conformation to a curved one. As the cytokinetic ring constricts, the protofilaments curve to accommodate the decreasing diameter and may actually provide the force required for constriction. The shift from a straight to a curved conformation is powered by GTP hydrolysis. It is assumed that the carboxy-terminal peptide of the FtsZ ring is attached to the inner face or the inside of the ring.

ZipA, the integral membrane protein that is an essential component for cell division in *E. coli*, contains an amino-terminal transmembrane anchor and a three-part cytoplasmic domain. ZipA presumably serves as a membrane anchor for FtsZ but may also stabilize and cross-link FtsZ polymers. The carboxy-terminal domain of ZipA binds to a short conserved peptide at the carboxyl terminus of FtsZ. The carboxy-terminal peptide is not required for polymerization of FtsZ or for formation of the Z ring, but it is essential for binding of FtsA and ZipA to FtsZ. FtsA is an actin homolog necessary for the cell division process. The structures of FtsA and ZipA appear to have the flexibility needed to permit the membrane-tethered ZipA to reach around the protofilament to bind the carboxy-terminal peptide on the inner face of FtsZ.

As discussed in Chapter 2, cell division does not take place until a round of replication has been completed. This observation suggests that cell division is triggered by events that occur at or near the termination of replication. The most important of these is decatenation of the replicated chromosomes. The separated chromosomes are then partitioned into the two daughter cells. Division occurs subsequent to or simultaneously with partitioning.

In *E. coli* several enzymes appear to participate in resolving chromosome dimers formed during replication. Two site-specific recombinases, XerC and XerD, act in concert at a site near the terminus of chromosome replication (*dif*) to separate chromosome dimers into monomers during the process of cell division. Mutations in *xerC* and *xerD* result in the development of filamentous cells containing abnormally partitioned cells. It has also been shown that the FtsK protein must be in place at the septal region for the Xer-mediated resolution of chromosomes to occur. Topoisomerase IV, encoded by *parE* and *parC*, is also responsible for decatenation of newly formed chromosomes. In *parE* and *parC* mutant cells, newly replicated chromosomes fail to be decatenated. The *mukB* gene encodes a protein (MukB) involved in chromosome partitioning in *E. coli*. A suppressor gene *smbA* (suppressor of *mukB*) is essential for cell proliferation in the temperature range of 22–42 °C. Cells lacking the SmbA protein cease macromolecular synthesis.

One concept of the placement of the division plane in *E. coli*, nucleoid occlusion, is based on the observation that cytokinesis takes place only between duplicated nucleoids that have become fully separated. The location of the division plane is directly related to the cellular position of the replicating chromosome. Recent evidence that the positioning of the FtsZ ring is affected by the positioning of the segregating nucleoid lends credence to this view. The positioning of the FtsZ ring is inhibited by unsegregated nucleoids in an *E. coli* mutant defective in topoisomerase IV at 42 °C (*parC*). At the permissive temperature (28 °C), the nucleoids separate normally and the

FtsZ ring is assembled at the center of the cell, and division occurs in a normal manner. At the higher temperature, separation of the nucleoids is incomplete, and FtsZ rings are formed on either side of the unsegregated nucleoids, resulting in either anucleate cells or cells containing the incompletely separated chromosomes. FtsZ rings form in many of the anucleate cells and initiate constriction, indicating that the division site is not specified by the nucleoid. If the mutant cells are shifted back to the permissive temperature, the nucleoids begin to separate and the FtsZ ring is assembled in the spaces between the nucleoids.

The *min* operon contains *minB*, *minC*, *minD*, and *minE* genes. The *minB* mutant is the original minicell-forming mutant isolated by Howard Adler and colleagues in 1967. Under normal circumstances, the septum in an *E. coli* cell is placed accurately at the midpoint. Potential division sites (PDSs) are also present at the poles, but the products of the *min* operon, MinC, D, and E, specifically repress usage of these sites. The MinC protein can inhibit FtsZ ring assembly at all PDSs, but requires the ATPase MinD for this function. Mutants lacking MinC or MinD produce numerous minicells, reflecting frequent aberrant septation near the cell poles. The MinE protein suppresses the MinCD-mediated division block specifically at midcell, ensuring proper placement of the septum.

Using fusion with a green fluorescent protein (GFP), it has been shown that the MinE protein accumulates in a ring structure at or near the midpoint of the cell. Assembly of the MinE ring also requires MinD, which serves to direct MinE to the cell membrane. MinD displays dynamic properties and accumulates alternately in either one of the cell halves in what appears to be a rapidly oscillating membrane association-disassociation cycle imposed by MinE (Fig. 17-2).

The DicB protein, encoded by *dicB* (see Table 17-1), is another cell division inhibitor unrelated to MinC, MinD, or MinE. It interacts with MinC to cause division inhibition by a MinD-independent mechanism.

As indicated at the outset of the discussion on cell division, the growth and division of the murein sacculus must be an orderly and coordinated process. Because of the intricacies involved, it has been suggested that a multienzyme complex orchestrates the growth of the murein layer. Experimental evidence supports the concept that murein hydrolases and synthases interact with each other. Lytic transglycosylases immobilized to a matrix specifically retained the bifunctional transglycosylase/transpeptidases PBP1A, PBP1B, and PBP1C, the transpeptidases PBP2 and PBP3, and the endopeptidases PBP4 and PBP7. This group of enzymes is needed to enlarge murein according to the three-for-one growth mechanism shown in Figure 17-3. This strategy has been named the “make-before-break” system by Arthur Koch.

In Figure 17-3, a dimer of a bifunctional PBP (PBP1A or PBP1B) could supplement a preexisting primer murein strand to form a murein triplet, which in turn is attached to the sacculus by transpeptidation catalyzed by a dimer of a transpeptidase such as PBP2 or PBP3. The docking strand could be removed specifically by the action of a dimer of an endopeptidase (PBP4 or PBP7) splitting the cross bridges to the left and right of the old strand, and a lytic transglycosylase, that depolymerize the polysaccharide strand. The model shown in Figure 17-3 suggests that a murein triplet may be synthesized in one step. As shown in Table 17-3, the division system is postulated to contain an additional monofunctional transglycosylase that synthesizes the middle strand of the triplet. The cell division process also differs from that of

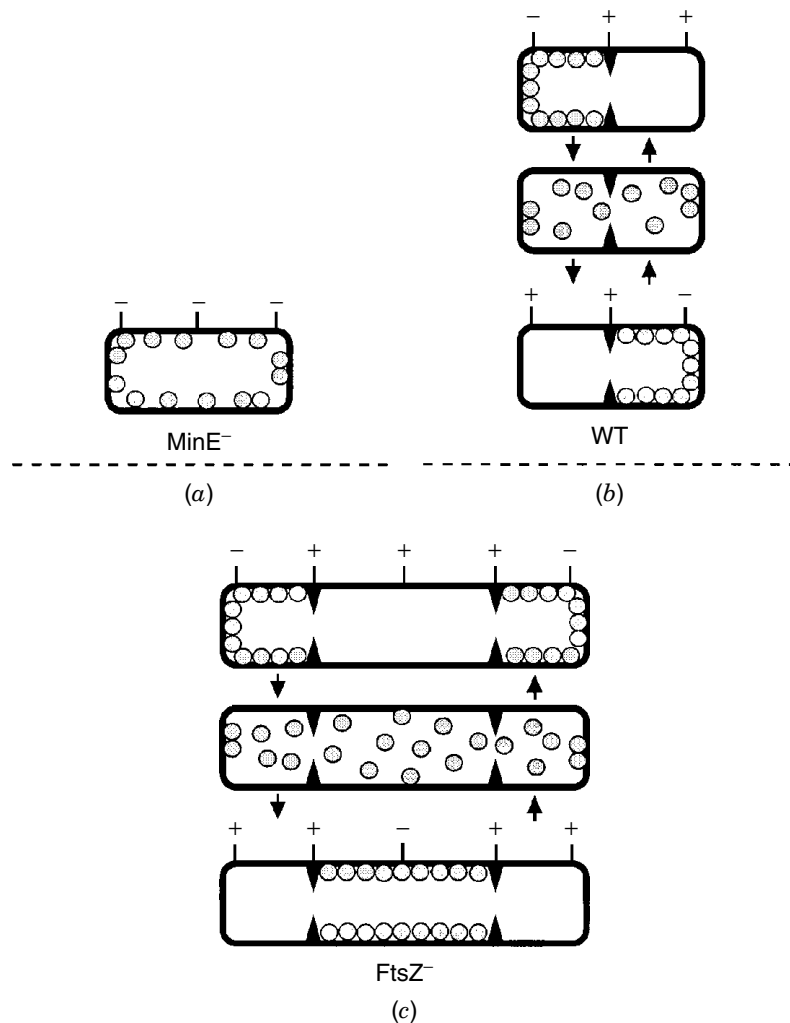


Fig. 17-2. Model for the MinD and MinE action in preventing aberrant septation events. MinD is represented by gray spheres, the MinE ring by filled triangles, and PDSs by either a minus sign (blocked by MinC/MinD action) or a plus sign (not blocked by MinC/MinD, available for assembly of FtsZ ring) sign. (a) In the absence of MinE, MinD is distributed evenly over the membrane. Provided MinC is present, this prevents septal ring formation at all PDSs, resulting in the formation of nonseptate filaments. (b) In wild-type (WT) cells, MinD oscillates from one side of the MinE ring to the other, alternately blocking division at each of the polar PDSs. For simplicity it is assumed that the MinC/MinD division block is relieved as soon as MinD leaves a PDS, although it may well remain refractive to FtsZ assembly for some period afterward. (c) In the absence of FtsZ, multiple MinE rings define three or more cell segments. As in wild-type cells, MinD oscillates between the segments flanking each MinE ring. (Source: From D. M. Raskin and P. A. J. de Boer, *Proc. Natl. Acad. Sci. USA* **96**:4971–4976, 1999.)

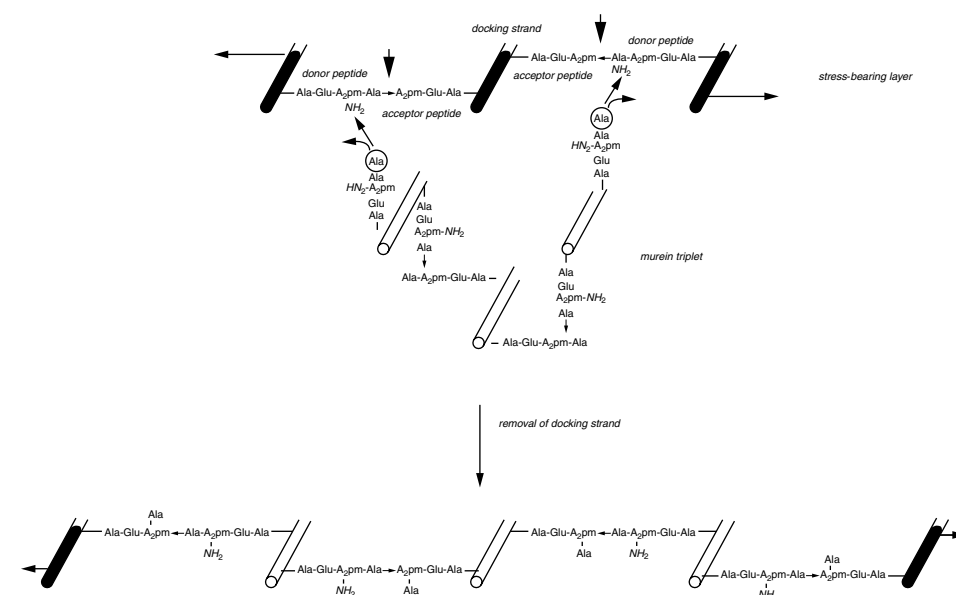


Fig. 17-3. Model for the growth of murein following a make-before-break strategy (three-for-one growth model). Three cross-linked glycan strands from the monolayered, stress-bearing murein are shown in gray. A triple pack of newly synthesized, cross-linked glycans still in a relaxed state is shown in outline. The murein triplet is covalently attached to the free amino groups present in the donor peptides of the cross-links on both sides of a strand, called the docking strand, which is substituted by acceptor peptides. Specific cleavage of the preexisting cross-links (arrowheads) results in the replacement of the docking strand by the murein triplet. The rods represent the glycan strands. A₂pm, diaminopimelic acid. (From J.-V. Höltje, *Microbiol. Mol. Biol. Rev.* **62**:181–203, 1998.)

TABLE 17-3. Elongation and Restriction Complexes in *E. coli* Morphogenesis

Elongation Complex	Constriction Complex
Lytic transglycolase	Lytic transglycolase
Endopeptidase	Endopeptidase
Transpeptidase	Transpeptidase
Transpeptidase-transglycolase	Transpeptidase-transglycolase
Penicillin-binding protein 2 (PBP2)	Penicillin-binding protein 3 (PBP3)
	Transglycosylase

elongation by the presence of the transpeptidase PBP3. The PBP2 transpeptidase, known to be responsible for the rod shape, would be a specific component of the elongation complex.

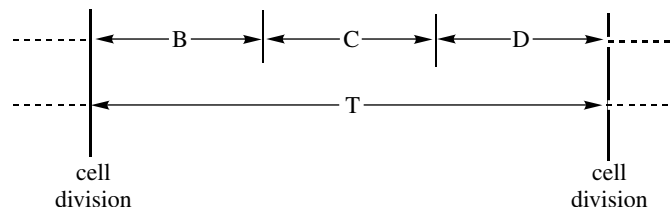
A comprehensive study of *E. coli* mutants lacking all possible combinations of 8 of the 12 penicillin-binding proteins (PBPs) revealed some interesting information (for a listing of the PBPs and their functions, see Table 7-4). Through genetic engineering it was possible to construct a set of 192 *E. coli* strains containing every possible

combination of deletions of PBPs 1A, 1B, 4, 5, 6, and 7, AmpC, and AmpH. In addition, the *dacD* gene was deleted from two septuple mutants, creating strains lacking 8 genes. The only deletion combinations that could not be produced were those lacking both PBP1A and PBP1B because this combination is lethal. Surprisingly, all other deletion mutants were viable even though, at the extreme, 8 of the 12 known PBPs had been eliminated.

Furthermore, when both PBP2 and PBP3 were inactivated by the β -lactam antibiotics mecillinam and aztreonam, respectively, several mutants did not lyse but continued to grow as enlarged spheres, so one mutant synthesized osmotically resistant peptidoglycan when only 2 of 12 PBPs—PBP1B and PBP1C—remained active. The studies show that from among these proteins, PBP2 and PBP3, plus either PBP1A or PBP1B, are the only PBPs required for laboratory growth of regularly dividing rod-shaped cells. This set of *E. coli* strains supplies genetic backgrounds in which the physiological roles of nonessential PBPs and other peptidoglycan-reactive enzymes may be observed and investigated in greater detail.

CELL DIVISION IN GRAM-POSITIVE COCCI

In its simplest aspect, the cell cycle of a bacterial cell can be envisioned as occurring in phases that include B, the interval between cell division and the initiation of DNA synthesis; C, the period of DNA synthesis (chromosome replication); and D, the period between termination of chromosome replication and the end of cell division. T, the total cell cycle, is equal to $B + C + D = 1$.



In dividing cells of *Enterococcus hirae*, an equatorial band develops at the initiation of cell division (Fig. 17-4). This fortuitous occurrence permits visual description of the cell cycle in this organism on rather precise terms. A cross-wall is assembled at each of these sites. As a constricting division furrow bilaterally splits this cross-wall, the two cleaved cross-wall layers separate and expand to form two new polar caps. As the formation of new poles nears completion, new sites must be initiated for continued cell growth. These nascent septa appear early in the cell cycle. The absence of significant turnover of peptidoglycan in *E. hirae* aids in distinguishing the preexisting wall surface from the newly extended areas.

The raised wall bands, at relatively fixed distances from the pole, are observed at all stages of the division cycle. With cells of increased length, an increased area of wall is observed between the bands and an equatorially located nascent septum. The wall bands mark the meeting of polar caps made during a previous cell cycle, with equatorially located walls synthesized during the most recent cell cycle. The events that occur during cell division are more or less continuous. However, for the

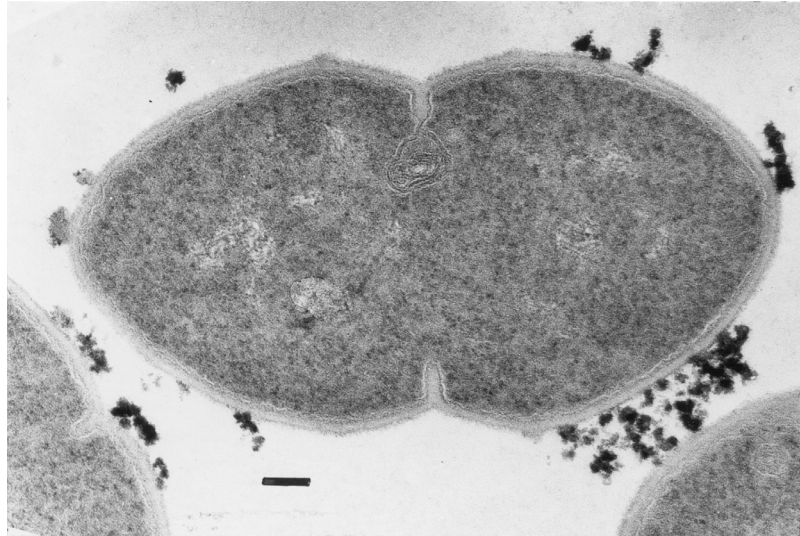


Fig. 17-4. Electron micrograph of dividing cells of *Enterococcus hirae*. Raised bands are observed around the circumference of the cell wall. Nascent septa appear at these sites early in the cell cycle. The wall bands mark the meeting of the polar cap, made during a previous cell cycle, with the equatorially located wall synthesized during the most recent cell cycle. Bar equals 0.1 μm . (Source: From Higgins and L. Daneo-Moore, *J. Cell Biol.* **61**:288–300, 1974.)

convenience of description they can be divided into stages as outlined here and in Figure 17-5:

- Stage 1:** Centripetal penetration of the wall is initiated under an equatorial band. A notch is formed in the external surface of the wall directly above the nascent cross-wall, and two external wall bands appear.
- Stage 2:** A thin cross-wall penetrates a short distance (70–80 nm) into the cell. The new wall bands are separated by the insertion of a newly synthesized peripheral wall.
- Stage 3:** The nascent cross-wall thickens somewhat as the new wall bands become separated still further.
- Stage 4:** When the two new hemispheres have reached adequate size and the wall bands are near the equators of the new daughter cells, the penetration of the equatorial cross-wall resumes. As it penetrates into the cytoplasm, the cross-wall continues to peel apart into two layers of wall that become the poles of the newly formed cells. In rapidly growing cells, the entire process begins again at the two new subequatorial bands.
- Stage 5:** Final separation of the two new daughter cells occurs. Although the separation may begin as early as the beginning of stage 4, the event that can be quantitated as a doubling in cell number is not completed until the end of stage 5.

In a rich growth medium, the mass doubling time of *E. hirae* is about 30 minutes. Since DNA synthesis takes longer ($C = 50\text{--}52\text{ min}$), a new round of chromosome

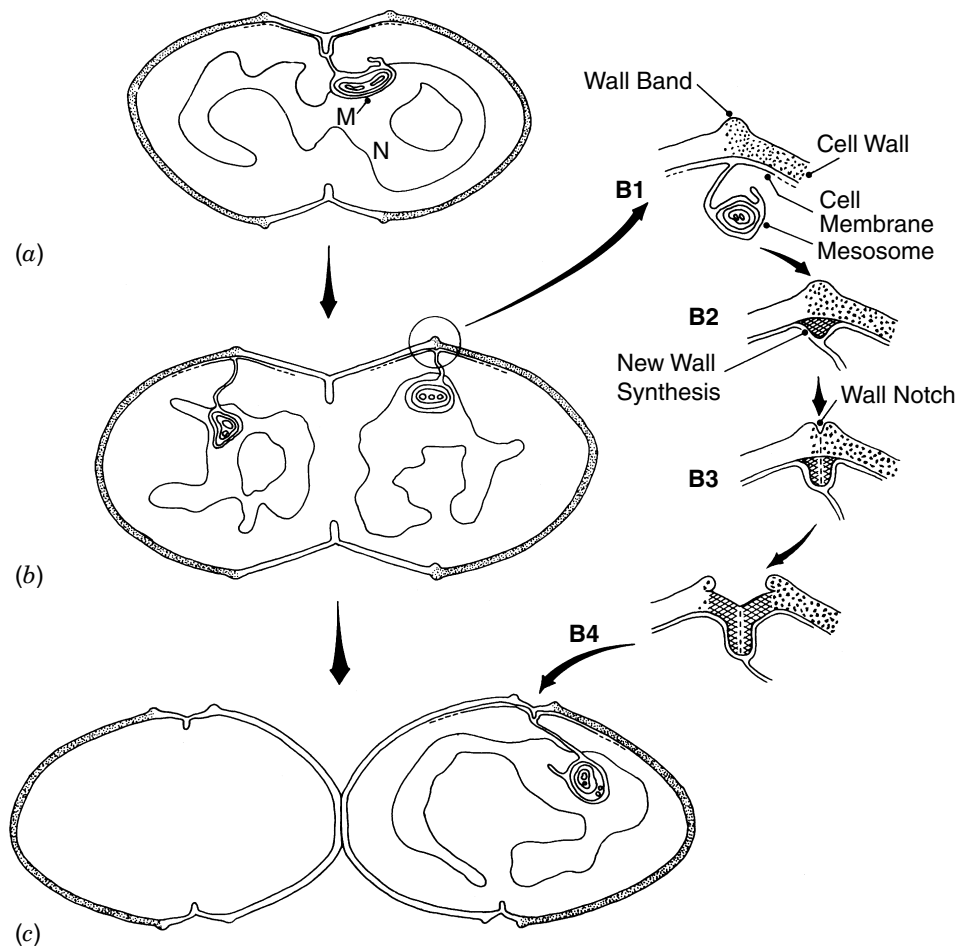


Fig. 17-5. Stages in the cell division cycle of *Enterococcus hirae*. The diplococcus in (a) is in the process of growing a new wall at its cross-wall and segregating its nuclear material (N) to the two nascent daughter cocci. In rapidly growing exponential phase cultures before completion of the central cross-wall, new sites of wall elongation are established at the equators of each of the daughter cells at the junction of the old polar wall (stippled) and the new equatorial wall beneath a band of wall material that encircles the equator (b). Beneath each band a mesosome (M) is formed while the nucleoids separate and the mesosomes at the central site are lost. The mesosomes appear to be attached to the plasma membrane by a thin membranous stalk (B1). Invagination of the septal membrane appears to be accompanied by centripetal cross-wall penetration (B2). A notch is then formed at the base of the nascent cross-wall, creating two new wall bands (B3). Wall elongation at the base of the cross-wall pushes the newly made wall outward. At the base of the cross-wall, the new wall peels apart into the peripheral wall, pushing the wall bands apart (B4). When a sufficient new wall is made so that the wall bands are pushed to a subequatorial position [e.g., from (C) to (A) to (B)], a new cross-wall cycle is initiated. Meanwhile, the initial cross-wall centripetally penetrates into the cell, dividing it into two daughter cocci. At all times the body of the mesosomes appears to be associated with the nucleoid. Doubling of the number of mesosomes seems to precede completion of the cross-wall by a significant interval. Nucleoid shapes and the position of mesosomes are based on projections of reconstructions of serially sectioned cells. (Source: From M. L. Higgins and G. D. Shockman, *Crit. Rev. Microbiol.* 1:29–72, 1971.)

replication must be initiated before the old round of synthesis is completed (**dichotomous replication**). It follows, therefore, that wall band splitting and initiation of chromosome replication do not occur simultaneously. The rounds of replication must have been completed early enough so that segregation can take place. Consequently, wall band splitting indirectly controls both cell division and chromosome replication. Thus, it may be concluded that growing streptococci respond to the alteration of nutritional conditions mainly by altering the rate at which they fashion the cell wall. Only to a small extent do they respond by shifting the time of wall band splitting to earlier portions of the cycle.

By comparing the cell cycle events of band splitting (BS) and cell division (CD), as shown in Figure 17-6, it can be concluded that the cell initiates wall band splitting when the growth zones cannot function rapidly enough to allow the increase of surface area required to accommodate continuing production of cytoplasm. Although initiation of new growth sites seems to be independent of normal chromosome replication, it has been shown that chromosome replication is necessary for the terminal events of growth site development that result in the division of a site into two separate poles. Mitomycin C, which inhibits DNA synthesis rapidly and with a minimal effect on the synthesis of other macromolecules, causes an eventual cessation of cell division. However, the growth sites enlarge until they reach about $0.25 \mu\text{m}^3$ of cell volume and do not increase further in size. This occurs regardless of whether they are formed before or after the inhibition of DNA synthesis. When those sites approach this $0.25 \mu\text{m}^3$ limit, new sites are initiated. These observations suggest that regardless of whether chromosome replication is inhibited, sites have the same finite capacity to enlarge to this size, and when this capacity is reached, new sites are initiated.

The genes governing peptidoglycan biosynthesis and cell division in *Enterococcus faecalis* and *Staphylococcus aureus*, corresponding to those in the 2 min region of *E. coli*, have been identified and sequenced. As shown in Figure 17-7, the gene clusters from *E. coli*, *B. subtilis*, *S. aureus*, and *E. faecalis* show some similarities

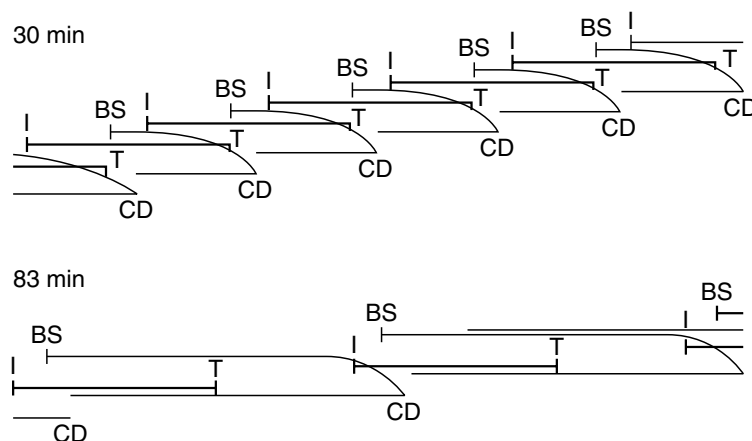


Fig. 17-6. Representation of the cell cycle of *Enterococcus hirae* at two growth rates. I, initiation of chromosome replication; T, termination of chromosome replication; BS, band-splitting events; CD, cell division events. (Source: From A. L. Koch and M. L. Higgins, *J. Gen. Microbiol.* **130**:725–745, 1984.)

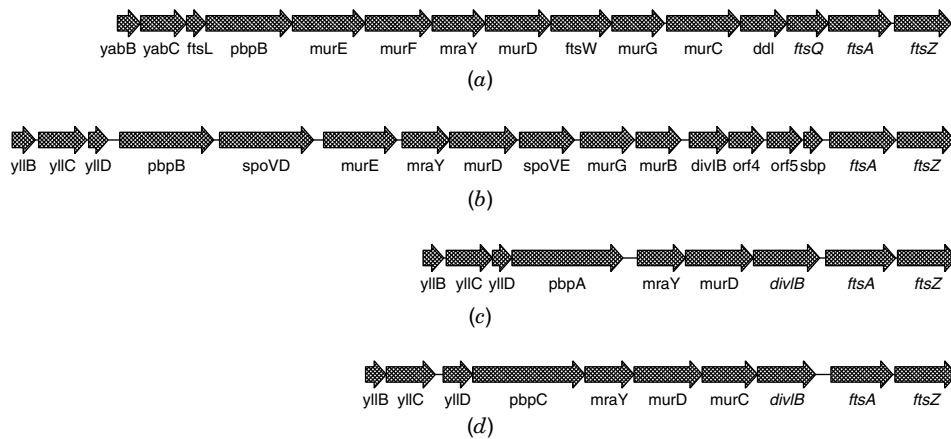


Fig. 17-7. Comparison of the clusters of genes coding for peptidoglycan synthesis and cell division in four bacterial species. (a) *E. coli*; (b) *B. subtilis*; (c) *S. aureus*; (d) *E. faecalis*. The arrows indicate open reading frames (ORFs) and directions of transcription. Gene designations are under the ORFs. Unidentified ORFs are designated as orfs. (Source: From M. J. Pucci, et al., *J. Bacteriol.* **179**:5632–5635, 1997.)

of organization including the presence of penicillin-binding proteins at the left ends and *ftsA* and *ftsZ* cell division genes at the right ends of the clusters. However, there are also some important differences, including the absence of several genes and the comparative sizes of the *divIB* and *ftsQ* genes. There is also a wide range of amino acid sequence similarities between gram-positive cocci and the other organisms. It should be emphasized that this comparison has been made only for genes present in the 2 min region of the *E. coli* chromosome map.

As indicated in Table 17-1, a substantial number of genes involved in peptidoglycan synthesis and cell division in *E. coli* map to other locations on the chromosome. Also, many of the *E. coli* genes encoding important PBPs map elsewhere on the *E. coli* chromosome map (see Table 7-4). Clearly, several genes essential to the synthesis of peptidoglycan and cell division in gram-positive cocci as well as in *E. coli* must be located at other sites on the chromosome.

A 50 k Da protein, PcsB, has been characterized and found to be present in all of 27 clinical isolates of group B streptococci tested. This protein shows significant similarity to a murein hydrolase from *Listeria monocytogenes*. Insertional inactivation of *pcsB* resulted in mutant strain Sep1, which exhibited a drastically reduced growth rate compared to the parental strain and displayed increased susceptibility to osmotic pressure and to various antibiotics. Electron microscopy revealed growth in clumps, cell separation in several planes, and multiple-division septa within single cells. The data suggest a pivotal role of PcsB for cell division and antibiotic tolerance in group B streptococci.

Resistance to ampicillin in *E. faecium* results from increased production of a low-affinity PBP, PBP5, thought to be intrinsic to all strains of this organism. In *E. hirae*, a close relative of *E. faecium*, increased production of PBP5 has been attributed to deletion of *psr*, a presumed repressor of *pbp5* expression. Mutations in *pbp5* of *E. faecium* result in a decrease in the penicillin-binding capability of PBP5.

Vancomycin is also an inhibitor of peptidoglycan synthesis. It binds to the terminal D-alanyl-D-alanine of pentapeptide precursors, preventing polymerization and cross-linking. Vancomycin resistance is due to the acquisition and expression of resistance operons resulting in the synthesis of precursors that terminate in D-alanine-D-lactic acid. The precursors bind glycopeptides with low affinity and cause destruction of normal pentapeptide precursors. Demonstration of the presence of a *vanB*-containing transposon integrated within a larger transferable element that also contains a mutated *pbp5* gene encoding high-level resistance to ampicillin provides a plausible explanation for the emergence of a number of high-level ampicillin-resistant strains of *E. faecium*.

In *Streptococcus pneumoniae* the class A high-molecular-weight PBPs are bimodular enzymes. In addition to a central penicillin-binding transpeptidase domain, they contain an N-terminal glycosyltransferase domain. Mutations in the *S. pneumoniae* genes for PBP1A, PBP1B, and PBP2A isolated by insertion duplication mutagenesis within the glycosyltransferase domain provide evidence that their function is not essential for cellular growth. PBP1B PBP2A and PBP1A PBP1B double mutants could be isolated but showed defects in positioning of the septum. Attempts to obtain a PBP2A PBP1A double mutant failed. All mutants with a disrupted *pbp2a* gene showed higher sensitivity to moenomycin, an antibiotic known to inhibit PBP-associated glycosyltransferase activity. This finding indicates that PBP2A is the primary target for glycosyltransferase inhibitors in *S. pneumoniae*. Since this organism is a major human pathogen causing pneumonia, meningitis, and ear infections, it has been the target of several studies directed at PBPs.

In all, six PBPs have been identified in *S. pneumoniae*. PBPs 1A, 2B, and 2A belong to class A while PBPs 2x and 2B are monofunctional class B proteins. Deletion of *pbp2x* and *pbp2b* is lethal for *S. pneumoniae*, while deletion of *pbp1a* is tolerated, probably due to compensation by PBP1B. The specific resistance to ceftriaxone and cefotaxime of strains of *S. pneumoniae* isolated from the hospital environment is mediated by modification of PBP2x and PBP1B. Gene transfer of *pbp1a*, *pbp2x*, and *pbp2b* from resistant strains conferred penicillin resistance on sensitive *S. pneumoniae* strains under laboratory conditions. Protection by moenomycin of the glycosyltransferase domain against trypsin degradation has been interpreted as an interaction between the glycosyltransferase domain and moenomycin. Studies on disruption of the activities of PBPs indicate that individually the *pbp1a*, *pbp1b*, and *pbp2a* genes are dispensable but that either *pbp1a* or *pbp2a* is required for growth in vitro. The results of many of these studies indicate the potential for acquisition by *S. pneumoniae* of high-level β -lactam resistance by interspecies gene transfer.

CELL DIVISION IN GRAM-POSITIVE BACILLI

The morphogenetic approach to the investigation of cell division that proved so useful in the study in gram-negative organisms has also been useful for studies with gram-positive bacilli. As shown in Figure 17-8, there is considerable similarity between the genes in the 133° region of the genetic map of *B. subtilis* and the genes in the *mra* (2 min) region of the *E. coli* genetic map that are involved in the synthesis of cell wall peptidoglycans and cell division. Mutants of *B. subtilis* have been found that form filaments at high temperature (40–50 °C) but grow normally at the permissive temperature (20–35 °C).

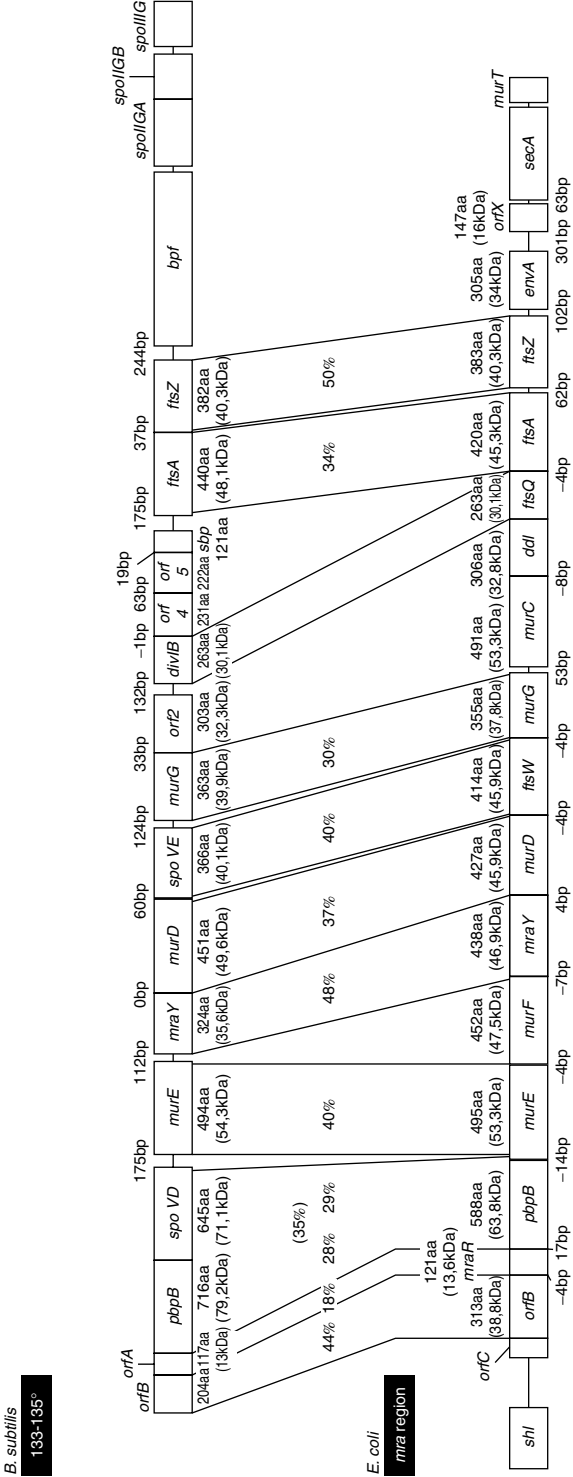


Fig. 17-8. Comparison of the 133° region of the *B. subtilis* genetic map with the *mra* (2 min) region of the *E. coli* genetic map. Similar pairs of genes are joined by dotted lines and similarities in predicting amino acid sequences are indicated as percent (%) identity. The gaps between adjacent coding regions are indicated in base pairs (bp). (From C. E. Buchanan, Ghuyssen and Hackenbeck, A. O. Henriques, and P. J. Piggot 1994. Cell wall changes during bacterial endospore formation. In Ghuyssen, J.-M. and R. Hackenbeck (Eds.). *Bacterial Cell Wall*, Elsevier, Amsterdam, The Netherlands, pp. 167–186.)

There are a number of similarities in the proteins involved in cell division in *B. subtilis* as compared to *E. coli*. However, in *B. subtilis* there are no homologs of ZipA or FtsN. FtsZ and three membrane-bound division proteins — DivIB (a homolog of FtsQ), DivIC (comparable to FtsL), and PBP2B (a homolog of PBP3) — localize at the division site. As in *E. coli*, FtsZ appears to assemble early prior to the appearance of FtsL, DivIC, DivIB, or PBP2B. However, there are a number of differences in the *B. subtilis* cell division system. Depletion of FtsL results in rapid degradation of DivIC, while DivIB and PBP2B are undetectable, suggesting that these proteins assemble at the division site after FtsL or at the same time. This instability of one division protein in the absence of another is not observed in *E. coli*.

When PBP2B is depleted at 37 °C, DivIB and DivIC localizations are significantly decreased and occur only at midfilament positions. PBP2B also fails to localize in *divIB* and *divIC* temperature-sensitive mutants at 48 °C. The interdependency of DivIB, DivIC, and PBP2B suggests that these three proteins localize together or that at the nonpermissive temperature the other two proteins are required to maintain PBP2B at the septal site. Immunofluorescence microscopy studies show that DivIC is required for DivIB localization. However, DivIC localization is dependent on DivIB only at high growth temperatures, at which DivIB is essential for division. FtsZ localization is required for septal recruitment of DivIB and DivIC, but FtsZ can be recruited independently of DivIB.

The high-molecular-weight PBP1 may also play a role in cell division in *B. subtilis*. Cells lacking PBP1 sometimes grow as filaments. PBP1 (encoded by *ponA*) is expressed predominantly during vegetative growth. Vegetative *B. subtilis* cells also contain the high-molecular-weight PBP2C and PBP4. However, PBP1 appears to be functionally more important. Cells lacking PBP1 have a reduced growth rate in a variety of growth media and are longer, thinner, and more bent than wild-type cells. PBP1-deficient cells require increased levels of Mg^{2+} , and under conditions of deficiency of this cation, they are either unable to grow or they grow as filaments. Using epitope tagging and immunofluorescence microscopy, PBP1 has been shown to be localized at division sites in vegetative *B. subtilis*. Growing *ponA* mutant cells display a significant septation defect. Also, while FtsZ localizes normally in most *ponA* mutant cells, a significant portion of these cells display FtsZ rings with aberrant structure or improper localization, suggesting that the lack of PBP1 affects FtsZ ring stability or assembly. This appears to be the first example of a high-molecular-weight PBP localized to the bacterial division septum.

Several genes that play roles in chromosome partitioning in *B. subtilis* include *smc*, *spoOJ*, and *spoIIIE*. Proteins involved in structural maintenance of chromosomes (SMC proteins) are found in eukaryotes, archaea, and prokaryotes. The SMC protein of *B. subtilis* may have a function analogous to that of MukB in *E. coli*. SMC protein has the ability to aggregate and/or renature single-stranded DNA in ATP-dependent reactions in vitro. A technique for visualizing specific regions on the chromosome in living cells that uses green fluorescent protein in combination with time-lapse microscopy has shown that the SMC protein is required for proper arrangement of the chromosome and for efficient segregation of replication termini but not for bipolar movement of the chromosome.

The SpoOJ protein is a member of the ParB family of partition proteins. It binds to multiple sites in the origin region of the chromosome and forms a large nucleoprotein complex that is visible by immunofluorescence microscopy or green

fluorescent protein fusion. This complex may play a role in pairing newly replicated sister origin regions or in structurally organizing the origin region. A double mutant, *smc spoOJ*, has a synthetic lethal phenotype in rich medium. SpoIIIE is involved in postseptational chromosome segregation during sporulation. However, it is also required for efficient segregation during vegetative growth when normal cell division or chromosome partitioning has been perturbed. During vegetative growth, SpoIIIE may provide a backup mechanism for chromosome partitioning when normal partitioning is defective. Mutants defective in nucleoid structure are unable to move chromosomes out of the way of the invaginating septum, and SpoIIIE is involved in repositioning these bisected chromosomes.

In *B. subtilis* the CodV and RipX proteins show 35% and 44% identity with the XerC and XerD proteins, respectively. The XerC and XerD proteins of *E. coli* are considered to act at the *dif* site to resolve dimeric chromosomes formed by recombination during replication. The *dif* gene, for deletion-induced filamentation, is a *recA*-independent recombination site at the terminus of the chromosome. In *B. subtilis*, genetic and biochemical evidence reveals that a 28 bp sequence of DNA (*Bsdif*), lying 6° counterclockwise from the terminus of replication (172°), is the site at which RipX and CodV catalyze site-specific recombination reactions required for normal chromosome partitioning. Cultures of *ripX* mutants contain a subpopulation of unequal-size cells held together in long chains. These chains include anucleate cells and cells with aberrantly dense or diffuse nucleoids, indicating a chromosome partitioning failure. In *B. subtilis* catenation nodes are removed from replicated chromosomes by ParC and ParE (subunits of topoisomerase IV). Both CodV and RipX are required for efficient recombination of substrates *in vitro*.

A number of genes involved in the cell division process in *B. subtilis* are primarily concerned with the unequal division of cells during the sporulation process. The role of these genes in sporulation is discussed in Chapter 19.

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CHAPTER 18

MICROBIAL STRESS RESPONSES

When supplied with sufficient nutrients and optimal growth temperature, pH, oxygen levels, and solute concentrations, microbes will grow at a maximum growth rate characteristic for the organism. Variations in any of these parameters can affect the maximum growth rate and, thus, can represent an **environmental stress** for the microbe. The ability of microbes to sense and respond (correctly) to impromptu alterations in the environment is crucial to their survival. In reality, conditions that allow for maximal growth rates outside the laboratory are few and far between. As a result, most bacteria live in a constant state of stress. This chapter covers the physiologic and genetic responses of certain well-characterized bacteria to various stress conditions.

OSMOTIC STRESS AND OSMOREGULATION

The concentration of solutes (e.g., salts, ions, metabolites) plays a critical role in microbial growth. In the laboratory, most microbes exhibit optimal growth in culture media of relatively low osmolarity. For many bacteria, **hypertonic** or **hyperosmotic** conditions result in water loss from the cytoplasm, causing the cell to shrink (**plasmolysis**). **Hypotonic** or **hypoosmotic** conditions result in an influx of water into the cytoplasm, which causes the cell to swell (**plasmolysis**) and perhaps burst in a process referred to as **osmotic lysis**. Microbial membranes are freely permeable to water and, therefore, water inside the cell is essentially at equilibrium with the outside of the cell. Microbes with cell walls can and do maintain a high cytoplasmic solute concentration relative to the outside. This translates into lower **water activity** (A_w) inside the cell.

$$A_w = x \div (x + c)$$

[c = osmolality of solute(s); x = moles of water per liter (55.6)]

The lower cytoplasmic A_w causes water to at least try and flow into the cell. This places significant pressure on the microbial cell wall as the cell volume attempts to increase at the same time the cell wall prevents the cell from swelling. This pressure placed on the cell wall by the cytoplasmic membrane is referred to as **turgor pressure** or **turgor**. Turgor is opposed by the tension of the cell envelope. The function of osmoregulatory mechanisms or osmotic stress responses is to maintain turgor within limits, allowing for maintenance of cell viability.

Movement of water occurs by diffusion and, in a much more rapid process, through water-selective channels called **aquaporins**. The AqpZ channel of *E. coli* has been shown to mediate rapid and large water fluxes in both directions in response to sudden osmotic upshifts or downshifts, although its role in the cell is not essential. Turgor is maintained by regulating the total osmotic solute pool in the cytoplasm and the relative level of solutes in the periplasm (in Gram-negative bacteria) immediately outside the cytoplasmic membrane. In low-osmolality media, cytosolic osmolality is largely due to ionic solutes (e.g., K^+ ions); in high-osmolality media it largely involves neutral solutes (e.g., trehalose).

Some of the mechanisms described below are also responsible for maintaining osmotic homeostasis. For this reason, mechanisms of osmoregulation such as K^+ ion influx/glutamate biosynthesis–coupled systems are present all the time in the cell.

High Osmolality

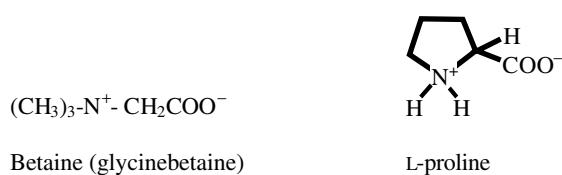
As the osmolality of the surrounding environment increases, turgor pressure drops and growth slows or halts. Macromolecular biosynthesis is inhibited and respiration rates decline. The most rapid response to this osmotic upshock is an **increase in K^+ ion influx** through, at least, three uptake systems in *E. coli*: **Trk**, **Kdp**, and **Kup**. Homologous systems to Trk and Kdp have been identified in *Salmonella enterica* and other bacteria. The Trk and Kdp systems appear to be the major systems for K^+ uptake under these conditions, since they can achieve sufficiently high rates of uptake. The **Trk system** is composed of three components: **TrkA** (peripheral membrane protein), **TrkE** (membrane associated), and either **TrkH** (in *E. coli*; membrane-spanning protein) or **TrkG** (*E. coli* and other bacteria; membrane-spanning protein). The Trk system binds NAD(H) via TrkA and may regulate K^+ ion uptake. The **Kdp system** (see Chapter 9) is also a three-component system composed of **KdpA** (membrane-spanning protein), **KdpB** (integral membrane protein), and **KdpC** (peripheral membrane protein). KdpB is a **P-type ATPase** and likely provides the energy to drive K^+ ion influx through this system. Kup is a single, large membrane-spanning protein possessing a significant cytoplasmic tail domain. In addition to influx, K^+ ion accumulation results from plasmolysis and the closing of stretch-sensitive K^+ ion efflux channels.

Concurrent with the increase in K^+ ion influx, as a result of high external osmolality, is a decrease in intracellular putrescine levels due to increased excretion. Putrescine is a divalent cationic polyamine; thus, a divalent putrescine is replaced by the monovalent K^+ ion, allowing cytoplasmic osmolality to increase with minimal effect on the intracellular ionic strength.

The major anionic compound involved in osmoregulation (**osmolyte**) is **glutamate**. Glutamate is synthesized and accumulates quickly following osmotic upshock and is dependent on K^+ ion uptake. In *E. coli* and other enteric bacteria, glutamate is synthesized by two enzymes: **glutamate dehydrogenase (GDH)** and **glutamate synthase (GS)**.

Many microbes also accumulate the **disaccharide trehalose** (composed of two glucose residues) as a **compatible solute** in response to osmotic stress as well as starvation, thermal stress, and desiccation stress. Trehalose is synthesized by the products of the *otsAB* operon. **OtsA** is a trehalose-6-phosphate synthase and **OtsB** is a trehalose-6-phosphate phosphatase. In the absence of exogenous osmoprotectants (see below), *E. coli* can accumulate trehalose to levels that may comprise as much as 20% of the cytoplasmic osmolality under conditions of high osmolality.

Several compounds, when present externally, can stimulate bacterial growth rates under hyperosmotic conditions. These compounds are **osmoprotectants**. They are zwitterionic in nature and resemble glycine betaine and proline.



Many potential osmoprotectants are found in plants and animals: betaine, betaine aldehyde, proline betaine, choline, choline-O-sulfate, stachydrine, and β -dimethyl-sulfonopropionate are present in plants, and carnitine is plentiful in animals. In *E. coli*, choline is converted to betaine aldehyde by the membrane-associated O_2 -dependent BetA choline dehydrogenase, and then to glycine betaine by the soluble NADP-dependent betaine aldehyde dehydrogenase. Ecotoine produced by many halophilic (salt-loving; marine) bacteria is an excellent osmoprotectant for *E. coli* and other bacteria. These compounds are taken up into the cell by two osmotically regulated permeases, ProP and ProU, in *E. coli* and *S. enterica* and homologs are found in other bacteria. ProP has relatively low affinity for proline and appears to function primarily in the uptake of betaine. The ProU permease also has relatively low affinity for proline but is a high-affinity betaine transporter. The major proline transporter is PutP, but it mainly functions in proline utilization and is not involved in osmoprotection (see Chapter 5 for its regulation).

Interestingly, these mechanisms of osmoprotection exhibit a hierarchy of function. Initially, the intracellular level of K^+ ions rapidly rises, maximizing by 20 minutes. Glutamate accumulation follows somewhat more slowly. The cell recovers from plasmolysis as turgor is restored. Trehalose levels rise and K^+ /glutamate levels decline after about 60 minutes. If an osmoprotectant such as betaine is added externally, trehalose concentrations decline and K^+ ion levels decrease even further. Restarting of macromolecular biosynthesis corresponds with a rise in trehalose levels.

Low Osmolality

Immediately following a significant decrease in osmolality (**hypoosmotic shock**), there is a rapid flux of water into the cell, increasing turgor. This results in stretching of the cell envelope and may lead to “cracks” in the membrane or may stretch existing pores or activate stretch-activated channels. The permeability of the membrane will be increased but only temporarily. Downshifts in osmolality can result in the extrusion of osmolytes and ions as well as the loss of amino acids, nucleotides, and other solutes from the cytoplasm.

Complex sugars, known as **membrane-derived oligosaccharides (MDOs)** that contain 6 to 12 glucose residues joined via $\beta(1 \rightarrow 2)$ and $\beta(1 \rightarrow 6)$ linkages, forming a branched structure, are found in the periplasm of Gram-negative bacteria. MDOs are substituted with *sn*-1-phosphoglycerol and phosphoethanolamine derived from the membrane phospholipids and also with *O*-succinyl ester residues. Synthesis of these compounds is induced by growth in conditions of low osmolality.

Osmotic Control of Gene Expression

EnvZ/OmpR is a paradigm of two-component systems (see Chapter 5). The outer membrane of *E. coli* K-12 contains several major outer membrane proteins. The relative amounts of two of them, coded by the *ompC* (47 min) and *ompF* (21 min) loci, are mediated by medium osmolality. **OmpC** (mwt 36,500 mw) and **OmpF** (mwt 37,000) are porin proteins that form aqueous pores in the outer membrane, allowing polar molecules (<600 da) to cross the outer membrane barrier. In media of low osmolality, OmpF protein is present in greater quantities than OmpC protein. In media of high osmolality, the OmpC porin predominates over OmpF. While the quantitative ratios of these proteins vary, their combined levels remain fairly constant.

Regulation of these genes (Fig. 18-1) is mediated by the regulatory *ompB* locus (21 min). The *ompB* region actually is comprised of two genes, *ompR* and *envZ*, arranged in an operon. **EnvZ** (50,300 Da) is a transmembrane sensor protein that undergoes **autophosphorylation** at histidine 243 under conditions of high osmolality. How EnvZ senses environmental change is unclear, since the periplasmic portion of the protein is apparently not required. Once phosphorylated, EnvZ will then transphosphorylate aspartate 55 in the receiver module of the true **DNA-binding regulator, OmpR** (27,400 Da), thereby increasing the level of OmpR~P in the cell. Phosphorylation of OmpR is required for the expression of both *ompC* and *ompF*. Presumably, conformational reshaping of OmpR by phosphorylation may increase DNA-binding affinity to the *ompC* and *ompF* promoters and enables interaction of OmpR with the α subunit of RNA polymerase, thereby activating transcription.

If OmpR~P is required for the transcription of both *ompF* and *ompC*, what could account for their reciprocal control? As shown in Figure 18-1, there are three OmpR-binding sites in front of *ompC* and *ompF*. At low osmolality, the OmpR~P present in the cell binds to *ompF* sites F1, F2, and F3, activating *ompF* transcription, but only binds to *ompC* site C1. At high osmolality, the higher level of OmpR~P resulting from EnvZ transphosphorylation now binds to C1, C2, and C3, activating *ompC* transcription. It is strange that this level of OmpR~P still binds at F1, F2, F3 but now represses *ompF*. It is proposed that when OmpR~P is bound to DNA, changes in osmolality affect OmpR~P-OmpR~P protein interactions, which in turn alter the ability of OmpR to activate transcription at *ompF* (Fig. 18-1). Osmotic downshift—that is, when osmolality shifts from high back to low—activates an EnvZ phosphatase that removes P from both OmpR and EnvZ, reversing the OmpC/OmpF expression ratio. A second phosphatase, SixA, may also be involved in this process.

A variety of studies indicate that alternative **phosphodonors** are capable of phosphorylating OmpR in the absence of EnvZ. One proven alternative phosphodonor is **acetyl phosphate**, although there may be additional cross-talk with other histidine kinases.

An additional control mechanism that appears to regulate the relative amounts of OmpC and OmpF involves a gene (*micF*) adjacent to *ompC*, which is transcribed

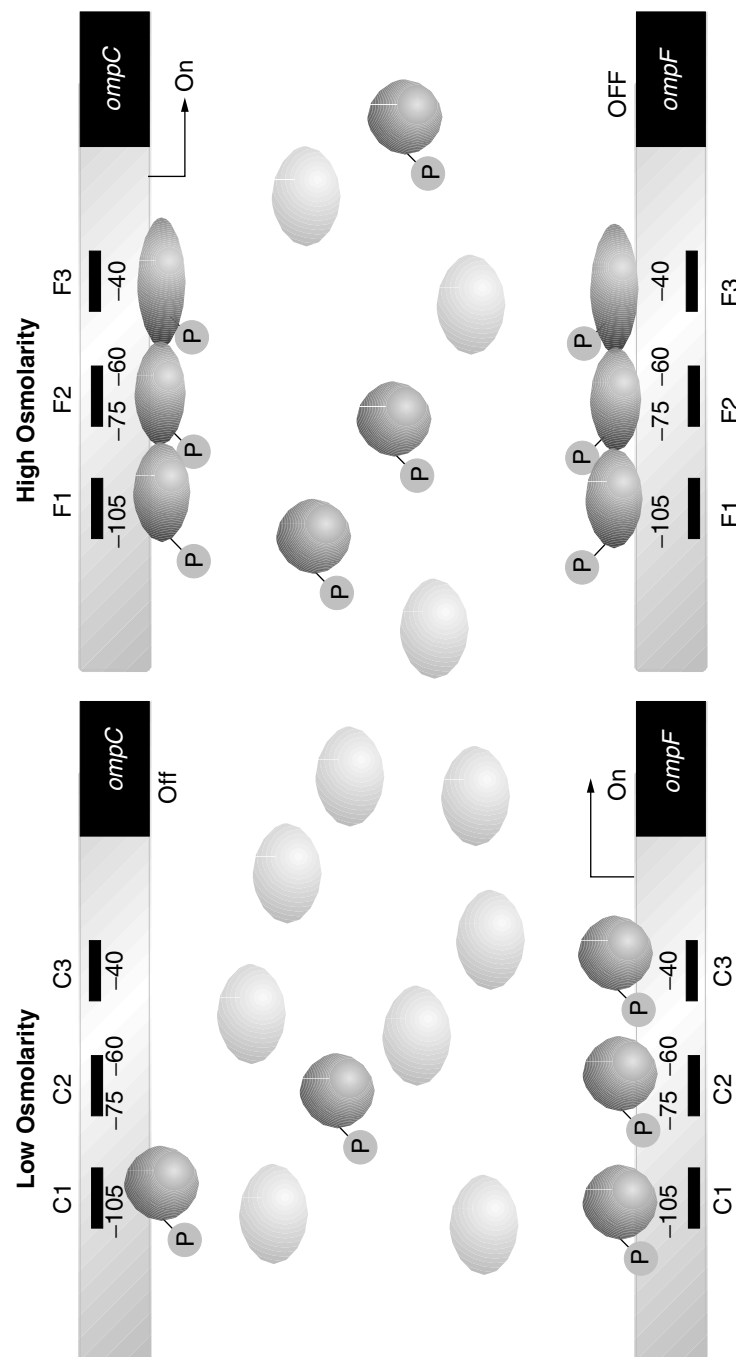


Fig. 18-1. Osmoregulation of major outer membrane proteins OmpF and OmpC.

in the opposite direction from *ompC*. The *micF* transcript (**micRNA**, 93 nucleotides) is complementary to the 5' end region of *ompF* mRNA. Under conditions of high osmolarity, the *micF-ompC* region is induced and transcribed divergently from a central promoter. The resulting micRNA (also referred to as **antisense RNA**) inhibits translation of *ompF* mRNA by hybridizing to it. This RNA–RNA interaction is proposed to cause premature termination of *ompF* transcription and/or destabilization of *ompF* mRNA. The end result is that when more OmpC protein is produced, less OmpF protein is synthesized.

The role of the EnvZ-OmpR system in cellular physiology extends well beyond its role in governing porin expression. Studies have connected OmpR with flagellar expression, cell division, fatty acid transport, microcin synthesis, curli fibers, *Salmonella* virulence, and acid tolerance. One mechanism by which OmpR may affect virulence is through its involvement in controlling cytotoxicity toward infected macrophages. In addition, OmpR controls the expression of another two-component regulatory system, SsrAB, which controls expression of genes in the *Salmonella* pathogenicity island SPI-2. Recent evidence indicates that OmpR autoregulates its own expression in response to acid pH, providing a partial explanation for why the SPI-2 genes are induced by low pH.

AEROBIC TO ANAEROBIC TRANSITIONS

Facultative microorganisms such as *E. coli* and *S. enterica* are capable of modifying their metabolism to accommodate growth under either aerobic or anaerobic conditions. The transition between aerobic and anaerobic metabolism is accompanied by alterations in the rate, route, and efficiency of pathways of electron flow. Figure 18-2 illustrates the basic pathways utilized by *E. coli* for aerobic versus anaerobic electron flow. Under anaerobic conditions without alternate electron acceptors, pyruvate is converted to formate, acetate, or ethanol, CO₂ and H₂ gas (mixed acid fermentation). However, the choices and energy yield become more plentiful when alternate electron acceptors are available. *E. coli*, even under aerobic conditions, synthesizes two distinct **cytochrome oxidases**—**cytochrome *o* (*cyo* operon)** and **cytochrome *d* (*cyd* operon)**—produced under high O₂ and low O₂ conditions, respectively. Under anaerobic conditions, at least five more terminal oxidoreductases can be produced (Table 18-1).

TABLE 18-1. Characteristics of Electron Transport Systems in *E. coli*

Electron Acceptor	δG^{Oa} (kJ/mol)	Terminal Respiratory Enzyme	Operon	Chromosomal Location (min)
O ₂	–233	Cytochrome <i>o</i> oxidase	<i>cyoABCDE</i>	10
O ₂	–233	Cytochrome <i>d</i> oxidase	<i>cydAB</i>	17
NO ₃ [–]	–144	Nitrate reductase	<i>narGHJI</i>	27
NO ₃ [–]	–144	Nitrate reductase	<i>narZYWV</i>	33
DMSO	–92	DMSO/TMAO reductase	<i>dmsABC</i>	20
TMAO	–87	TMAO reductase	<i>torA</i>	28
Fumarate	–67	Fumarate reductase	<i>frdABCD</i>	94

^aFree energy calculated by using NADH as an electron donor to the indicated electron acceptor.

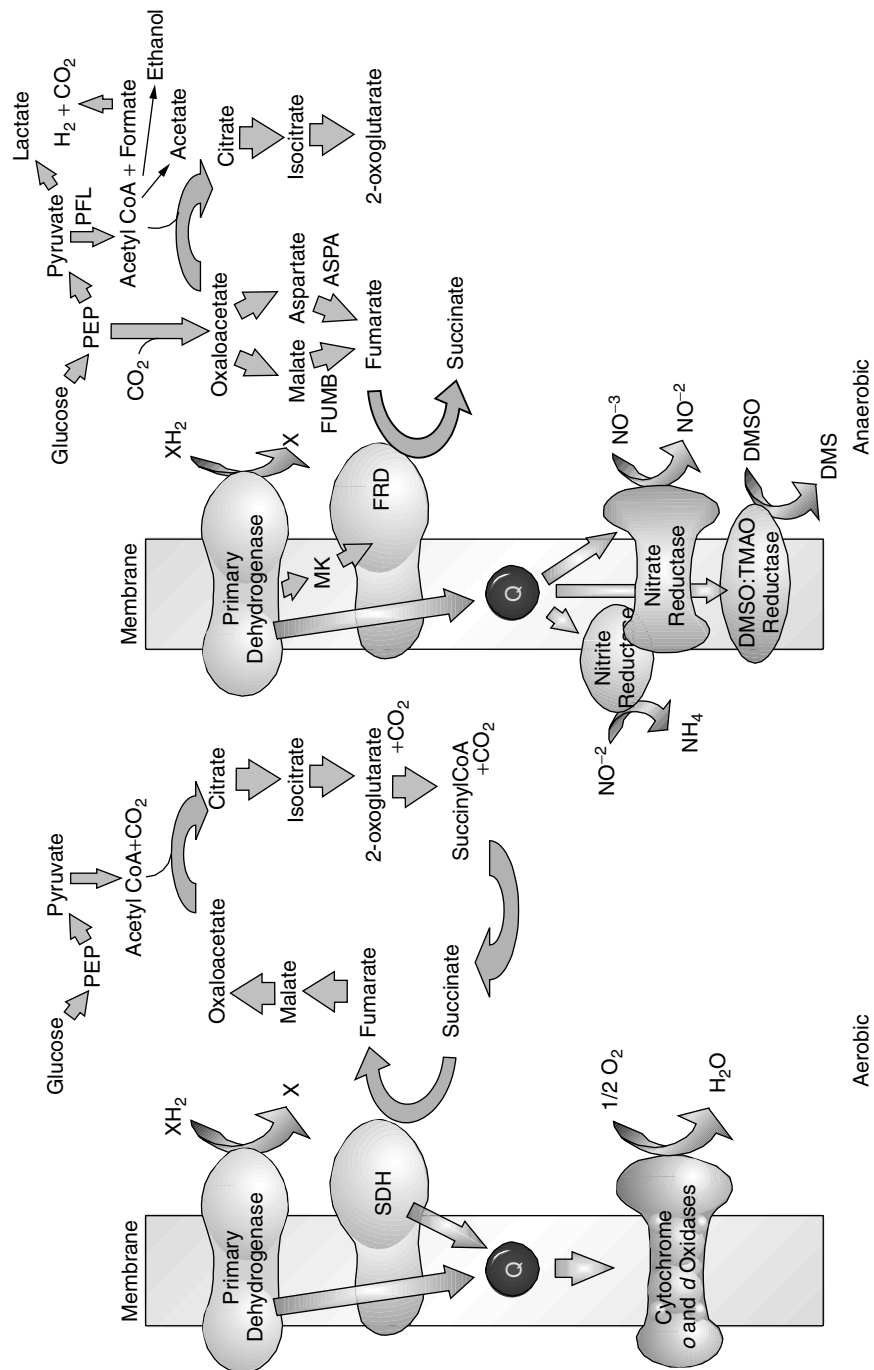


Fig. 18-2. Basic aerobic and anaerobic metabolic pathways of *E. coli*.

E. coli controls the production of the various respiratory pathway enzymes, first, in response to aerobic and anaerobic growth conditions and, second, to the availability of alternate electron acceptors. Clearly there is a hierarchy or preference for substrate use with the following order: oxygen > nitrate > DMSO > TMAO > fumarate. When several e^- acceptors are present simultaneously, the more energetically favored acceptor will be used first. An interesting question is how does the cell regulate such a complex system. There are three basic regulators described that sense changes in oxygen level or redox conditions. Their overlapping control circuits are illustrated in Figure 18-3.

Formate Nitrate Regulation

The **Fnr protein** (formate nitrate regulation) regulates over a hundred genes in response to the presence or absence of oxygen. The manner in which Fnr senses oxygen appears to involve an **oxygen-sensitive (4Fe-4S) center** (Cys-X₃-Cys-X₂-Cys-X₅-Cys). Active Fnr is a dimer and, therefore, contains two such centers. Exposure to oxygen causes partial disassembly of the two (4Fe-4S) centers to form two (2Fe-2S) centers. This disassembly converts Fnr dimers, which bind DNA, to monomers that are not active (Fig. 18-4). Under anaerobic conditions, Fnr represses *cyoABCDE*, *cydAD* and *narZYWV* (Fig. 18-3) but transcriptionally activates *sdhCDAB* (**succinate dehydrogenase**), *frdABCD* (**fumarate reductase**), *dmsABC* (**DMSO reductase**), and *narGHJI* (**nitrate reductase**). Fnr bears significant sequence homology to the CRP protein and, like CRP, alters transcription through DNA bending and direct interactions with the α -subunit of RNA polymerase. In contrast to CRP, there is no evidence for cAMP involvement with Fnr.

Nitrate Response

Nitrate is the preferred **electron acceptor** for anaerobic cells because of the high midpoint potential of the nitrate/nitrite couple. In addition to *fnr*, the ***narXL* operon** and ***narQ narP*** are required to regulate respiratory gene expression in response to nitrate availability. DNA sequence analysis indicates *narX narL* and *narQ narP* are two-component regulatory systems that mediate a series of complex transcriptional adjustments in response to a dynamic ratio of two alternate electron acceptors: nitrate, and its reduction product, nitrite. **NarQ** and **NarX** are **histidine kinases** that sense nitrate and transmit a signal to their cognate response regulators **NarP** and **NarL**, respectively. This system contributes to the induction of *frd*, *dms*, and ***narGHIJ* operons** but has no effect on *cyo*, *cyd*, or *narZYWV*.

The control of the nitrite reductase operons *nir* and *nrf* by NarP and L offers a glimpse of the complex adjustments that occur in response to the cell's needs. Nitrate, at low concentration, induces nitrate reductase and the high-affinity Nrf nitrite reductase via the NarXL and NarQP systems. Thus, the cell will derive maximum energy yield from the nitrate present. However, at high nitrate concentrations, nitrate reductase is still expressed, but NarL will actually repress the **Nrf nitrite reductase** while inducing the low-affinity **NirB nitrate reductase** system. This differential regulation avoids generating too much proton motive force and energy that would result from using the high-affinity nitrite reductase under high nitrate conditions when plenty of energy is derived from nitrate reductase. Nevertheless, the low-affinity nitrite reductase is still necessary to detoxify the nitrite produced.

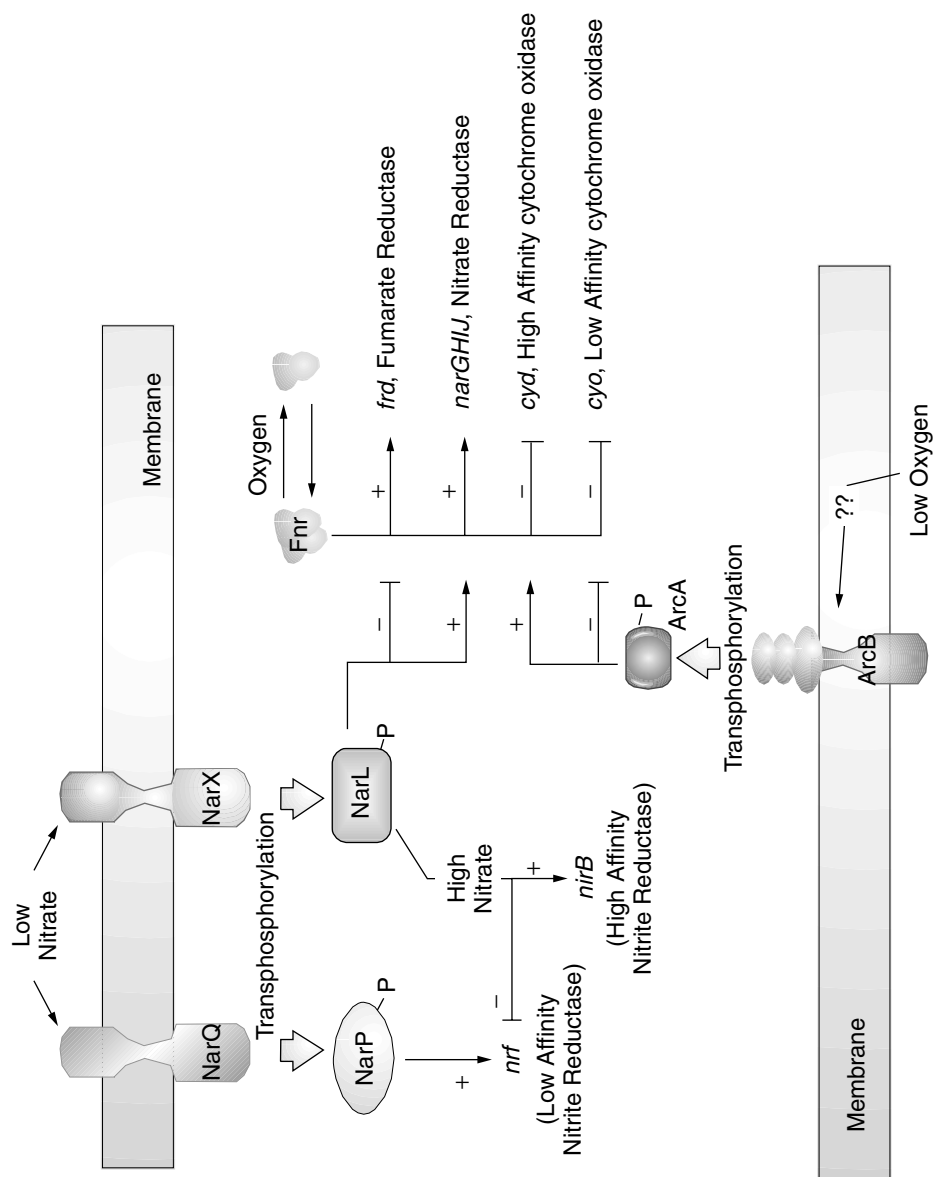


Fig. 18-3. Regulatory scheme for the control of aerobic and anaerobic respiratory pathways in *E. coli*.

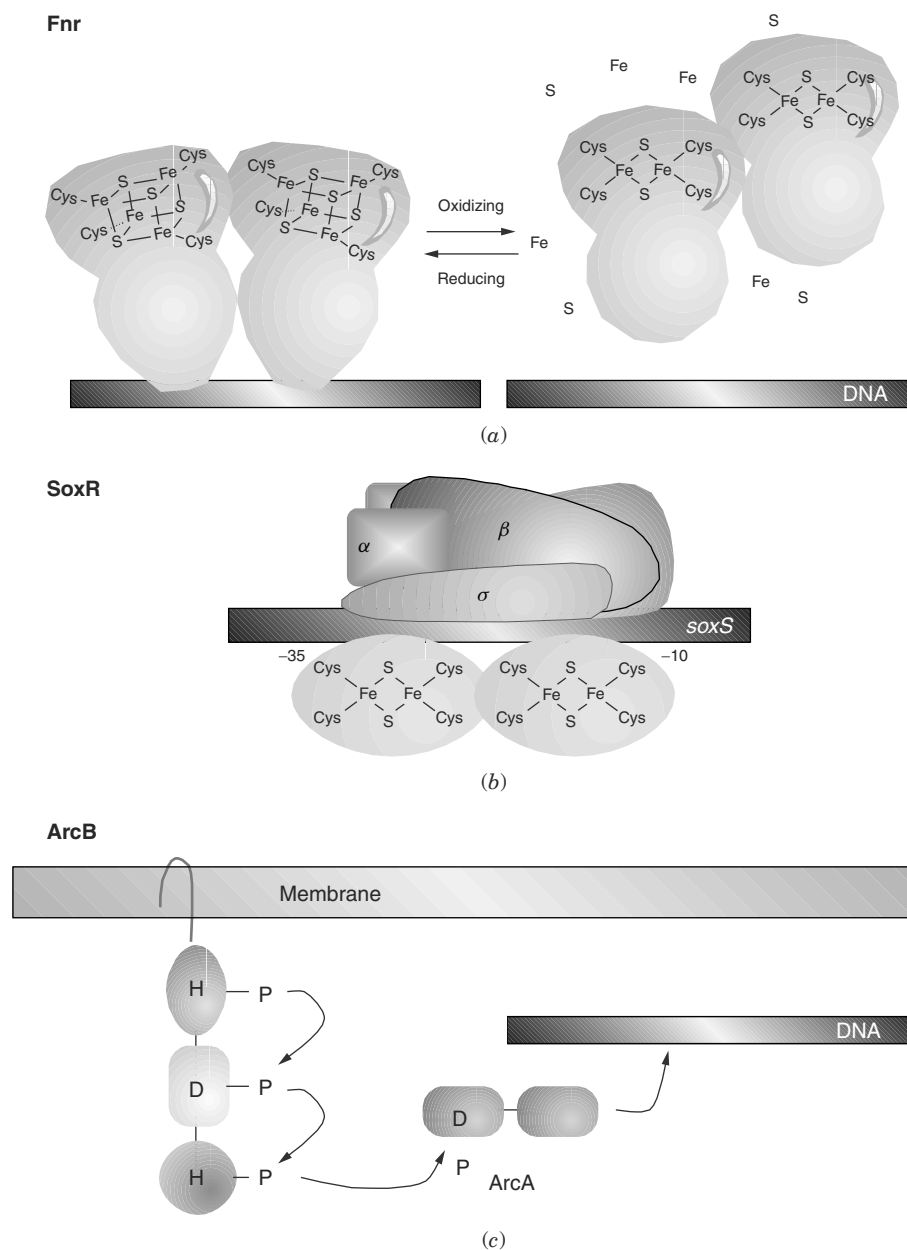


Fig. 18-4. Schematic representation of Fnr, SoxR, and ArcB function.

ArcAB System

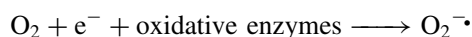
The system that controls *cyo* and *cyd* includes the *arcA* and *arcB* products (*arc*, meaning anoxic redox control) comprising a two-component regulatory system. The system actually regulates a total of 30 genes. **ArcB** functions as the membrane-bound sensor/transmitter that communicates with **ArcA**, the receiver/regulator. ArcB is an

unorthodox histidine kinase in that it contains three phosphotransfer signaling domains (Figs. 18-3 and 18-4). The first domain, the **histidine-kinase** module, phosphorylates itself at a conserved histidine (H292). The phosphate is subsequently transferred, first to an aspartate residue (D576) in the receiver module and then to a second histidine (H717) in the phosphotransfer domain located at the carboxyl end of ArcB. The phosphate is subsequently transferred to an aspartate residue (D54) within the amino terminus of ArcA.

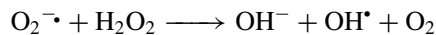
The Arc system regulates the coordinate synthesis of tricarboxylic acid cycle enzymes. ArcA~P is generally a repressor of aerobically expressed target genes [e.g., *acn* (aconitase), *cyo*, *glbA* (citrate synthase), *sdh* (succinate dehydrogenase)] and an activator of anaerobically expressed target genes [*cyd*, *hya* (hydrogenase 1), *pdu* (propanediol degradation)]. The **SixA phosphohistidine phosphatase** modulates the system by removing phosphate from ArcB residue H717, downregulating phosphotransfer. The nature of the sensed low oxygen signal, how it is sensed, and how the signal is transduced to activate ArcB kinase are unclear. A model in which ArcB senses proton motive force has been refuted, since the only amino acid (H47) with a physiologically relevant pK within the transmembrane region has proven dispensable. Other models invoke the ArcB-sensing redox state of the cell and/or metabolites (e.g., lactate, acetate) generated by anaerobic metabolism. Figure 18-3 presents a model depicting the regulatory interactions of Nar, Fnr, and Arc that govern selected oxygen-regulated genes.

OXIDATIVE STRESS

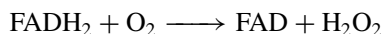
Generation of energy by the electron transport chain is dependent on the catalytic spin pairing of triplet oxygen. During this process, oxygen species that are toxic to DNA, protein, and lipid components of the cell are formed through both enzymatic and spontaneous chemical reactions. The reaction of oxidative enzymes with molecular oxygen can generate superoxide (superoxide anion):



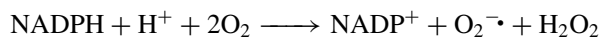
Superoxide is relatively unreactive with DNA and proteins. However, it may interact in a number of enzymatic as well as spontaneous chemical reactions to produce more highly reactive oxygen derivatives such as **hydrogen peroxide** and **hydroxyl radicals**:



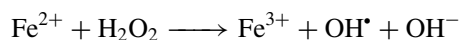
Autooxidation of reduced FAD or reduced flavoprotein gives rise to hydrogen peroxide:



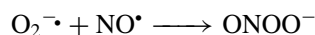
The enzyme **NADPH oxidase** can generate superoxide anion and hydrogen peroxide:



Superoxide can release Fe^{2+} and Fe^{3+} from various cellular compounds and ultimately give rise to hydroxyl radicals through the **Fenton reaction**:

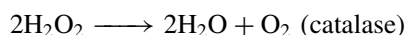
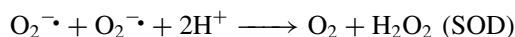


Reaction of superoxide anion with **nitric oxide** forms **peroxynitrite anion**:

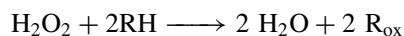


Peroxynitrite is highly reactive with various proteins. Methionine, cysteine, tyrosine, and tryptophan residues of proteins are especially vulnerable. Peroxynitrite can also give rise to other toxic derivatives that are highly reactive with biological compounds.

Most aerobic organisms are protected from the toxicity of superoxide and hydrogen peroxide by the enzymes **superoxide dismutase (SOD)** and **catalase**:

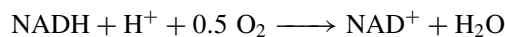


Peroxidases can also catalyze the reduction of hydrogen peroxide by organic reductants such as **glutathione** or **ascorbic acid**:



E. coli produces a cytoplasmic **Mn-SOD (SodA)** and **Fe-SOD (SodB)** that protect DNA and proteins from oxidation. Mutants deficient in these enzymes display enzyme inactivation, growth deficiencies, and DNA damage. A periplasmic **Cu/Zn-SOD (SodC)** protects the periplasmic and membrane constituents from exogenous superoxide. Several DNA repair systems also aid in recovery from oxidative damage (see Chapter 2).

The presence of SOD and catalase in anaerobic organisms would appear to be self-defeating since oxygen is a product of both reactions. Anaerobes must rely on other mechanisms to eliminate oxygen and prevent the formation of toxic oxygen species. One method used by organisms that do not carry out oxygen-dependent respiration involves a unique flavoprotein, **NADH oxidase**, which catalyzes the direct four-electron reduction of oxygen to water:



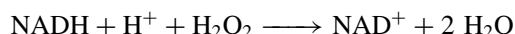
This enzyme actually allows streptococci and other lactic acid bacteria to use oxygen directly in the metabolism of carbohydrates without the inherent problems of oxygen toxicity caused by the formation of reactive oxygen derivatives.

A superoxide reductase system that provides protection against oxidative stress is present in the anaerobic sulfate-reducing bacterium *Desulfovibrio vulgaris* and in the hyperthermophilic anaerobe *Pyrococcus furiosus*. It has the advantage of eliminating superoxide without the formation of molecular oxygen. This system consists of the

nonheme iron-containing proteins **rubrerythrin (Rbr)** and **rubredoxin oxidoreductase (Rbo)** that reduce superoxide to hydrogen peroxide without dismutation:



This system functions in conjunction with an **NADH peroxidase** that reduces hydrogen peroxide to water:



These and other mechanisms used by anaerobic or facultative organisms to protect against oxygen damage are discussed later in reference to organisms that utilize primarily fermentation pathways in their metabolism.

Regulation of the Oxidative Stress Response

The natural by-products of aerobic metabolism are the reactive compounds superoxide ($\text{O}_2^{\cdot-}$) and hydrogen peroxide. These two species can lead to the generation of hydroxyl radicals ($\text{OH}^{\cdot}$), which can damage biological macromolecules. The oxidation stress modulon includes at least 80 proteins induced during exposure to superoxide. About half are also induced by H_2O_2 . Research efforts have uncovered two regulons within this stress response, although there are certainly more. The two known systems, **OxyR** and **SoxRS**, used by *E. coli*, are outlined in Figure 18-5.

The **OxyR regulon** comprises nine of the proteins induced by H_2O_2 . All are controlled by the positive regulator OxyR. OxyR is a class I type activator that, once bound to a target DNA sequence, interacts with the α C-terminal domain of RNA polymerase (see Chapter 5). The protein is activated by oxidation, which results in the formation of a disulfide bond between cysteine residues C199 and C208. Under noninducing conditions these cysteines are reduced by glutaredoxin I. For most OxyR-dependent genes, the oxidized but not the reduced form of OxyR binds target DNA sequences. However, the reduced form of this regulator can bind to the *oxyR-oxyS* target operator/promoter region but will not activate transcription until OxyR itself is oxidized. Some of the enzymes whose expression is regulated by OxyR include **catalase (*katG*)**, **glutathione reductase (*gorA*)**, **glutaredoxin I (*grxA*)**, and **alkyl hydroperoxide reductase (*ahpC* and *ahpF*)**. The likely in vivo function for alkyl hydroperoxide reductase would be the detoxification of lipid and other hydroperoxides produced during oxidative stress.

In *Salmonella*, reduced-OxyR was found to act as a repressor of the ***narZYWV* locus**, which encodes a stress-inducible aerobically expressed nitrate unresponsive **nitrate reductase (NR-Z)**. In the presence of H_2O_2 , the oxidized-OxyR no longer represses *narZYWV* transcription and the operon becomes derepressed to about one-third of its maximal induced level of expression. Other loci have also been shown to be repressed by OxyR. Thus, OxyR, like many other response regulators, can function as both a transcriptional activator and repressor protein, depending on the target gene.

A second oxidative stress regulon comprises nine proteins induced by superoxide but not hydrogen peroxide. This regulon is under positive transcriptional control by the ***soxRS* loci**. Genes under *soxR* control include those responsible for **Mn^{2+} -containing superoxide dismutase (*sodA*)**, the DNA repair enzyme **endonuclease**

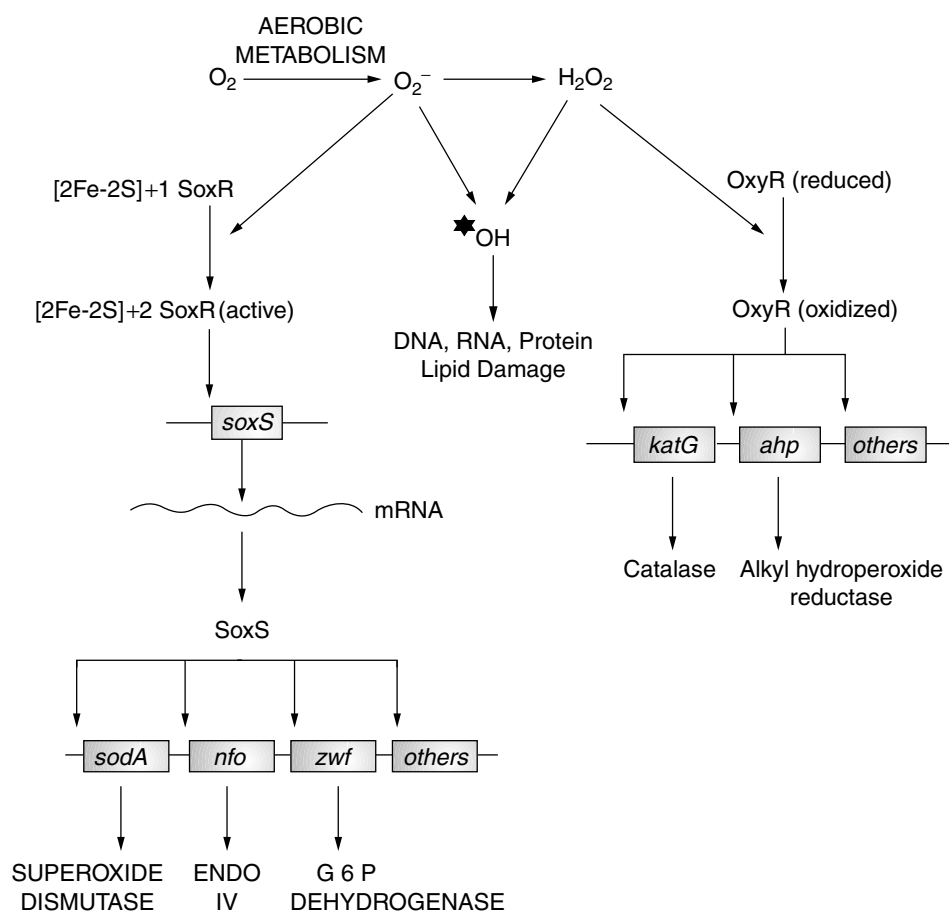


Fig. 18-5. The *E. coli* response to oxidative damage.

IV (*nfo*), glucose-6-phosphate dehydrogenase (*zwf*, production of NADPH), *fldA* (flavodoxin 1), NADPH:flavodoxin oxidoreductase (*fpr*), and fumarase C (*fumC*). The current model for this system is that preexisting SoxR protein senses oxidative stress and then triggers expression of the *soxS* gene. The SoxS product then activates transcription from the promoters of the other members of the regulon.

As with Fnr, SoxR utilizes an Fe sulfur center to sense oxidative stress conditions in the cytoplasm. SoxR contains two stable (2Fe-2S) centers anchored to four cysteine residues near its carboxy terminus (Fig. 18-4). These (2Fe-2S) centers remain reduced (+1 state) under normal physiological conditions but rapidly oxidize (+2 state) when challenged with oxidative stress. It is believed that iron-sulfur centers are especially sensitive to redox reactions with superoxide. The change in redox state of the protein induces an active conformation of SoxR but does not change binding affinity.

Both oxidized and reduced forms of SoxR bind to the *soxS* promoter. Thus, oxidation, while not affecting binding, must enable SoxR to interact with RNA polymerase, stimulating open complex formation. There is some evidence that oxidized SoxR is reduced by the NADPH:flavodoxin oxidoreductase/flavodoxin couple

with the oxidation of NADPH. The genes encoding these enzymes are themselves induced by SoxR, which suggests a mechanism of SoxRS autoregulation—that is, as the system is induced by oxidized SoxR (e.g., aerobic growth), the levels of the oxidoreductase/flavodoxin couple increase, which will reduce SoxR and downregulate the system.

pH STRESS AND ACID TOLERANCE

Microbes can grow over a wide range of hydrogen ion concentrations (pH). For example, **acidophilic bacteria** will grow in acidic sulfur springs where a pH of 1 is common. In contrast, **alkalophilic bacteria** prefer to grow in environments such as soda lakes where pH conditions rise as high as pH 11. Most bacteria in the human sphere, however, prefer to grow at pH values closer to neutral and are called **neutrophilic**. They generally grow over a range of pH 5 to pH 9 and include *E. coli*, *S. enterica*, *Streptococcus lactis*, *Bacillus subtilis*, and many others. Although the neutrophiles do not generally grow under conditions of extreme acid or base, they can survive these exposures to various degrees if they are allowed to go through an adaptive transition whereby pH gradually changes.

It is well established that one way microbes respond to an acidifying pH is by producing enzymes that can convert acidic metabolites to neutral ones or neutral metabolites to alkaline products. Good examples of these types of enzymes are glutamate decarboxylase, lysine decarboxylase, and arginine decarboxylase of *E. coli*, all of which exhibit increased expression at external acidic pH. The concept of **pH homeostasis** or the ability of the microbe to control its internal pH is discussed in Chapter 9. It has been demonstrated that the major mechanisms typically used by Gram-negative organisms to control internal pH during growth involves the modulation of the primary proton pumps as well as the K^+/H^+ and Na^+/H^+ antiporters. However, additional adaptive mechanisms are engaged to survive under pH conditions outside the growth range.

Exposures of *S. enterica* and *E. coli* to a nonlethal acidic pH between pH 5 and 6 results in the induction of sets of genes whose products can protect the cell when exposed to potentially lethal pH conditions from pH 4 to pH 2. These inducible systems protect the cell at external pH values where the nonadaptive pH homeostasis mechanisms fail. The genetic and physiologic changes that occur in the cell are referred to as the **acid tolerance response** (ATR). Full induction of the ATR results in the increased expression of at least 50 newly synthesized or existing proteins called **acid shock proteins**, some of which are also heat shock induced.

One system important to acid tolerance in some organisms is the Mg^{+2} -dependent proton translocating ATPase (see “Energy Production” in Chapter 9). In Gram-positive organisms such as *Enterococcus faecalis*, the ATPase, which normally harnesses proton motive force to generate ATP, can work in reverse, hydrolyzing ATP to extrude a proton from the cell. As opposed to Gram-negative organisms that struggle to keep internal pH near 7.8, the streptococci are more flexible. Their only requirement is to maintain a pH of 0.5 to 1 unit—a goal the ATPase alone could accomplish.

The superior acid resistance of *E. coli* and *Shigella* over that of *S. enterica* is of particular interest from a medical view, since it helps explain the differences in infectious dose required by these organisms to overcome the **gastric acid barrier** and

cause disease (about 10 organisms for *E. coli* and *Shigella* vs 10,000 for *Salmonella*). The primary reason for this difference is the production of **glutamate decarboxylase**, an enzyme missing from *S. enterica*. Besides the enzyme proper, this system includes a specific antiporter that links export of γ -amino butyric acid (the decarboxylation product of glutamate) to the import of more glutamate. The pH optimum for this enzyme is around pH 5 and thus it does not play a role until internal pH falls to that level. Although initially thought to have a part in consuming and extruding intracellular protons, how this system really helps the cell survive pH 2 environments is not clear. Another organism that uses this system is the Gram-positive *Lactococcus lactis*.

In addition to glutamate decarboxylase, *Lactococcus* uses an **arginine deiminase** (ADI) pathway to generate ATP during acid stress. ADI converts arginine to citrulline and ammonia, ornithine transcarbamylase converts the citrulline to ornithine and carbamyl phosphate, and carbamate kinase cleaves carbamyl phosphate to ammonia and CO₂, generating an ATP in the process. The system also requires an arginine/ornithine antiporter to rid the cell of the end-product ornithine while importing new arginine substrate. The ammonia generated can alkalize the environment, but since this system also helps survival of dilute cultures, the ATP generated must substantially aid in survival.

Helicobacter pylori, an important cause of gastric ulcers and cancer, utilizes another potent acid resistance system to survive in the stomach. The organism constitutively produces a powerful **urease**, a nickel-containing metalloenzyme that converts urea to carbon dioxide and ammonia. Contrary to what would be expected, the pH optimum of urease is around 7.5, raising the question of why cells grown at pH 7.5 have very little urease activity. There are two reasons. Although the urease is constitutively produced, the degradation of transcripts encoding nickel-incorporating enzymes appear subject to pH control. In addition, UreI is a membrane transporter of urea that only becomes activated at acid pH. Therefore, urea cannot gain access to intracellular urease except at low pH. Like the glutamate decarboxylase system, it is not entirely clear how urease provides acid resistance. The ammonia produced does not seem to be sufficient protection, since the system works even at low cell density—a condition where the small amount of ammonia produced will not change external pH. *Yersinia enterocolitica* is another Gram-negative enteropathogen that uses urease to gain safe passage through the stomach.

Much remains to be learned regarding how different bacteria cope with acid or alkaline threats to survival. What is clear is that the microbial world has taken full advantage of what is available in the environment to counter this form of stress and that successful strategies have been shared among species.

THERMAL STRESS AND THE HEAT SHOCK RESPONSE

As with pH, microbes exhibit a wide range of temperatures at which they can grow (Table 1-4). For example, bacteria can be isolated from hot springs where temperatures can reach as high of 90 °C or so (**thermophilic bacteria**) and from frozen tundra or polar caps where temperatures are below 0 °C (**psychrophilic bacteria**). Most bacteria, however, prefer to grow at milder temperatures such as 20 ° to 40 °C (**mesophilic bacteria**). This latter group includes *E. coli* and *S. enterica* serovars as well as other animal and human pathogens.

Upon a shift from 30° to 42°C, *E. coli* and other bacteria transiently increase the rate of synthesis of a set of proteins called **heat shock proteins (HSPs)**. Many of these HSPs are required for cell growth or survival at more elevated temperatures (**thermotolerance**). Among the induced proteins are **DnaJ** and **DnaK**, the **RNA polymerase σ^{70} subunit (*rpoD*)**, **GroES**, **GroEL** (see “Protein folding and Chaperones” in Chapter 2), **Lon protease** (see “Proteolytic Control” in Chapter 2), and **LysU**. There are nearly 50 heat shock-inducible proteins identified in *E. coli*. They can be subdivided into those regulated by alternative σ factors and two component regulatory systems.

The **σ^H regulon** provides protection against cytoplasmic thermal stress. The *E. coli* ***rpoH* locus** (formerly called *htpR*) encodes a 32 kDa σ factor, alternatively called σ^H or σ^{32} , which redirects promoter specificity of RNA polymerase. The σ^H protein regulates the expression of 34 heat shock genes. The simple explanation for how heat shock increases expression of the σ^H regulon is that heat shock first causes an elevation in σ^H levels, which in turn increases expression of the σ^H target genes.

Although the principle is simple, the controls governing σ^H production are complex. First, a temperature upshift from 30° to 42°C results in the increased translation of *rpoH* message. *Cis*-acting mRNA sites within the 5' region of *rpoH* message form temperature-sensitive secondary structures that sequester the ribosome-binding site. At higher temperatures, these secondary structures melt, thereby enabling more efficient translation of the *rpoH* message. In addition to the increased translation of *rpoH* message, the σ^H protein itself becomes more stable, at least transiently. The mechanism regulating proteolysis centers on whether σ^H associates with RNA polymerase. During growth at 30°C, σ^H can be degraded by several proteases including **FtsH**, **HslVU**, and **ClpAP**. However, if σ^H is bound to RNAP, σ^H is protected from degradation.

The cell uses the **DnaK-DnaJ-GrpE chaperone team** to interact with σ^H at low temperature, sequestering σ^H from RNA polymerase (Fig. 18-6). Failure to bind RNAP facilitates degradation of the σ factor. Upon heat shock, there is an increase in the number of other unfolded or denatured proteins that can bind to DnaK or DnaJ. This reduces the level of free DnaK/DnaJ molecules available to bind σ^H , allowing σ^H to bind RNAP, which protects σ^H from degradation. As the cell reaches the adaptation phase following heat shock, the levels of DnaK and DnaJ rise (both are induced by σ^H) and can again bind σ^H , redirecting it toward degradation. Nevertheless, even though σ^H degradation resumes, translation of *rpoH* remains high at the elevated temperature and σ^H continues to accumulate, although at a slower rate.

In addition to translational and proteolytic controls, production of σ^H is regulated at the transcriptional level via a feedback mechanism. There are four promoters driving *rpoH* expression, three of which are dependent on σ^{70} , the **housekeeping σ factor**. The gene encoding σ^{70} , ***rpoD***, is also a heat shock gene induced by σ^H . So increased production of σ^H increases σ^{70} , which increases transcription of *rpoH*. The fourth *rpoH* promoter is recognized by another σ factor, σ^E , encoded by *rpoE*. The heat shock response is also triggered by a variety of environmental agents such as ethanol, UV irradiation, and agents that inhibit DNA gyrase. Induction by all of these stimuli occurs through σ^H . How can all of these seemingly diverse stresses activate *rpoH*? The only explanation that appears reasonable is the accumulation of denatured or incomplete peptides. There is a potential **alarmone** that has been implicated in signaling expression of this global network. The molecule is **diadenosine 5', 5'''-P¹, P⁴-tetraphosphate**

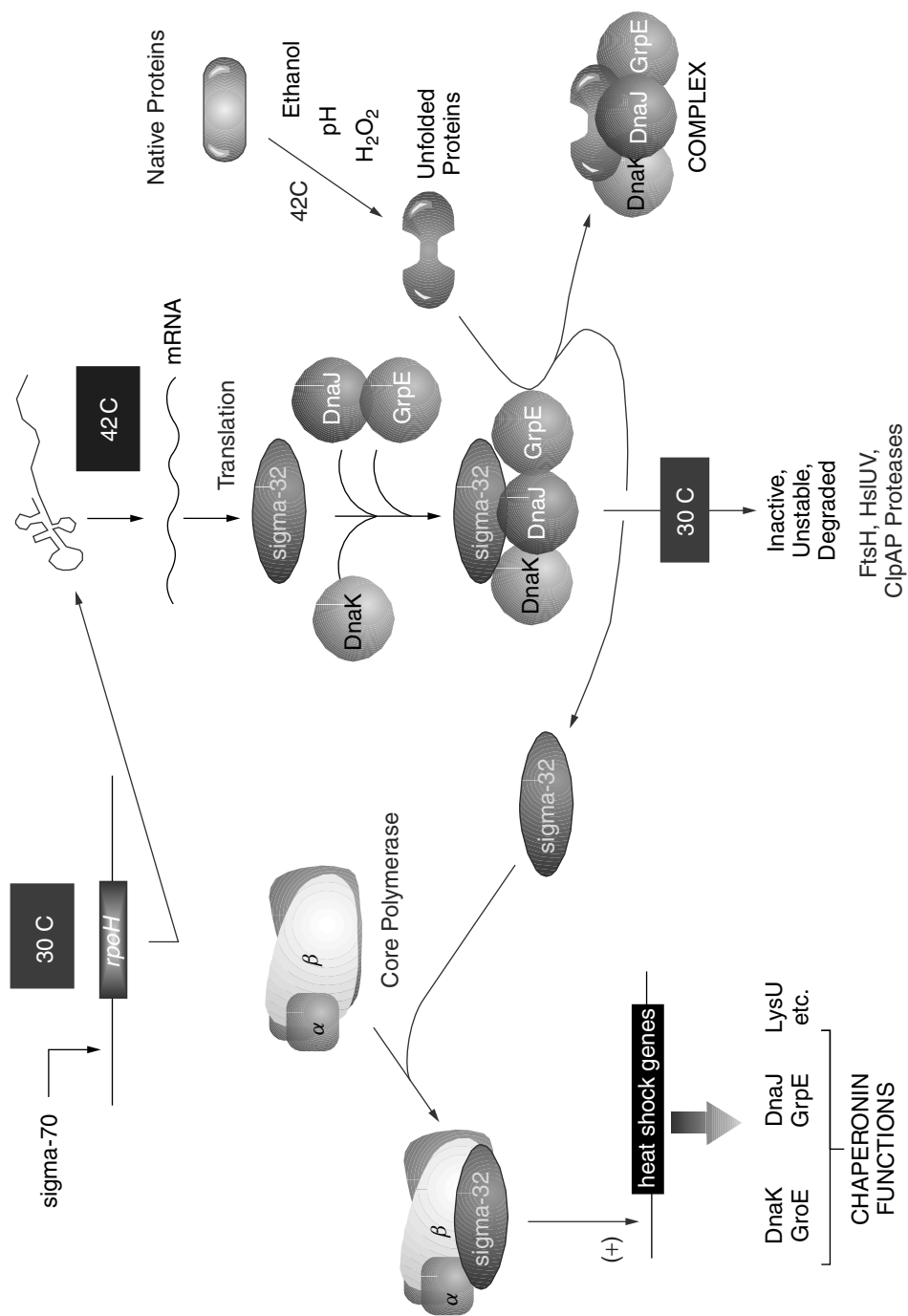


Fig. 18-6. Modulation of the heat shock response.

(**AppppA**), which is made by some aminoacyl-tRNA synthetases (e.g., *lysU*) at low tRNA concentrations. How this may influence the response is not known.

The σ^E **regulon** provides protection against extracytoplasmic stress. The hallmark of the Gram-negative cell is the existence of two membrane-bound subcellular compartments: the cytoplasm and the periplasm. Conditions in each of these compartments differ markedly. The cytoplasm is energy rich, reducing (low redox potential), and osmotically stable, whereas the periplasm lacks ATP, is oxidizing, and is in contact with the external milieu. Since optimal cell growth requires that the cell senses and responds to changes in these disparate subcellular compartments, it is not surprising that the stress responses in *E. coli* and *S. typhimurium* are compartmentalized into cytoplasmic and extracytoplasmic responses. The extracytoplasmic response pathways involve two partially overlapping signal transduction cascades: the σ^E and **Cpx systems**. These pathways are induced following the accumulation of misfolded proteins in the periplasm as a result of stresses such as high temperature, pH extremes, or carbon/energy starvation.

σ^E is a member of the **extracytoplasmic function (ECF) subfamily** of σ factors. In *E. coli*, σ^E is responsible for the transcription of up to genes, including *rpoH* (σ^H), *degP* (*htrA*) encoding a periplasmic protease for the degradation of misfolded proteins, *fkpA* encoding a periplasmic peptidyl prolyl isomerase, and *rpoE rseABC* operon (see below).

The gene encoding σ^E , *rpoE*, is the first member of an operon followed by the genes *rseA*, *B*, and *C* (*rse*, meaning regulators of σ^E). **RseA** is a transmembrane protein whose cytoplasmic C-terminal domain interacts with σ^E , acting as an anti- σ factor. The periplasmic face of RseA binds to the **periplasmic RseB**. Extracellular stress in some way signals increased proteolysis of RseA by the **periplasmic protease DegS**, thus relieving the anti- σ effect of RseA on σ^E . It has been proposed that RseB and perhaps other periplasmic proteins involved in protein folding protect RseA from degradation by binding to the RseA periplasmic domain, capping the target site of DegS. Stress-induced misfolding of periplasmic proteins would titrate the RseA cap proteins off of RseA, rendering the anti- σ factor vulnerable to attack by DegS. The result would be increased activity of σ^E leading to increased levels of σ^E protein and RseA anti- σ (since they form an operon). The increased amount of σ^E will drive further expression of genes whose products handle the periplasmic damage while the increased level of RseA will enable the cell to downregulate the system once the capping proteins are again free to bind and protect RseA from DegS degradation.

A second system dedicated to protecting the periplasmic perimeter of the cell is the CpxRA two-component system (see Chapter 5) with **CpxA** playing the role of membrane-localized sensor histidine kinase and **CpxR** as the cytoplasmic response regulator. CpxA responds to envelope stress by autophosphorylation followed by phosphotransfer to CpxR. CpxR~P activates expression of *dsbA* (**disulfide oxidoreductase**), *ppiA* and *ppiD* encoding **peptidyl-prolyl-isomerases**, and, in *E. coli*, in conjunction with σ^E , *degP* (*htrA*). In addition, CpxR~P activates transcription of *cpxP* encoding a small protein that negatively regulates the CpxAR regulon, probably by binding to a periplasmic domain of CpxA. The ability to autoactivate and then repress (via CpxP) enables a temporary amplification of the Cpx response that may be important to rescue cells from transitory stresses. **PrpA** and **PrpB** are **type I serine/threonine phosphatases** that also participate in the ECF pathway at least in part by affecting the phosphorylation level of CpxR.

B. subtilis has four classes of heat shock genes. All of them are not described here, but one is particularly interesting because of the mechanism used to control the genes. **Class I heat shock genes** include the major **chaperones DnaK-DnaJ-GrpE** and **GroEL-GroES**. Their transcription requires the housekeeping σ factor, σ^A (σ^{70}) and is negatively controlled by a repressor called **HrcA**. HrcA binds to a class I gene operator, a well-conserved 9 bp inverted repeat with a 9 bp spacer, called **CIRCE**. The **CIRCE/HrcA regulon** is normally repressed by the **HrcA repressor** but can be heat induced by inactivating the repressor. The molecular switch involves the **chaperone GroE**. Unlike σ^{32} in *E. coli*, the GroE chaperones bind to and facilitate folding of HrcA and thereby modulate repressor function. Titration of GroE by stress-induced misfolded proteins results in lower HrcA repressor activity.

NUTRIENT STRESS AND THE STARVATION — STRESS RESPONSE

Nutrient starvation and other environmental stresses are routine occurrences for most bacteria. Situations of true “feast” or nonstress conditions are few and far between for microbes outside the laboratory. Thus, microbes are most frequently found in a state of nutrient starvation or stress-induced slow growth or nongrowth. Fortunately this is the case, because given bacterial growth rates achieved under nonlimiting conditions in the laboratory, if microbial growth were nonlimiting in natural or animal host environments, the consequences would be deadly.

Starvation-Stress Response

When *E. coli*, *Salmonella*, and many other nondifferentiating microbes are starved for an essential nutrient such as a carbon-energy (C) source, they respond by inducing the expression of up to 50 or so new proteins or preexisting proteins. The genetic and physiologic reprogramming that occurs is the **starvation-stress response (SSR)**. The function of the SSR is to allow for the long-term starvation survival of the bacteria and to provide a general cross-resistance to a variety of other environmental stresses including extremes in temperature, pH, and osmolarity as well as exposure to reactive oxygen and nitrogen species and antimicrobial peptides/proteins.

It should be noted that a distinction must be made between starved cells and so-called stationary-phase cells. Typically, **stationary-phase cells** populate cultures that have stopped growing following exponential growth in rich or nonlimiting media, in contrast to **starved cells**, which populate cultures that have ceased growing in response to exhaustion of one or more defined nutrients. For stationary-phase cells, the condition that limits growth is not necessarily defined nor is it typically limited to a single stress; for starved cells, the limitation/stress that restricts growth is defined. Another key difference is that stationary-phase cultures normally achieve a much higher cell density compared with starved cultures, which can have a significant effect on overall cellular responses and long-term survival. Furthermore the genes/proteins expressed in stationary-phase cells may or may not overlap with those expressed in starved cells. The SSR refers specifically to the response of starved cells.

In general, for Gram-negative bacteria, starved cells are morphologically and physiologically very different from log-phase cells. The initial response to carbon-energy source limitation is to try and avoid the stress by increasing expression or expressing

new uptake or scavenging systems to be able to utilize any nutrients that may become available. Persistence of the starvation-stress eventually results in a cell that is smaller, much more hardy, and metabolically efficient. This is mediated by the accumulation of at least two cellular nucleotides: **cyclic 3', 5' adenosine monophosphate (cAMP)**; see “Catabolite Control” in Chapter 5) and **guanosine 3', 5'-bis(diphosphate) (ppGpp)**; see “Stringent Control” in this chapter). In addition, at least two alternative σ factors, σ^S (or σ^{38} ; see Chapters 2 and 5) and σ^E (see above under heat shock response), encoded by the *rpoS* and *rpoE* genes, respectively, are key SSR regulators. These regulators are general factors controlling responses to starvation stress in a more global manner. However, this is not sufficient for the regulation of some stress-responsive genes. Thus, additional regulators may be involved in the regulation of specific genes, indicating a discriminating complexity for the SSR and other stress responses.

Examples of some regulatory proteins include **Fis (factor for inversion stimulation)** first characterized for its role in flagellar phase variation; see “Phase Variation” in Chapter 5), **FadR (regulator of fatty acid metabolism)**, which binds to medium/long-chain fatty acyl-CoA molecules and represses fatty acid biosynthetic genes and activates fatty acid degradation genes; see Chapter 13), **Lrp (leucine-responsive protein)**, which controls certain aspects of amino acid metabolism; see “Serine/Glycine Family” in Chapter 15), **OxyR** (see above), **SpvA/SpvR (regulators of the *Salmonella* plasmid virulence or *spv* genes)**; see “Paradigms of Bacterial Pathogenesis” in Chapter 20), **PhoP (response regulator that controls virulence factors)**; see “Paradigms of Bacterial Pathogenesis” in Chapter 20), **IHF (integration host factor)**; see Chapter 5), and **H-NS (“Nucleoid Structure”** in Chapter 2).

Some of the physiologic changes that occur during SSR include the expression of new or higher-affinity nutrient utilization systems to scavenge the environment for carbon-energy sources and other nutrients; the degradation of cellular RNA, proteins, and fatty acids; the reduction in the number of ribosomes; altering of the amounts and types of lipid components in the cytoplasmic membrane; an increase in the relative amounts of lipopolysaccharide or LPS in the outer membrane of Gram-negative bacteria; and the condensation of chromosomal DNA in order to protect it from damage.

Stringent Control

When bacteria experience conditions that limit the availability of one or more amino acids (shift from a rich medium to a minimal medium) or exhaust their primary carbon source, growth stops temporarily and rapid adjustments in metabolism are made. These include decreasing the rates of RNA accumulation (particularly stable RNAs such as rRNA and tRNA) and DNA replication, as well as reducing the biosynthesis of carbohydrates, lipids, nucleotides, peptidoglycan, and glycolytic intermediates. The transport of many macromolecular precursors into the cell is also shut down. This set of responses, characterized best as a response to amino acid starvation, is referred to as the **stringent response** or **stringent control**. The stringent response collectively enhances cellular viability during periods of amino acid or energy limitation and allows rapid recovery and reinitiation of growth when conditions improve.

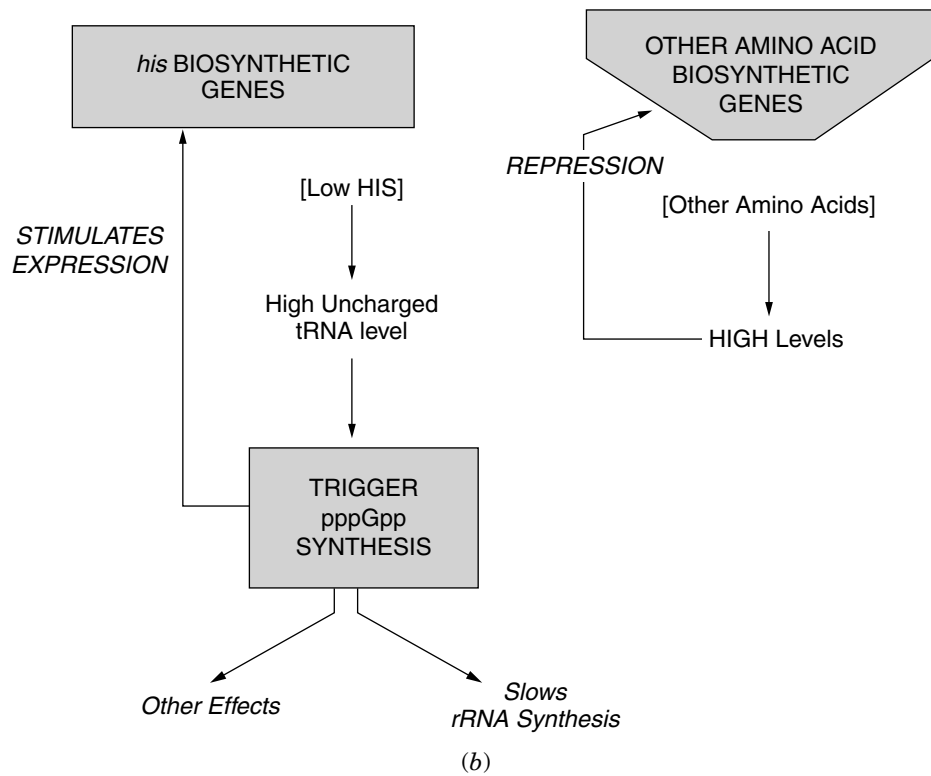
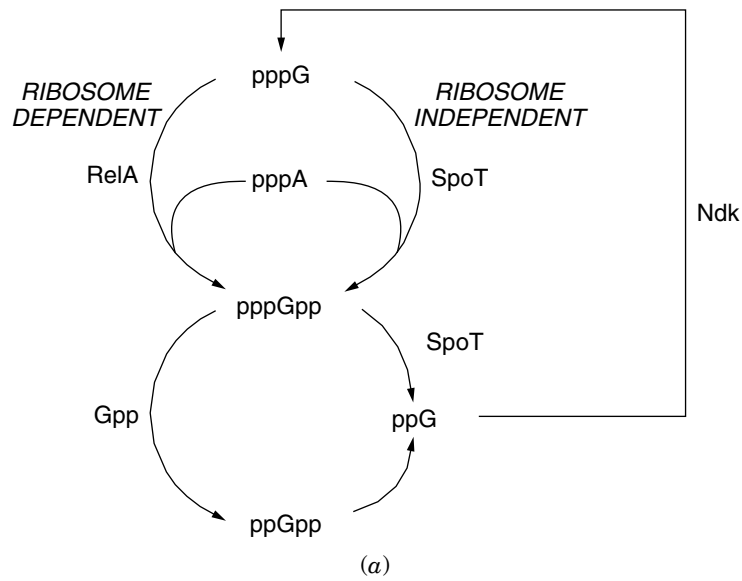
For rapidly growing cells, a major amount of the available energy is used for ribosome synthesis. Therefore, blocking ribosome synthesis under amino acid starvation

conditions is a major mode of energy conservation. Conditions causing a stringent response lead to an abrupt change in the rate of ribosome synthesis caused by inhibiting the transcription of genes encoding ribosome-associated components such as rRNA, tRNA, and mRNA for ribosomal proteins. When starved for amino acids, bacterial cells rapidly accumulate millimolar concentrations of two unusual nucleotides: guanosine pentaphosphate (guanosine 5'-triphosphate-3'-diphosphate, pppGpp) and guanosine tetraphosphate (guanosine 5'-diphosphate-3'-diphosphate, ppGpp). These nucleotides, first referred to as Magic Spot II and Magic Spot I, respectively, accumulate in *E. coli* and *Salmonella* as well as other bacteria during amino acid limitation.

Starvation for a specific amino acid will lead to an increase in the corresponding uncharged tRNA species. As the ratio of charged to uncharged tRNA falls, ribosomes stall on mRNAs when encountering the codon for that amino acid. Synthesis of pppGpp is triggered by the repeated binding of the uncharged tRNA molecules to the stalled ribosome. On the ribosome, the product of the *relA* gene (RelA, stringent factor), a ribosome-bound pyrophosphotransferase (pppGpp synthetase I) present on about 1% of ribosomes, catalyzes the formation of pppGpp from GTP and ATP, as shown in Figure 18-7. A second, ribosome-independent route to pppGpp also exists involving the *spoT* product (pppGpp synthetase II). SpoT is responsible for basal levels of (p)ppGpp. The enzyme pppGpp phosphohydrolase (*gpp*) is the major pppGpp hydrolase, degrading pppGpp to ppGpp. SpoT appears to have two functions: one in the synthesis of pppGpp and one in the degradation of ppGpp to GDP plus pyrophosphate.

A major result of the stringent response is a reduction in the rate of stable RNA accumulation — a response strongly correlated with the rise in (p)ppGpp concentration in the cell. Several studies suggest that RNA polymerase is a target of ppGpp action and that the regulated process during stringent control is transcription itself. The probable target is the β -subunit of RNA polymerase (see Fig. 2-16c in Chapter 2). Both transcription initiation and polymerase pausing (important in certain regulatory mechanisms and transcription termination) are postulated as the major targets of ppGpp action. Thus, ppGpp would reduce the affinity of RNA polymerase for rRNA promoters or inhibit elongation of polymerase. Either effect will obviously lower the amount of rRNA available for ribosome synthesis, which can subsequently affect the synthesis of ribosomal proteins. On the other hand, ppGpp has also been found to stimulate transcription of several amino acid biosynthetic operons (e.g., *his*) and possibly other genes. Thus, it appears that ppGpp can act either as a negative effector or a positive effector depending on the target gene.

Fig. 18-7. Stringent control. (a) Ribosome-dependent and -independent pathways for the biosynthesis of guanosine 5'-diphosphate-3'-diphosphate. ppG, guanosine 5'-diphosphate; pppG, guanosine 5'-triphosphate; pppA, adenosine 5'-triphosphate; adenosine 5'-monophosphate; ppGpp, guanosine 5'-diphosphate-3'-diphosphate; pppGpp, guanosine 5'-triphosphate-3'-diphosphate; pp_i, inorganic pyrophosphate. Ndk, nucleoside 5'-diphosphate kinase; Gpp, pppGpp 5'-phosphohydrolase. (b) Role of (p)ppGpp in maintaining an efficiently balanced pool of amino acids. In this example, histidine levels have fallen below optimum. The ensuing accumulation of ppGpp stimulates expression of the *his* operon while slowing translation and ribosome synthesis. As growth slows, the other amino acids accumulate and repress their own synthesis. Restoration of histidine levels shut off (p)ppGpp synthesis, allowing the resumption of normal ribosome synthesis rates.



In vivo, the synthesis of (p)ppGpp on the ribosome is controlled by the charging of total tRNA species as a function of intracellular concentrations of the 20 amino acids. The ratio of charged to uncharged tRNA appears to be of central importance rather than the overall level of charged tRNA. It would appear that ppGpp is a component of a sensing mechanism that functions to adjust the synthesis, for example, of histidine biosynthetic enzymes with respect to the need for histidine relative to the total amino acid concentration of the cell as well as in the growth medium. Thus, along with the operon-specific attenuator mechanism that responds to the need for histidine, specifically altering the level of ppGpp enables the organism to sense how the supply of histidine, in this example, compares with the availability of all the amino acids in the cell—a kind of fine-tuning mechanism that maintains the correct relative levels of each amino acid.

EXTREMOPHILES

Although, the stress responses described above are important for the survival of most bacteria during exposure to the various stresses, some bacteria view stresses such as extremes in pH or temperature as a lifestyle choice. These bacteria survive and thrive in conditions that would send other microbes to their maker. **Acidophiles** typically survive and grow at an external pH (pH_o) below 4.0; **alkalophiles** grow only at a pH_o of 8.0 or above. Similarly, **thermophiles** grow at temperatures generally around 50° to 65°C and **psychrophiles** grow at temperatures as low as 5° to 10°C.

Despite growing at pH 2–4, **acidophilic bacteria** maintain their cytoplasmic pH at 6.0 or higher. Consequently, there has been considerable interest in determining the mechanism whereby this large pH differential (ΔpH) is maintained. It is generally agreed that maintenance of this large transmembrane ΔpH is energy dependent and that its maintenance requires a transmembrane electrical potential ($\Delta \Psi$) that is positive inside (the reverse of that found in neutralophilic bacteria). A reverse transmembrane potential has been observed in both *Bacillus acidocaldarius* and *Thiobacillus acidophilis*. If this transmembrane potential is abolished, the ΔpH collapses. Despite the large ΔpH , the cytoplasmic pH is extremely stable. The cytoplasmic buffering capacity of *T. acidophilis* is responsible for the cytoplasmic pH homeostasis in metabolically comprised cells. When a large influx of H^+ occurs, the cytoplasmic buffering capacity prevents drastic changes in pH. In addition, the resultant increase in positive membrane potential due to this influx of H^+ eventually leads to cessation of further H^+ influx.

Studies with a heterotrophic, mesophilic, obligate acidophile provide additional insights into the nature of changes in membrane potential under active and inactive conditions (Fig. 18-8). Starving cells of this acidophile continue to show a ΔpH of about 1.7 but exhibit changes in membrane potential ($\Delta \Psi$) and proton motive force ($\Delta \rho$) that are just the opposite of those seen under conditions of optimal nutrition. The linkage of the transient H^+ influx with the rise of $\Delta \Psi$ and the cytoplasmic buffering capacity play central roles in acidophilism. It is considered that the same impermeant cellular macromolecules can account for both. Thus, the $\Delta \Psi$ represents a Donnan potential (electrochemical potential across a membrane at electrochemical steady state) that is offset in active cells by energy-dependent H^+ extrusion.

Many bacteria display optimal growth in alkaline media. Although a number of these are *Bacillus* species, alkalophilic strains of *Micrococcus*, *Pseudomonas*, *Clostridium*,

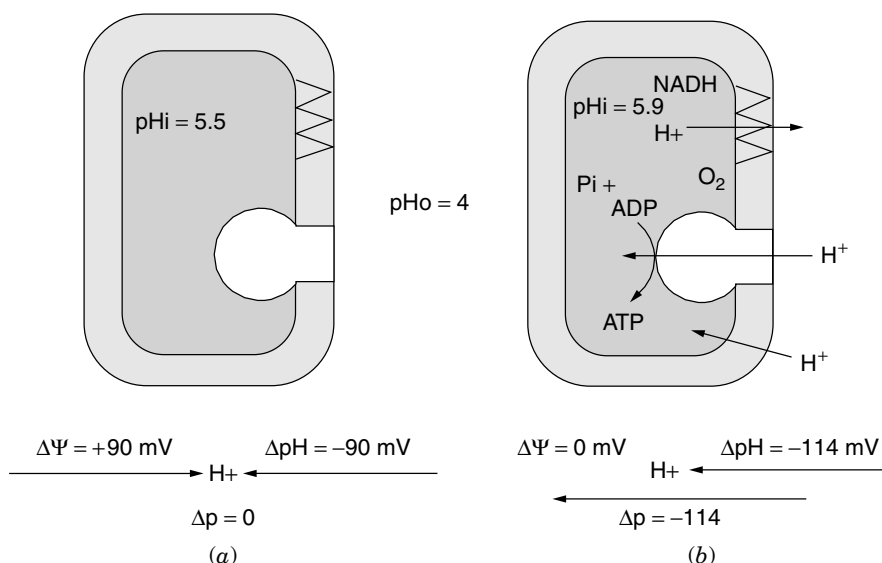


Fig. 18-8. Model for bioenergetic parameters in an obligately acidophilic bacterium at an outside pH (pH_o) of 4.0 in inactive (a) and respiring (b) cells. Left and right arrows indicate the forces impelling the H^+ ion flux into and out of the cells and are drawn to scale. There is no H^+ ion flux in inactive cells. In active cells H^+ ion influx equals H^+ ion efflux. (Source: From Goulborne et al., 1986.)

the photosynthetic bacterium *Ectothiorhodospiras*, and many others have been reported. These **alkalophiles** can grow only at pH levels of 8.0 to 11.5. A true alkalophile maintains a cytoplasmic pH of 9.0 or lower even at an external pH of 11.0, thus growing optimally under conditions in which the cytoplasm is more acidic than the external medium. The alkalophiles *B. firmus* and *B. alcalophilus* both depend on the activity of a Na^+/H^+ antiporter to achieve acidification of the cytoplasm relative to the exterior. A nonalkalophilic strain of *B. firmus* isolated after mutagenesis of the alkalophilic strain lacks the Na^+/H^+ antiporter. The inability of *B. firmus* to grow at neutral pH is not due to excessive acidification but is related to a failure of respiratory activity to generate a transmembrane electrical potential that is high enough to maintain certain cellular functions such as Na^+ /solute symport and motility. Nonalkalophilic mutant strains of *B. alcalophilus* exhibit loss of Na^+/H^+ antiport activity and Na^+ coupling of solute transport as well as lower amounts of membrane-bound cytochromes and a membrane-bound chromophore.

Thermophilic organisms have been studied intensively in an effort to determine the mechanisms whereby these organisms not only survive but also prefer higher temperatures for growth. Chemical differences in the lipids found in thermophiles, higher metabolic rates facilitating rapid resynthesis of heat-denatured cellular components, and macromolecules with higher thermostability have all been suggested as mechanisms for greater heat tolerance in thermophiles. Thermophilic organisms contain lipids with higher melting points than those found in mesophiles, suggesting that the temperature at which the major lipid components of the cell melt may establish the upper limit for growth. In yeast, a direct correlation has been established between the growth temperature limits and the degree of unsaturation in mitochondrial lipids.

The lower the temperature limit of growth, the greater the degree of lipid unsaturation. The membrane lipid compositions of thermophilic yeasts are distinguished by the higher percentage (30–40%) of saturated fatty acids as compared with mesophilic and psychrophilic species. Psychrophilic yeasts contain approximately 90% unsaturated fatty acids, 55% of which is linolenic acid ($C_{\alpha-18:3}$).

Certain lines of evidence indicate that thermophilic organisms possess an intrinsic thermostability that is independent of any transferable, stabilizing factors. Membranes with greater heat stability, more rapid metabolic and growth rates, and factors that impart structural stability and greater inherent heat stability to individual protein macromolecules have been implicated as contributing factors. The greater heat stability of protein molecules from thermophilic organisms appears to reside in their ability to bind certain ions more tightly, thereby enhancing the establishment of more stable conformations. Proteins from thermophiles also contain increased levels of hydrophobic amino acids as compared to those from mesophiles. Although macromolecules from thermophiles are more thermostable, they are, for the most part, physicochemically similar to their mesophilic counterparts with regard to molecular weight, subunit composition, allosteric effectors, amino acid composition, and major amino acid sequences.

Certain attributes of protein thermostability are due to subtle changes in structure and to alterations in hydrogen bonding, hydrophobic interactions, and other noncovalent activities. For example, the thermostability of the tryptophan synthetase of *E. coli* is increased by amino acid substitutions that alter the hydrophobicity of the molecule in the absence of gross changes in conformation. In the thermophile *B. stearothermophilus*, a marked increase in thermostability is engendered by the presence of lysine in place of threonine in the plasmid-encoded enzyme that inactivates kanamycin. The nucleotide sequence of the plasmid encoding this enzyme in *B. stearothermophilus* differs by only one base from the plasmid coding for the identical enzyme in the mesophilic *S. aureus*. The lysine substitution permits increased electrostatic bridging with little significant change in the three-dimensional structure.

Isolation of thermophilic mutants of *B. subtilis* and *B. pumilus* and transformation of the thermophilic trait to mesophilic strains provides direct genetic evidence that the thermophilic trait is a phenotypic consequence of at least two unlinked genes. Additional studies of this type should provide information that could shed light on the specific nature of thermophily.

Psychrophilic organisms—those that exhibit an optimal temperature for growth at 15°C or lower and a maximum temperature for growth at 20°C—are defined as **obligate psychrophiles** to differentiate them from organisms that can grow at low temperatures but are actually mesophiles in terms of their optimum growth temperature. Although some confusion still exists with regard to the proper nomenclature and cardinal temperatures that differentiate psychrophiles from organisms that are **psychrotolerant** or **psychrotropic**, those organisms that grow well at temperatures at or below the freezing point of water are considered to be true psychrophiles.

The psychrophilic yeast *Leucosporidium* grows well at –1°C. In this organism there is a positive correlation between the growth temperature and the unsaturated fatty acid composition of the cell lipids. At subzero temperatures (–1°C) with ethanol as the substrate, 90% of the total fatty acid is unsaturated, with linolenic acid (35–50%) and linoleic acid (25–30%) predominating. At temperatures close to the maximum for growth, linolenic acid accounted for less than 20% of the total fatty acid, and oleic acid

(20–40%) and linolenic acid (30–50%) were the major components. Marked changes also occurred in the cytochrome composition of the cell, $a + a_3 : b : c$ were 1 : 1 : 2.9 at 8 °C with glucose as the substrate, whereas at 19 °C they were 1 : 2.3 : 16.7. Thus, it can be concluded that changes in membrane structure and composition are fundamental to temperature adaptation in psychophilic yeasts.

The psychophilic bacterium *Micrococcus cryophilus* also undergoes alteration in its lipid composition with changes in growth temperature. Cultures of this organism continue to grow without a lag following a sudden increase in temperature from 0° to 20 °C (upshift) or a reciprocal decrease (downshift). The growth rate changes gradually to that typical of cultures grown isothermally at the final temperature. After a temperature downshift, the phospholipid acyl chain length begins to change immediately, whereas there is a delay following upshift. However, the final fatty acid composition is attained within the same number of cell division times after an upshift or downshift. It appears, therefore, that this psychophile is more stressed by a sudden increase in growth temperature than by a sudden decrease.

Studies on the viscosity and phase transition temperatures of lipids isolated from psychophilic, psychotropic, and mesophilic organisms are able to adjust their lipid-phase transition temperature to the growth temperature. By comparison, a psychophilic *Clostridium* synthesizes lipids that have the same phase transition temperature after growth at different temperatures. This lack of growth temperature–inducible regulation of lipid-phase transition temperatures appears to be a molecular determinant for psychophily in this organism. Comparisons of the properties of triosephosphate isomerase purified from psychophilic, mesophilic, and thermophilic clostridia indicate that the purified enzymes have the same molecular weight, subunit molecular weight, and susceptibility to the active site-directed inhibitor, glycidol phosphate. However, their temperature and pH optima, as well as stabilities to heat, urea, and sodium dodecyl sulfate (SDS), differ markedly.

SUMMARY

Our understanding of stress responses has made great strides over the last several years, especially in the responses of microbes other than *E. coli* and *Salmonella*. However, the ever-increasing realization of the complexity and interrelationships of stress responses clearly indicates that we still have a great deal to uncover. Since most microbes spend the vast majority of their life under environmental stress, further knowledge of stress responses is critical to the complete understanding of microbial physiology. Information obtained may contribute to the development of new vaccines, new disease therapies, new agricultural strategies, new food safety procedures, and new antimicrobial agents. Thus, the study of stress responses continues to be an exciting area of basic and applied biomedical and biological research.

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CHAPTER 19

BACTERIAL DIFFERENTIATION

Development or differentiation among eukaryotic microbes is well known and diverse, especially in the production of asexual conidia and sexual spores, but is not discussed here. In comparison, prokaryotic differentiation, although not as widespread, is beginning to show interesting diversity. The reasons for differentiation of developmental cycles in prokaryotes are reflected in the surprising number of examples including the differentiation into (1) a variety of “resting” cell forms that are more resistant to environmental stresses such as endospores (*Bacillus*) and myxospores (myxobacteria), (2) a number of cell forms specifically “designed” for dispersal of the bacteria within certain niches such as swarmer cells (*Caulobacter*), (3) cell forms performing specific functions such as heterocysts (*Anabaena*), and (4) cell forms designed to establish a symbiotic relationship with another organism for their mutual benefit such as the nitrogen-fixing nodules associated with legumes (*Rhizobium*) (see Chapter 14). This chapter highlights three interesting paradigms of bacterial differentiation: *Bacillus* endospore formation, myxospore/fruiting body formation in myxobacteria, and stalked and swarmer cell formation in *Caulobacter*.

BACILLUS ENDOSPORE FORMATION

As shown in Table 12-1, six separate genera of bacteria produce a developmental form called an **endospore**. All of these spores are resistant to heat, cold, radiation, and other adverse environmental conditions. However, the endospores produced by members of the *Bacillaceae* are probably the most resistant to both natural and artificially imposed hostile environmental conditions. The primary function of endospore formation appears to be the survival and dissemination of the species. Since endospore formation in bacteria is triggered by starvation, this phenomenon represents a special example of the stress responses discussed in Chapter 18. Although the production of endospores is displayed by several genera of bacteria (see Table 19-1), a much greater amount of

TABLE 19-1. Characteristics of Endospore-Forming Bacteria^a

Genus	G + C (mol%)	Shape	Distinguishing Metabolic Traits
<i>Bacillus</i>	33–66	Rods	Catalase positive ; most strict aerobe
<i>Sporolactobacillus</i>	38–40	Rods	homolactic fermentation ; facultative anaerobe or microaerophilic
<i>Clostridium</i>	24–54	Rods or filaments	Strict anaerobe
<i>Desulfotomaculum</i>	37–50	Rods or filaments	Sulfate reduction and strict anaerobe
<i>Sporosarcina</i>	40–42	Cocci in tetrads or packets	Strict aerobe
<i>Thermoactinomyces</i>	52–55	Branched filaments	Strict aerobe

^aThe major distinguishing factor that separates each spore former from the others is in bold type.

Source: From Slepecky, 1993, p. 1–21. In *Biology of Bacilli: Applications to Industry*. R. H. Doi and M. McLaughlin (Eds.). Butterworth-Heinemann, Boston, MA.

time and effort has been expended in the study of this process in the genus *Bacillus*, particularly *Bacillus subtilis*. For this reason, we emphasize the comparative biology, physiology, and genetics of bacterial endospore formation by this organism.

On a practical basis, the ability to produce resistant spores enables many organisms to survive autoclaving, radiation, or chemical processes for the preservation of foods or sterilization of material for medical procedures. Great care must be taken to establish the conditions required for their elimination. Concerted efforts to understand the developmental and regulatory factors governing sporulation may provide insights into other developmental processes in higher forms.

Life Cycle of *Bacillus*

All endospore-forming bacteria undergo a life cycle that includes vegetative growth in the presence of adequate nutrition and favorable environmental conditions. At the beginning of the stationary phase, when nutrients become limiting and pH and other conditions in the culture medium change, the cells have a variety of overlapping genetic networks available with which they can respond to this changing environment. Among the processes that can be activated under growth-restricting conditions are establishment of motility and competence for transformation. The decision to undergo sporulation appears to be a response of last resort in an effort to provide for survival.

A series of molecular switches appears to control the decision-making process to determine which of these responses will be followed. As the cells enter the stationary phase of growth, nutritional deprivation can trigger entry into sporulation. This process can be visualized by light and electron microscopy as a series of complex morphological changes that result in the formation of a highly resistant dormant form called an endospore. When conditions become favorable again, the spore can undergo activation, germination, and outgrowth into a metabolically active cell capable of entering the vegetative growth cycle, or, if conditions are unfavorable, the spore can proceed directly into another round of sporulation (the **microcycle**). The response of spores to activation by environmental factors that lead to germination and outgrowth need to be considered along with spore formation, since the genetic and regulatory machinery that governs these processes must be built into the spore as it is being formed.

Stages of Sporulation

The morphological events in the sporulation process as they occur in *Bacillus subtilis* are shown in Figure 19-1 and are as follows:

Stage 0 (vegetative cells): Cells undergoing normal vegetative growth are defined as being in stage 0 with regard to the sporulation cycle.

Stage I (axial filament): This stage has not been recognized universally, since mutants arrested in stage I were not found until recently. But since mutants blocked at stage I have been isolated, it is useful to consider this as a valid stage.

Stage II (septum formation): The first sign of sporulation is the formation of an asymmetrically sited division septum resulting in two distinct cells: the mother cell and the prespore. Each of the two cells receives a portion of the nuclear DNA. Intermediary stages designated Ii, Iii, and Iiii are based on the isolation of mutants blocked at these points between stages II and III.

Stages Iii, Iiii, Iiiii (prespore engulfment): The prespore is engulfed by the mother cell to form a protoplast.

Stage III (forespore development): The engulfed protoplast is now referred to as the forespore. In most electron micrographs the forespore is rather amorphous in appearance, presumably because no peptidoglycan layer has, as yet, been formed to provide a defined shape.

Stage IV (cortex formation): Shortly after forespore engulfment, a dense, narrow band of peptidoglycan, the cortex layer, is formed between the inner and outer membranes of the forespore. At this point, the beginnings of the spore coat can barely be discerned. As time progresses, the cortex becomes striated and multilayered.

Stage V (coat formation): Overlapping the cortex development process is the deposition of the multilayered spore coat. Discontinuous segments of coat material are observed. These segments coalesce into a continuous dense layer. Deposition of coat material at discontinuous points is more characteristic of coat formation in *Clostridium*. In *Bacillus*, deposition of coat material is a continuous process. However, there does not appear to be any fundamental difference in synthesis of the coat material in the two species.

Stage VI (maturation): Following the deposition of the cortex layers and formation of the spore coat, the final stage of maturation occurs. Observed as a whitening under phase-contrast microscopy, this process is associated with the synthesis of dipicolinic acid and calcium uptake into the mature spore. The characteristic properties of resistance, dormancy and germinability, appear at this stage.

Stage VII (release of the mature spore): The mother cell is lysed and the mature spore is released as a dormant form.

It should be emphasized that the cytological events outlined here and depicted in Figure 19-1 occur as an overlapping continuum rather than as separate and distinct events. These morphological stages serve as useful reference points. Mutants that permit the sporulation process to proceed to one of these stages but prevent any further development beyond this point are designated spo0, spoII, spoIII, spoIV, and so on. A temporally and spatially controlled program of gene expression drives the

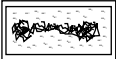
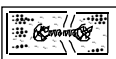
Stage	Description	Morphology	Genes or enzymes involved; other properties
0	Vegetative cell		Non-refractile; normal vegetative growth stage
I $\sigma^H + \sigma^A$	Axial filament prespore		CitC ⁻ , isocitrate dehydrogenase deficient mutants, blocked at stage I; DNA replic.; FtsZ ring formed
IIi $\sigma^E + \sigma^F$	Asymmetric Septation		Forespore develops; alanine dehydrogenase; <i>spoOA</i> , <i>spoOK</i> , <i>spoOH</i> ; chromosome pumped into forespore.
IIii σ^E	Septal membrane Rearrangement		<i>spolIA</i> , <i>spolIP</i> , <i>spolIG</i> , <i>spolIB</i> + <i>spoVG</i>
IIiii σ^E	Forespore engulfment		<i>spolID</i> , <i>spolIA(P)</i> ; <i>spoIVCA</i> + <i>spoVCB</i> + <i>spolIIC</i> required for generation of σ^K
III $\sigma^K + \sigma^G$	Forespore development		<i>spolIIA</i> , <i>spolIIE</i> , <i>spolIIJ</i> , <i>spolIID</i> , <i>spolIIG</i> , <i>spoIVB</i> , <i>spoIVF</i> , <i>spoVB</i> , <i>spoVD</i> , <i>spoVE</i> ; alkaline phosphatase; glucose dehydrogenase; aconitase, HR catalase
IV σ^K	PGCW and cortex synth.		<i>cotD</i> , <i>cotT</i> , <i>gerE</i> ; refractility, ribosidase, adenosine deaminase, dipicolinic acid
V σ^K	Coat synthesis		<i>cotE</i> , <i>gerE</i> , <i>cotA</i> , <i>cotB</i> , <i>cotC</i> ; cysteine incorporation, chemical & UV resistance develops
VI	Maturation		Alanine racemase, heat resistance
VII	Mother cell lysis		Mature spore is released from mother cell.
VIII	Free spore		Refractile; resistant to heat, radiation, etc.
	Germination		Non-refractile; sensitive to environmental factors

Fig. 19-1. Diagrammatic representation of the life cycle of *Bacillus subtilis*. As discussed in the text, the stages are designated by Roman numerals. The stage originally designated as I has not been universally recognized because mutants arrested at this stage were not found. However, mutants blocked at this stage have recently been described, so it will be considered as a valid stage. The cycle proceeds as a continuum and the stages are primarily for convenient reference to indicate a stage at which mutants block the progress of sporulation. The diagram provides a representative sample of mutants blocked at the stage indicated. The approximate stages at which various enzymatic activities and other properties develop are also shown. The diagram represents a composite of several diagrams from a number of the reviews cited at the end of this chapter.

sporulation process. Many of the genes required for the eventual response to activation, germination, and outgrowth must be built into the spore during the sporulation process, since they must be available for action within minutes of their activation. The program that regulates and orchestrates these events is discussed in further detail below.

PHYSIOLOGICAL AND GENETIC ASPECTS OF SPORULATION

Sporulation Genes

Over 125 gene products govern the complex morphological and biochemical changes that take place during sporulation. The genes involved in sporulation are scattered around the chromosome and are interspersed with many other genes that have no known role in sporulation. More than 60 genetic loci, designated *spo*, have been identified and named according to the morphological stage at which the sporulation is blocked. A partial list of these genes is shown in Table 19-2. Other genes that participate in sporulation include those coding for a number of small acid-soluble proteins (SASPs) of the spore core, or for the proteins in the spore coat, designated *cot*. Some of the genes involved in normal cell division, such as *ftsA* and *ftsZ*, are also involved in the asymmetric division process that initiates sporulation. Finally, those genes regulating

TABLE 19-2. A Partial List of Sporulation Genes and Their Function in *Bacillus subtilis*

Stage	Gene Designation	Function
I	<i>citC</i>	Isocitrate dehydrogenase
IIi	<i>spoOA</i>	Response regulator; phosphorylation initiates sporulation via phosphorelay system
	<i>spoOK</i>	Regulates phosphorelay system
	<i>spoOH</i>	Encodes σ^H
IIii	<i>spoIIG</i>	Pro σ^E and activating protease
	<i>spoIIB</i> + <i>spoVG</i>	Prespore engulfment; septum formation
IIiii	<i>spoIID</i>	Septal peptidoglycan hydrolysis
	<i>spoIIA(P)</i>	Processing of pro σ^E
III	<i>spoIIIA</i>	Prespore engulfment
	<i>SpoIIIE</i>	Controls σ^F expression
	<i>spoIiJ (kinA)</i>	Histidine protein kinase; activates SpoOF
	<i>spoIIID</i>	Controls σ^E -dependent genes
	<i>spoIIIG</i>	Structural gene for σ^G
	<i>spoIVB</i>	Production of σ^K ; signal peptide
	<i>spoIVF (spoIIIF)</i>	Processing of pro- σ^K
	<i>spoVB</i>	Cortex synthesis
	<i>spoVD</i>	Cortex synthesis
	<i>spoVE</i>	Cortex synthesis
IV	<i>cotD</i>	Coat synthesis
	<i>cotT</i>	Coat synthesis
	<i>cotA</i>	Coat synthesis
	<i>cotB</i>	Coat synthesis
	<i>cotC</i>	Coat synthesis
	<i>gerE</i>	Germination

spore germination (*ger*) are included, since they must be incorporated into the spore genome during the sporulation process.

Initiation

Endospore formation is initiated in response to starvation for carbon, nitrogen, or phosphorous. The very early stages of sporulation are reversible and transfer to fresh culture medium results in resumption of vegetative growth. Thus, sporulation seems to be initiated by the accumulation of factors that inhibit vegetative growth and derepress the spore genome. Sporulation is more efficient in a dense population of cells. At low cell density, sporulation efficiency is improved if the cells are suspended in medium previously conditioned by growth of cells to high density. This finding indicates that a substance produced by cells at high density is necessary for efficient sporulation. Production of this extracellular differentiation factor (EDF-1) or pheromone is governed by *spoOA*, *spoOB*, and *spoOH*.

Even if starvation conditions are imposed, sporulation will occur only if the cell has reached a certain stage in the vegetative growth cycle, the tricarboxylic acid (TCA) cycle is functional, and at least one external pheromone is present in sufficient concentration. Mutants blocked in the TCA cycle—for example, a mutant with a deletion of *citC*, the gene encoding isocitrate dehydrogenase—cause a block at stage I of sporulation. For this reason, it is useful to retain stage I as a valid part of the sporulation cycle. Once these conditions are met, a controlled program of sporulation gene expression largely replaces the pattern of vegetative gene expression. The temporal changes in gene expression that occur during sporulation are controlled by the sequential appearance or activation of sporulation-specific transcription factors (designated sigma, σ) that bind to the core RNA polymerase and direct it to transcribe only from specific promoters. Sigma factor binding also confers the capacity to recognize new classes of these factors as they appear.

The first visual recognition of the initiation of sporulation involves a special asymmetric cell division. Asymmetric placement of the septum separates the cell into the mother cell and the forespore. The forespore goes on to become the spore while the mother cell serves to nurture the cell until the spore is fully matured.

At this point, the sporulation gene program splits and two distinct programs become active: one in the mother cell and the other in the forespore. The mother cell produces certain proteins that are incorporated into the developing spore from the outside, and the forespore generates other proteins that complete the process by being added from the inside. The transcriptional regulator, SpoOA, plays a key role during the initiation stage. The SpoOA protein contains an aspartyl residue in the N-terminal region that is the target of phosphorylation by a histidine protein kinase, a component of the phosphorelay system (Fig. 19-2).

This complex system of regulation of stationary-phase gene expression involves at least two histidine protein kinases: KinA and KinB. They initiate the phosphorelay system by phosphorylating SpoOF. Other genes involved in this system include the *spoOB* operon, which contains *obg*, a GTP-binding protein essential for vegetative growth. It has been suggested that the *obg* gene product may be the protein that senses the decrease in GTP levels that occurs following nutrient deprivation and provides the link to sporulation events via the phosphorelay system. The experimental results suggest that a threshold level of activated SpoOA (SpoOA~P) induces sporulation gene

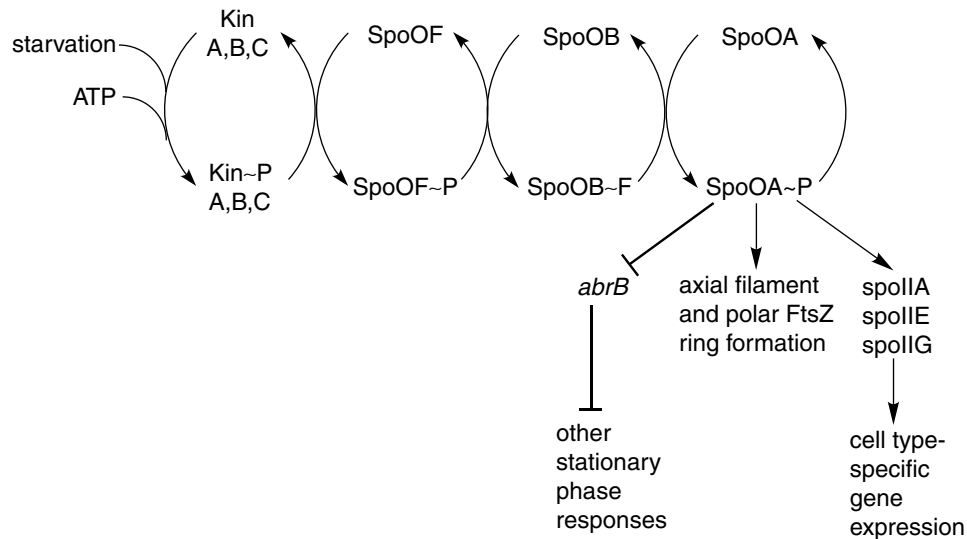


Fig. 19-2. Production of SpoOA~P by the phosphorelay system. The sensor kinases KinA, KinB, and KinC autophosphorylate on a histidine. Phosphate is transferred to SpoOF, then to SpoOB, and finally to SpoOA. Low levels of SpoOA~P are sufficient to repress transcription of *abrB*, derepressing expression of many of the stationary-phase response pathways negatively regulated by AbrB. Higher levels of SpoOA~P stimulate axial filament formation, polar septation, and transcription of genes (e.g., *spoIIA*, *spoIIE*, and *spoIIG*) required for cell type-specific gene expression. (Source: From Burkholder and Grossman, 2000.)

expression and that those cells able to induce the expression of early genes directly activated by SpoOA~P go on to produce mature spores. Genes that are under the apparent control of SpoOA cause the site of formation of the FtsZ ring to shift from midcell to polar sites as discussed further below.

Many stationary-phase genes are under the control of σ^H (SpoOH). This transcription factor is weakly expressed from a σ^A promoter in midlogarithmic phase but is greatly increased after the initiation of sporulation. Among the various genes controlled by σ^H is the sporulation-specific operon *spoIIA*, which produces the important σ^F .

As discussed in Chapter 17, FtsZ and FtsA are cell division proteins in *E. coli* that are recruited to the division site where they are involved in the formation of a ring that remains associated with the invaginating septum until septation is completed. Homologs of the *E. coli* cell division genes *ftsA* and *ftsZ* have been identified in *B. subtilis*. In *B. subtilis*, the *ftsA* gene was originally designated *spoIIG* or *spoIIN*. Both *ftsA* and *ftsZ* are required for vegetative septation as well as for the asymmetric septation that occurs in the initial stage of sporulation. The *ftsA* and *ftsZ* genes of *B. subtilis* constitute a simple operon expressed from promoter sequences immediately upstream of *ftsA*.

Both *ftsA* and *ftsZ* are transcribed from a distinct promoter, p2 (*ftsAp*₂), during sporulation but not during exponential growth. Transcription from p2 is dependent on RNA polymerase containing σ^H but does not require expression of other sporulation loci (*spoOA*, *spoOB*, *spoOE*, *spoOF*, or *spoOK*). Introduction of a *cat* cassette in the middle of promoter p2 does not affect vegetative growth but prevents postexponential symmetrical division and spore formation. A bifunctional protein, SpoIIE, interacts

with FtsZ and is involved in the SpoOA-dependent switch in the positioning of FtsZ rings from midcell to polar locations.

Two proteins involved in chromosome partitioning, Soj (ParA) and SpoOJ (ParB), also appear to be involved in regulating the initiation of sporulation in response to chromosome structure or partitioning. Soj affects sporulation and SpoOA~P-dependent gene expression by decreasing the level of SpoOA protein. Soj negatively regulates transcription of *spoOA* and associates with the *spoOA* promoter region in vivo. Expression of *spoOA* from a heterologous promoter restores SpoOA levels and partly bypasses the sporulation and gene expression defects of a *spoOJ* null mutant.

Proteins homologous to the MinC and MinD proteins known to exist in *E. coli* have been identified in *B. subtilis*. The last two open-reading frames of the *divIVB* locus in *B. subtilis* are homologous to *minC* and *minD* of *E. coli*. However, the presence of an equivalent to MinE is less clear. The DivIVA protein displays some characteristics of a MinE-like protein and appears to have a function analogous to that of MinE in *E. coli*. The MinCD complex appears to play a role in asymmetric septum formation during the sporulation process.

Transition from Stage II to Stage III

Initiation of the prespore and mother cell programs of gene expression in the correct compartments is crucial to the success of sporulation. A sensitive and accurate signal from prespore to mother cell after septation is important in orchestrating further development of the spore. Two sporulation-specific transcription factors, σ^F in the prespore and σ^E in the mother cell, specify different programs of gene expression in the two compartments. Formation of the asymmetrically located septum during the initial stage of sporulation results in enclosure of the origin-proximal 30% of the chromosome in the prespore compartment. One copy of the chromosome is translocated from the mother cell into the prespore by an active process requiring SpoIIIE.

The SpoIIIE protein of *B. subtilis* is a DNA-dependent ATPase capable of tracking along DNA in the presence of ATP. The amino-terminal part of the protein mediates its location to the division septum. SpoIIIE acts as a DNA pump that actively moves one of the replicated pair of chromosomes into the prespore. A second copy of the chromosome remains in the mother cell.

The SpoIIR locus is near the origin of the chromosome and is transcribed only by RNA polymerase containing σ^F . The SpoIIR protein is the only σ^F -directed gene required for the activation of the transcription program directed by σ^E in the mother cell and ensures that σ^E is not activated until the septum is formed. Rapid activation of σ^E following septation may be important in preventing further septation. Normally, the *spoIIR* locus is situated at 324°, near the origin of replication (0/360°). Relocation of *spoIIR* to regions not in the origin-proximal part of the chromosome substantially reduces sporulation efficiency. This finding suggests that it is extremely important that the chromosomal location of *spoIIR* and septum positioning coincide with the partitioning of the chromosome during the asymmetric cell division.

The two unequal-size compartments, the mother cell and the forespore, are under the control of compartment-specific transcription factors. In the mother cell, σ^E is the first to be synthesized at the onset of sporulation. At the same time, σ^F is produced in the prespore. Neither σ^E nor σ^F become active until compartmentalization is complete. The inhibitor, SpoIIAB, keeps σ^F in an inactive state while σ^E is formed as an inactive

proprotein (pro- σ^E). The binding of SpoIIAA to SpoIIAB instead of σ^F relieves the inhibition of σ^F by SpoIIAB. In the mother cell, SpoIIAA is kept inactive by phosphorylation. However, in the prespore, SpoIIE, a membrane-bound phosphatase, activates SpoIIAA.

Activation of pro- σ^E is accomplished by cleavage of a 27 amino acid sequence from the amino terminus catalyzed by the protease SpoIIGA. This protein is present in the preseptal cell but is maintained in an inactive state until septum formation is complete. SpoIIR triggers removal of the amino acids from pro- σ^E by SpoIIGA. Binding of SpoIIE to the forespore face of the septum has been suggested as an explanation for how the activation of σ^F is limited to the forespore. As the sporulation process continues, the transcription factor σ^K is activated in the mother cell and σ^G is activated in the forespore.

Shortly after the onset of differential gene expression in the mother cell and forespore compartments, the septum between them begins to migrate around the forespore until the leading edges of the membrane meet on the distal side of the forespore and fuse, releasing the forespore into the cytoplasm of the mother cell (see Fig. 19-1). This engulfment process results in the forespore being entirely enclosed within the mother cell and being bounded by two membranes: its own original cytoplasmic membrane and a membrane derived from the engulfing mother cell membrane. It is between these two membranes that the spore cell wall (the cortex) is synthesized.

The multilayered spore coat is assembled around the forespore within the mother cell. It is considered that thinning or removal of the peptidoglycan between the septal membranes allows the mother cell membrane to move around the forespore and eventually engulf it. Several gene products (SpoIIM, SpoIIP, SpoIID, SpoIIB, and SpoVG) have been implicated in the engulfment process. However, a *spoIIB spoVG* double mutant sporulates poorly and is blocked at an early stage of engulfment, with little or no thinning of septal peptidoglycan.

In *SpoIIB* mutants the septal peptidoglycan is incompletely degraded throughout the septum. When the forespore grows, it breaks the weakened septal peptidoglycan, resulting in broad bulges of the forespore into the mother cell. Engulfment is eventually completed, but the process proceeds more slowly. These observations suggest that *SpoIIB* facilitates the rapid and spatially regulated breakdown of septal peptidoglycan in the normal sporulation process.

Mutations in the σ^E -dependent *spoIID* gene cause a morphological block at intermediate stage IIIi. In stage IIIii the edges of the septum begin to migrate toward the proximal pole of the cell. The edges of the septal membranes meet at the pole of the cell and fuse, completing the engulfment of the prespore within the mother cell cytoplasm (stage III). This final step is prevented by the *spoIIAC* (P) mutations, indicating that a σ^F -dependent gene is necessary for completion of engulfment.

Forespore Development

Once engulfment of the prespore is complete, it is referred to as a **forespore**. At this point, synthesis of σ^G , the product of the *spoIIIG* gene, begins with the resultant activation of the σ^G regulon in the forespore. The σ^G regulon contains a large number of genes that orchestrate many changes in the properties of the developing spore. Regulation of σ^G occurs at the transcriptional and posttranscriptional level. However, the genes involved in this stage vary considerably with regard to their regulation.

Mutations in *spoIIIA* and *spoIIJ* allow engulfment to be completed but prevent further progression beyond stage III. The onset of the late stages of forespore development are dependent on events occurring in the mother cell at the level of *spoIIIG* transcription and σ^G activity. Specific intercommunication between the mother cell and the forespore appears to be necessary to couple the activation of σ^G in the forespore to the program of gene expression in the mother cell under the control of σ^K .

Three groups of σ^G -dependent genes are of major importance in the later stages of forespore development. These include the *ssp* genes encoding the small acid-soluble proteins (SASPs) in the spore core, the genes encoding sporulation-specific penicillin-binding proteins (PBPs), and the germination (*ger*) genes.

The structures of the nucleoids in the mother cell and forespore are quite different. Throughout the later stages of sporulation, the mother cell nucleoid retains the diffuse lobular structure of the vegetative cell nucleoid, while the forespore nucleoid is initially rather condensed and then assumes a ring-like structure. The development of the ring-like appearance of the forespore nucleoid is the result of the synthesis of a group of forespore-specific DNA-binding proteins termed the α/β -type SASPs, which saturate the forespore chromosome. These proteins also saturate the dormant spore chromosome. Sporulation of a *B. subtilis* strain lacking the majority of the α/β -type SASPs (termed $\alpha^- \beta^-$) exhibit a number of differences from that of wild-type strains, including delayed forespore accumulation of dipicolinate, overexpression of forespore-specific genes, and delayed expression of at least one mother cell-specific gene activated late in sporulation. These α/β -type SASPs appear to have global effects on gene expression during sporulation and spore outgrowth.

In addition to the role in the sporulation and germination processes, binding of the α/β -type SASPs to the outside of the DNA helix appears to straighten and stiffen the DNA while changing the DNA to an A-like helix. The properties of DNA containing bound SASPs change dramatically. DNA containing bound SASPs displays increased resistance to a variety of chemicals and appears to be inert to the action of wet heat and potentially mutagenic agents. Spore killing by dry heat and radiation occurs largely through DNA damage, and any deficiency of α/β -type SASPs renders spores more susceptible to their effects than wild-type spores. SASP binding to DNA suppresses formation of all cyclobutane pyrimidine dimers following UV irradiation and permits only the formation of thymine-thymine adducts whose repair is apparently much more error-free than repair of the cyclobutane dimers normally produced by UV irradiation.

Final Stages of Sporulation

The final stages of sporulation are controlled by the sporulation-specific transcription factor σ^K in the mother cell. The synthesis of σ^K results in activation of genes that participate in the formation of the cortex and the spore coat and other factors involved in maturation, as well as release of the free spore. Regulation of σ^K synthesis must be timed so that it occurs only after the appearance of σ^E and sealing off the forespore from the mother cell. A site-specific recombinase in the mother cell, encoded by the *spoIVCA* gene, brings together two partial coding sequences to form a circular DNA molecule. Another level of regulation involves processing of the inactive precursor, pro- σ^K , by proteolysis to form σ^K .

Spore Cortex Synthesis

The peptidoglycan cortex of bacterial endospores is required for maintenance of spore core dehydration, heat resistance, and dormancy. The spore cortex peptidoglycan differs in structure from that of the vegetative cell walls in that many of the peptide side chains are removed either partially, leaving single L-alanine side chains, or completely, with the formation of muramic- δ -lactam structures. The decrease in the number of side chains results in a very low degree of cross-linking of the peptidoglycan strands in the spore cortex, a property that may contribute to its flexibility and ability to undergo dehydration during the sporulation process.

Synthesis of the spore cortex involves the activity of several genes. The *spoVD* and *spoVE* genes map in a region of the chromosome containing a cluster of genes involved in cell wall synthesis and cell division. Both SpoVD and SpoVE proteins show amino acid sequence homology to PbpB (FtsI) and FtsW of *E. coli*. The *spoVD* gene is regulated independently of other genes in the cluster, is sporulation specific, and the gene product is involved in cross-wall synthesis, suggesting that its action may result in the production of a unique spore peptidoglycan. The presence of a σ^E -dependent promoter immediately upstream of *spoVE* allows expression during sporulation. The gene product appears to be similar to that of FtsW, suggesting a role in peptidoglycan synthesis. Several other genes, including *spoVB* (*spoIIIF*), *spoVG*, *gerI*, and *gerM*, seem to be involved in cortex synthesis, but their role is less well defined.

Penicillin-binding proteins PBP2B and PBP3 increase during sporulation. Their map location immediately upstream of *spoVD* suggests that the gene for PBP2B may be a functional homolog of *pbpB* of *E. coli* and indicates a role in septum formation. A *B. subtilis* mutant, *dacB*, lacks the sporulation-specific D,D-carboxypeptidase penicillin-binding protein PBP5* and shows a significant increase in spore peptidoglycan cross-linking as compared to the wild type, but displays normal spore coat dehydration. A *cwlD* mutant produces spore peptidoglycan with no muramic- δ -lactam and shows a 2-fold increase in cross-linking, but has near-normal spore core dehydration and normal spore heat resistance. A *cwlD dacB* double mutant produces spore peptidoglycan with increased cross-linking but with little change in spore core hydration as compared to the wild type. Double mutants also produce novel muropeptides containing glycine with no significant changes in spore resistance or core hydration. Structural analysis of *B. subtilis* spore peptidoglycan during sporulation reveals that there is a gradient of cross-linking that spans the spore cortex peptidoglycan. However, the loss of this gradient in some mutant strains does not alter spore core dehydration, indicating that the gradient is not required for core dehydration.

Spore Coat Protein Synthesis

The multilayered spore coat is composed of at least 15 polypeptides plus an insoluble protein fraction. This coat provides a high degree of resistance to a variety of environmental factors, including mechanical stress, and degradative proteins, such as lysozyme. Coat formation involves the orderly assembly and cross-linking of upward of 20 coat proteins.

Genes for at least 7 coat proteins, designated *cotA* through *cotF* and *cotT*, have been identified. Most of the well-characterized *cot* genes have σ^K -dependent promoters even though they show considerable variation in their time of expression. The *cotE* gene has a σ^E -dependent promoter that permits earlier expression, but it also has a σ^K -dependent

promoter allowing expression to continue once σ^K appears. The CotA, CotB, and CotC proteins are responsible for the formation of outer coat proteins. The CotD, CotE, and CotT proteins are involved in the formation of inner coat proteins. Mutations at *cotE* are deficient in proteins encoded by *cotA*, *cotB*, and *cotC* genes, indicating that CotE is deposited on the outside surface of the inner coat, where it serves as a basal protein on which the other proteins assemble.

Mutations in the *spoIVA* locus abolish cortex synthesis and interfere with the synthesis and assembly of the spore coat. The phenotypic properties of *spoIVA* mutants suggest a role for SpoIVA at an early stage in the morphogenesis of the spore outer layers. In the absence of SpoIVA, cortex synthesis is virtually absent and coat proteins accumulate in swirls in the mother cell cytoplasm rather than on the surface of the outer prespore membrane. The dual defects in both coat and cortex synthesis could be explained if coat deposition was dependent on completion of the cortex and SpoIVA was essential for cortex synthesis. The *B. subtilis* genome sequencing project revealed six open-reading frames—*yabD*, *yabE*, *yabF*, *yabG*, *yabH*, and *yabJ*, in the region between *dnaA* and *abrB* on the *B. subtilis* chromosome. The σ^K RNA polymerase transcribes the *yabG* gene during sporulation and mutant spores display altered-coat protein composition. YrbA (also called SafA, for SpoVID-associated factor A) associates with SpoVID to form a complex during the early stages of assembly of the spore coat. YrbA is one of several coat proteins that engender resistance of spores to lysozyme and may also play a role in germination.

The spores of *Bacillus* and *Clostridium* species contain as much as 10% of their dry weight as dipicolinic acid (DPA; pyridine-2,6-dicarboxylic acid). This compound is synthesized in the mother cell compartment during the later stages of sporulation but accumulates only in the developing forespore. Most of the DPA is in the core where it is chelated with Ca^{2+} and other ions. Since DPA is found only in dormant spores of *Bacillus* and *Clostridium* species, there has been considerable conjecture as to the role of DPA in the unique properties of spores.

Spores of a strain of *B. subtilis* containing a double mutation, $\Delta ger3spoVF$, lack a major germinant receptor and cannot synthesize DPA. The DPA-less spores have normal cortical and coat layers, as observed by electron microscopy, but their core region appears to be more hydrated than the spores with DPA. Analysis of the resistance to various agents of spores produced by wild-type, $\Delta ger3$, and $\Delta ger3spoVF$ revealed that (1) DPA and core water content play no role in spore resistance to dry heat, desiccation, or glutaraldehyde; (2) an elevated core water content is associated with decreased spore resistance to wet heat, hydrogen peroxide, formaldehyde, and the iodine-based disinfectant Betadine; (3) the absence of DPA increases spore resistance to UV radiation; and (4) wild-type spores are more resistant than $\Delta ger3$ spores to Betadine and glutaraldehyde.

The final step in sporulation is lysis of the mother cell and release of the fully formed spore (Fig. 19-1). If conditions are suitable, the spore can germinate and undergo outgrowth, converting back to a growing vegetative cell.

ACTIVATION, GERMINATION, AND OUTGROWTH OF BACTERIAL ENDOSPORES

The bacterial spore evolved as a means of surviving periods of low nutrient supply and other harsh environmental conditions. At the same time, the dormant spore must possess

the ability to respond quickly and specifically to conditions favorable to reentry into the vegetative growth cycle. Inherent in the sporulation process, then, is the incorporation of an efficient mechanism for germination and outgrowth. This section examines the environmental factors that influence activation of the spore coat and its entry into the vegetative growth cycle.

Activation

The process of spore activation in natural environments is unknown. Under laboratory conditions, activation is most commonly achieved by heating (usually at 65–70 °C for 30–45 min). Exposure at low pH or low temperature, reducing agents, or a number of chemical agents may be equally effective in activating bacterial endospores. Activation can be distinguished from germination by a number of criteria. Activated spores still retain their resistance to heat, are refractile, are resistant to staining, and still contain large amounts of dipicolinic acid. Activation is generally considered nonessential. However, the percentage of germinating spores is greatly increased by activation. Indeed, it has been found that the number of germinating spores is often very low unless some form of activation is employed. The mechanism of the activation process is not known with certainty. Some activating agents appear to increase the permeability of the spore to germinants. Activation is reversible, whereas spores are committed to germinate following brief exposure to an effective germinant.

Germination

Germination occurs in response to specific germinants that act as triggers, leading to rapid changes in the structure and physiology of the spore. As already stated, the spore is committed to germinate even if the germinant is removed. Commitment precedes the detectable changes in the spore: loss of heat resistance, ion fluxes, release of Ca^{2+} and dipicolinic acid, hydrolysis of cortex peptidoglycan, rehydration of the core protoplast, and resumption of metabolic activity. During the early stages of germination, the light-scattering ability changes rapidly (within 6–8 min) from phase-bright to phase-gray to phase-dark.

Effective germinants include (1) L-alanine and alanine analogs; (2) L-alanine plus inosine or other ribonucleotides; (3) sugars plus inorganic ions; (4) inorganic ions alone; or (5) asparagine, glucose, fructose, and KCl (AGFK). Spores of different species of bacilli may respond to one or all of these combinations. D-Alanine competitively inhibits the germinant action of L-alanine. Alcohols and methyl anthranilate inhibit germination triggering. Some inhibitors of germination are selective. For example, azide inhibits AGFK germination, whereas phenyl methyl sulfonyl fluoride inhibits only L-alanine germination. Inhibition of germination by protease inhibitors, such as tosyl arginine methyl ester, suggests a role for proteolytic activity in the germination process. Other inhibitors appear to be somewhat nonspecific in their action.

Germination Monitoring. Qualitative tests for monitoring spore germination include loss of optical density, loss of heat resistance, and changes in appearance under the phase-contrast microscope. A plate test that is widely used to distinguish Ger^+ and Ger^- colonies involves detection of dehydrogenase-linked metabolism that is resumed upon germination by reduction of 2,3,5-triphenyltetrazolium chloride (a tetrazolium

dye) added to the plating medium. Reduction of the tetrazolium dye stains the germinated colonies red, whereas ungerminated spores do not stain (appearing white). A high correlation between the absence of or marked reduction in tetrazolium reduction and a measurable defect in spore germination aids in the selection of germination-defective mutants.

Germination Loci. Genetic analysis of spore germination in *B. subtilis* has identified a number of germination (*ger*) loci. Germination mutants are grouped according to their phenotype. Spores of *gerA* and *gerC* mutants fail to germinate in response to L-alanine but are able to respond to AGFK. Mutants that respond normally to L-alanine but are not stimulated in response to AGFK are designated *gerB*, *gerK*, and *fruB*. Mutant spores that fail to germinate in response to either of these germinants have been designated *gerD* and *gerF*. The *gerE* gene may code for or regulate a protease required for the processing of polypeptides in the spore coat, during germination. These differences in response to germinants suggest the presence of at least two alternate systems for their detection and response. Mutants in a number of other loci (*gerE*, *gerJ*, *gerM*, *spoVIA*, *spoVIB*, *spoVIC*, *cotD*) affect both the germination properties of the spore and the overall structure of the cortex or coat. One novel gene, *yaaH*, encodes a spore protein produced in the mother cell compartment and is required for the L-alanine-stimulated germination pathway. Germination of *yaaH* mutant spores in AGFK is almost the same as that of wild-type spores.

Models of Spore Germination. Several models for the mode of action of triggering germinants have been proposed. Models that include metabolism of the germinants infer that the germinants are converted to some intracellular metabolites. Models based on the premise that the germinants are not metabolized propose that the germinants act allosterically on a receptor protein. Proponents of this latter model suggest that the germinants may alter membrane permeability or trigger proteolytic or cortex lytic enzyme activity.

Metabolic Changes During Germination. Dormant spores contain reduced levels of a number of important metabolites, particularly high-energy compounds such as ATP, sugar phosphates, reduced pyridine nucleotides, ribonucleotide triphosphates, and tRNA. Many other compounds, such as adenine nucleotides, ribonucleotides, pyridine nucleotides, and other RNA species, are present at levels approximating those in growing cells. One potential high-energy phosphate donor, 3-phosphoglycerate (PGA), is present in the dormant spores of some, but not all, strains at very high concentrations. Accumulation of ATP occurs within a few minutes after germination begins (150-fold increase in the first 5 min). This initial increase in ATP occurs at the expense of endogenous energy sources, particularly the large PGA pool.

Dormant spores contain PGA mutase, enolase, and pyruvate kinase, the three enzymes required for conversion of PGA to ATP. The dormant spore also contains sufficient enzymes of the glycolytic pathway and the hexose monophosphate shunt, since neither protein nor RNA syntheses are required for high-energy phosphate production through at least 40 minutes of germination. The mechanisms whereby these enzymes remain inactive in the dormant spore and are activated on germination are not known with certainty. After the exhaustion of the PGA pool, energy metabolism apparently occurs primarily via the hexose monophosphate shunt, since many enzymes

of the TCA cycle are absent in germinating spores. The glucose dehydrogenase gene (*gdh*) is expressed only during sporulation.

The synthesis of RNA begins by the second minute of germination in *B. megaterium* and constitutes a major use for the ATP generated at this stage. Synthesis of RNA during the first 20 minutes of germination can be completely accomplished from nucleotides stored in the dormant spore. Approximately 85% of the ribonucleotides required for RNA synthesis during the first 25 to 50 minutes of germination are derived from RNA degradation. This hydrolytic activity is necessary for rapid RNA synthesis early in germination, since nucleotide biosynthesis is not demonstrable at this time.

Chromosome replication (DNA synthesis) does not begin until late in spore germination. Degradation of RNA supplies the nucleotides for DNA synthesis as well as for RNA synthesis. There is a temporal relationship between the increase in deoxyribonucleotide levels and the increased rate of DNA synthesis.

Germination-specific enzymes that hydrolyze spore-specific cortex peptidoglycans have been identified and studied in *B. subtilis* and *C. perfringens*. Cortex hydrolysis during germination is initiated by attack of spore cortex-lytic enzyme on intact spore peptidoglycan and leads to removal of cross-linkages. This step is followed by further degradation of the polysaccharide portion of the modified peptidoglycan by cortical fragment-lytic enzyme. The spore peptidoglycan hydrolase, SleC, which is active during germination, apparently exists in an inactive form (pro-SleC) on the outside of the cortex layer in the dormant spore. During germination, a germination-specific protease (CspC) cleaves amino acid residues from the pro-SleC, converting it to its active form. Apparently several serine proteases (CspA, CspB, CspC) are the products of genes positioned just upstream of the 5' end of the *sleC* gene in *C. perfringens*.

A defect in the *cwlD* gene of *B. subtilis* causes a block in the formation of muramic acid lactam structure and leads to a lack of germination. Another cell wall hydrolase, CwlJ, also has an effect on germination. Spores deficient in *cwlJ* respond to both L-alanine and AGFK, but the refractility of spore suspensions decreased more slowly than in the case of the wild-type strain, and the mutant spores released less dipicolinic acid than the wild-type strain during germination.

A germination-specific *N*-acetylmuramoyl-L-alanine amidase is encoded by *sleB* in *B. subtilis* and *B. cereus*. Immunoelectron microscopy with anti-SleB antiserum and a colloidal gold-immunoglobulin G complex revealed that the enzyme is located just inside the spore coat layer in the dormant spore. The amidase appears to exist in a mature form lacking a signal sequence, indicating that *SleB* is translocated across the inner membrane of the forespore by a secretion signal peptide and is deposited in the cortex layer between the forespore inner and outer membranes. This peripheral location of the spore-lytic enzyme in the dormant spore suggests that spore germination may be initiated at the exterior of the cortex. Spores deficient in *sleB* do not release hexosamine at a significant level. However, a doubly deficient mutant, *cwlJ sleB*, produces spores that are unable to germinate but exhibit initial germination reactions such as partial decrease in refractility and slow release of dipicolinic acid.

Protein synthesis begins within the first 2 or 3 minutes of germination and can be accomplished solely from stored nitrogen reserves through the first 60 minutes of germination. The free amino acid content of dormant spores is limited but increases rapidly in germinating spores as a result of hydrolysis of spore protein. The proteins degraded in this process are the SASPs formed during sporulation (see earlier discussion). In *B. megaterium* and *B. subtilis*, a number of SASPs have been

identified. Proteolytic activity is necessary for protein synthesis early in germination, since *de novo* biosynthesis of most amino acids is not demonstrable at this time.

Metabolism of small molecules during spore germination can be divided into two general categories or stages: stage I, or the turnover stage, from 0 to 70 minutes of germination; and stage II, or the synthesis stage, from 70 minutes and beyond. By the onset of stage II, the germinating spore gains the capacity for *de novo* biosynthesis of all small molecules required for the formation of DNA, RNA, and protein. Endogenous reserves of energy, amino acids, and nucleotides are exhausted by this time. During stage I, small molecules are derived primarily from stored reserves. Stage I consists of three overlapping substages — Ia (0–15 min), Ib (0–20 min), and Ic (0–70 min) — each representing the period when endogenous energy, nucleotide, and amino acid reserves, respectively, are sufficient for most of the metabolic and biosynthetic requirements.

Outgrowth

Although germinating spores lose the characteristics of the intact spore rather quickly, the germinants are by no means comparable to vegetative cells. Germinated spores have cytological structures typical of vegetative cells, are sensitive to environmental conditions, and are metabolically active. However, they lack many of the activities and properties of vegetative cells. The transition from germinated spore to vegetative cell has been termed **outgrowth** and involves a number of metabolic changes. Generally, if a suitable nutritional environment is provided, the germinated spore will continue rapidly into the outgrowth process. If nutritional requirements are lacking, the process may stop altogether, or the cells may proceed to sporulate and enter into **microcycle sporulation**. In the microcycle, asymmetric division, rather than symmetrical cell division, begins immediately after the elongation of the cell is initiated and the cells proceed to sporulate. However, one round of DNA replication must occur prior to reentry into the sporulation cycle.

The SASPs that saturate the spore chromosome and protect the spore from damage by UV radiation, heat, and peroxides are cleaved during spore germination by a germination protease and provide the amino acids for protein synthesis. This rapid degradation of SASPs is essential to allow for DNA transcription and DNA replication during spore outgrowth. If SASPs bind too tightly to DNA or their degradation is impaired, then spore outgrowth is inhibited. A mutant SASP, SspC, has a very high affinity for DNA and confers UV resistance on spores lacking α/β -type SASP. However, outgrowth of the germinated spores was blocked by 90% compared to wild-type spores, and the mutant form of SspC persisted in the germinated spores while wild-type SspC was almost completely degraded (see ref. by Moir and Smith, 1990). The persistence of the mutant form of SspC on DNA appears to interfere with transcription during spore outgrowth.

Isolation of mutants of *B. subtilis* that are temperature sensitive only during spore outgrowth suggests that there are functions that are specific to this stage of the life cycle. Studies of these mutants show that unique RNAs are synthesized during spore outgrowth. It has been possible to isolate and determine the chromosomal location of these genes from a cloned library of *B. subtilis* DNA using hybridization with labeled RNA prepared from outgrowing spores in the presence of a large excess of competing vegetative RNA.

MYXOBACTERIAL DEVELOPMENTAL CYCLE

Myxobacteria, with the exception of newly identified marine forms, are aerobic soil or terrestrial bacteria that prefer relatively neutral or nonextreme growth environments. These bacteria are intriguing for (1) their ability to form metabolically inactive, environmentally resistant endospore-like forms, called **myxospores**, organized into sometimes elaborate and beautiful macroscopic structures called **fruiting bodies**, (2) their social behavior; and (3) their predatory nature. The social behaviors of these bacteria are important aspects of both their vegetative growth and developmental cycle. In particular, the cell-to-cell interactions, the cell signaling, and the gliding motility that mediate such social behaviors make these organisms especially fascinating. Most of the basic knowledge of the life cycle of myxobacteria has come from the study of *Myxococcus xanthus*, and, thus, this organism has become the prototype myxobacterium. Therefore, unless indicated otherwise, our discussion of the myxobacterial life cycle comes from studies on *M. xanthus*.

Life Cycle of Myxobacteria

The life cycle of *M. xanthus* is illustrated schematically in Figure 19-3. As with endospore-forming bacteria, the life cycle consists of a vegetative cycle, during which growth and cell division occurs, and a developmental cycle, during which myxospores and fruiting bodies are formed.

The vegetative cycle is itself very interesting. In nature or in the laboratory, when supplied with sufficient nutrients and favorable growth conditions, the bacteria can grow with generation times as low as 3.5 hours. Even in the vegetative state, an important aspect of myxobacterial growth involves cell-to-cell interactions that create localized high cell densities. Myxobacteria, as a group, also excrete a wide assortment of hydrolytic enzymes, allowing them to grow on a variety of macromolecules including proteins, simple and complex polysaccharides, peptidoglycans, and nucleic acids. In

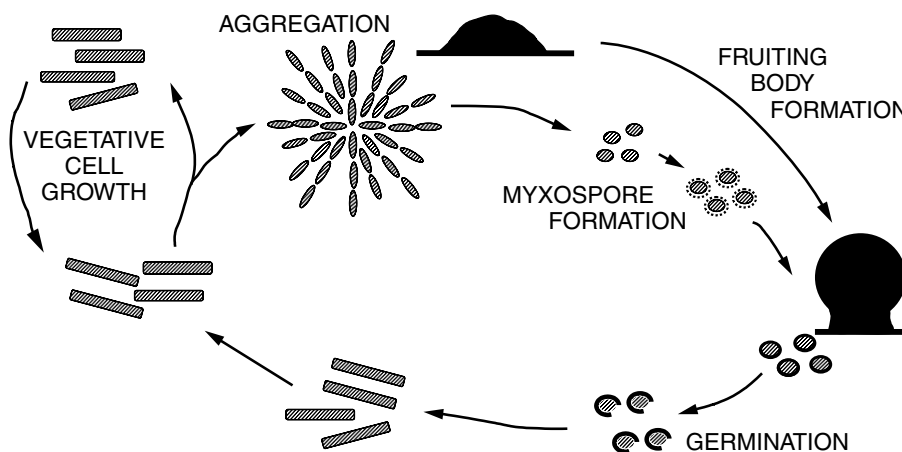


Fig. 19-3. Schematic representation of *Myxococcus xanthus* life cycle. Structures are not drawn to scale. Fruiting bodies can be a few hundredths of a millimeter in diameter and vegetative cells are 5 to 7 by 0.7 μm . (Source: Modified from Dworkin, 2000. In Y. V. Brun and L. J. Shimkets (eds.), *Prokaryotic Development*, ASM Press: Washington, DC.)

addition, many myxobacteria produce a broad spectrum of antibiotics. Thus, like their closest relatives, the bdellovibrios, myxobacteria are capable of killing and lysing a number of other bacteria, fungi, and protozoa. In fact, myxobacteria are thought to prey on other living cells, and this activity, coupled with their tendency to “socialize” (staying close together in groups and maintaining a localized high cell density), has caused researchers to call this behavior the “**wolf-pack effect**.” Studies with *M. xanthus* have shown that cells, although capable of individual movement (**adventurous** or **A-motility**), stay together and move in swarms (**social** or **S-motility**). This type of behavior is believed to optimize feeding on macromolecules by increasing the local concentration of hydrolytic enzymes.

Vegetative cells will continue to divide until nutrient downshift or limitation is sensed. At this point, cells can either enter a stationary-phase condition of very slow growth, or if a solid surface is present and a high cell density is achieved, cells will enter into the developmental stage. Entry into either a stationary phase or the developmental stage is a biologically costly process for myxobacteria, since both are accompanied by a considerable loss in viable cells. Thus, the choice to enter into the developmental stage is not automatic. It is only when the three conditions of a decline in available nutrients are met, a solid surface is present (to allow for **gliding motility**), and there is high cell density that cells will enter into the developmental mode. In a process not completely understood but involving cell-signaling mechanisms and cell-to-cell interactions, cells begin to aggregate, forming **aggregation centers**. Interestingly, aggregation coincides with an extensive loss of viable cells in the population due to lysis in a process called **developmental autolysis**. The surviving cells go on to form myxospores and fruiting bodies. When conditions become favorable once again, the myxospores can germinate to produce metabolically active, highly motile, vegetative cells.

Aggregation and Fruiting Body Formation

Aggregation. Nutrient limitation in the presence of a solid surface and at a high cell density triggers the first steps in the developmental cycle of *M. xanthus*. Unlike the responses of endospore-forming and non-spore-forming bacteria to nutrient depletion, the initial stage of the developmental cycle of myxobacteria involves an energy-intensive aggregation. The initiation and progression of aggregation involves both intracellular signals in individual cells and intercellular signals. The intracellular signal is believed to be mediated by *relA*-dependent (p)ppGpp accumulation, since *M. xanthus* relies on primarily an amino acid-based diet, and depletion of one or more amino acids will trigger a stringent response.

Two additional factors are proposed to play a role in determining entry into the developmental cycle: the products of the *socE* and *csgA* genes. The *socE* gene product is unknown and does not exhibit homology to any other proteins in the databases. The *csgA* gene encodes two products, a 17 kDa and a 25 kDa protein, which function as the C-signal (see below). The *socE* gene is highly expressed in vegetative cells and becomes depleted in nongrowing starved cells as a result of inhibition by ppGpp or the stringent response. It is the depletion of the *socE* product that is believed to be important in SocE's role in regulating the induction of the developmental cycle.

In contrast, the *csgA* gene is poorly expressed in vegetative cells and is induced some 5-fold during development. It is partially repressed by the *socE* product and positively regulated by ppGpp and interestingly another intercellular signal encoded by the *bsgA*

gene, which encodes the B-signal (see below). The C-signal functions throughout the developmental cycle, but early on, when at low concentrations, it is needed for growth arrest and directional motility for aggregation. Thus, it is the induction of the stringent response and the balance between SocE and CsgA that appear to be important in the decision to enter into development.

One of the effects of inducing a *relA*-dependent stringent response is the production of the earliest intercellular signal, the A-signal. **A-signal** appears to be a mixture of both amino acids and peptides that are generated by proteases secreted by the myxobacteria. Generation of the A-signal involves at least three genes: *asgA* (*asg*, meaning A-signal gene), *asgB*, and *asgC*. Other signals include (1) the **B-signal**, which has not been identified as of yet but is dependent on the *bsgA* gene product; (2) the **C-signal**, which is a 17 kDa and a 25 kDa protein encoded by the *csgA* gene; (3) the **D-signal**, the least understood of the extracellular signals, which is believed to be a mixture of fatty acids and is dependent on the *dsgA* gene; and (4) the **E-signal**, which appears to be long branched-chain fatty acids and requires the *esgA* and *esgB* gene products. Response to, and for some the production of, these extracellular signals (A-, B-, C-, D- and E-signals) requires high cell densities. High cell densities are essential for the effective exchange and response to cell signals as well as the necessary direct cell-to-cell interactions.

Direct cell-to-cell interactions involve the production of cell surface appendages called **fibrils**. Fibrils are filamentous structures up to 50 μm long and are composed of polysaccharides and closely associated **integral fibril proteins (IFPs)**. Fibrils are important for cell-to-cell cohesion and the characteristic social behavior associated with the developmental cycle. This was confirmed genetically by demonstrating that strains lacking fibrils, *dsp* mutants, are also defective in several social functions. In particular, fibrils are proposed to function in the normal cell-to-cell positioning required for efficient intercellular signaling. Part of this function appears to lie in their ability to allow individual cells to perceive adjacent cells and analyze cell density. Important to this ability is a fibril-associated ADP-ribosylation activity.

The requirement of a solid surface for the induction of development in *M. xanthus* reflects the need for gliding motility (e.g., S-motility) in myxobacteria aggregation and fruiting body formation. As with other bacteria that exhibit gliding motility, the underlying mechanism is not completely understood. This motility is central to formation of the aggregation centers and cellular alignment within the developing fruiting body. Although the mechanism of S-motility is not well understood, it requires the ability to produce polar type IV pili. Mutants in pilus biosynthesis, *pil* mutants, as well as in the S-motility genes, *sgl* and *tgl* mutants, exhibit defects in fruiting body formation, further supporting the need for gliding or S-motility in the developmental cycle.

S-motility is not a random process. In both vegetative swarming and developmental aggregation, cells respond chemotactically, moving toward attractants (nutrients) and away from repellents. One such attractant is phosphatidylethanolamine (PE), both dilauroyl and dioleoyl PE. At least two signal transduction systems play a role in developmental aggregation: the **Frz** and **Dif systems**. Both are composed of components showing homology to the Che systems of enteric bacteria. Each acts at different stages of development. The Dif system functions early in the aggregation process, since *dif* (defect in fruiting) mutants fail to aggregate beyond the early stages of fruiting body formation. The Frz system functions in both vegetative swarming activity and at later stages in fruiting body formation. Mutants in the *frz* genes exhibit

as a “frizzy” phenotype and form defective aggregates but not discrete mounds. Both of these systems are proposed to sense and respond to developmental signaling molecules and perhaps other extracellular factors to control developmental aggregation/motility.

Fruiting Body Formation. The induction of the developmental cycle ultimately leads to the formation of a myxospore-filled fruiting body in *M. xanthus* and other myxobacteria. The size and complexity of fruiting bodies ranges considerably among the myxobacteria (Figure 19-4) from very simple sacs of myxospores (*Nanocystis exedans*) to elaborate branched tree-like structures (*Stigmatella* or *Chondromyces*). Why there is such a difference in fruiting body morphology is unclear but may reflect the habitats that the particular genera or species typically populates and the effect on dispersal of myxospores into the environment.

As mentioned, fruiting body formation is initiated by the presence of three conditions: nutrient limitation, solid surface, and high cell density. These conditions lead to accumulation of the intracellular signal (p)ppGpp, which in turn regulates the transcription of genes necessary for the production of the first developmental signal, A-signal. Based on evidence to date, the generation of these two signals is required for the initiation of development and fruiting body formation. Within a few hours following the onset of starvation, cells begin to aggregate, forming aggregation centers. Microscopic examination shows cells moving in spiral patterns within stacked monolayers, eventually forming a hemispheric mound. In the outer densely packed portions are vegetative cells moving in a single direction, clockwise or counterclockwise, around the fruiting body periphery. A less densely packed inner core contains nonmotile myxospores. Based on experiments using *lacZ* fusions to several

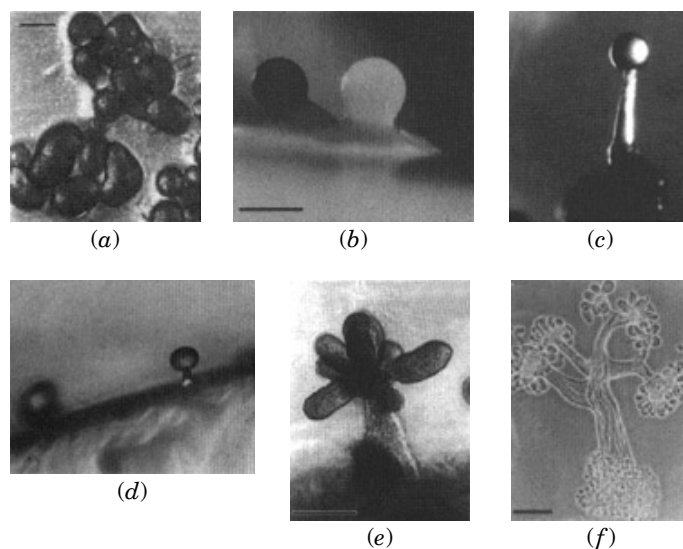


Fig. 19-4. Examples of fruiting bodies produced by the myxobacteria. Fruiting bodies of (a) *Cystobacter fuscus* (bar, 100 μ m); (b) *Myxococcus fulvus* (bar, 100 μ m); (c) *Myxococcus stiptatis* (fruiting body about 170 μ m high); (d) *Mellitangium* sp. strain Hp (fruiting body about 40 μ m high); (e) *Stigmatella aurantiaca* (bar, 50 μ m); and (f) *Chondromyces crocatus* (bar, 100 μ m). (Source: Modified from Dworkin, 2000. In Y. V. Brun and L. J. Shimkets (eds.), *Prokaryotic Development*, ASM Press: Washington, DC.)

developmental genes, it is proposed that C-signal, produced in the outer regions by the circling cells, induces myxospore formation, and these cell forms ultimately get moved into the inner domains by the movement of the undifferentiated cells in the outer areas of the fruiting body.

The myxospores of *M. xanthus* have been studied in the most detail and their characteristics have been extrapolated to those of other genera/species. Myxospores are metabolically quiescent and exhibit significant resistance to high temperatures, UV irradiation, and desiccation, especially compared to vegetative cells. Although generally not as resistant as endospores, they are sufficiently environmentally resistant to survive the extremes they would typically encounter in their natural milieu.

Genetics of *Myxococcus xanthus* Development

Some of the genes involved in *M. xanthus* development are listed in Tables 19-3, 19-4, and 19-5.

Genes Involved in Signaling. As mentioned, the primary carbon energy source of myxobacteria is amino acids. The limitation of any amino acid induces development and fruiting body formation. It is well known in enteric bacteria that amino acid starvation induces the stringent response. Thus, it is not surprising that studies showed that *relA*-dependent (p)ppGpp accumulation is an early intracellular signal essential to development and that *relA* mutants are defective in myxospore and fruiting body formation. Two additional early signals are dependent on the *socE* and *csgA* genes. The *socE* gene is an essential gene and does not exhibit homology to anything in the various

TABLE 19-3. A Partial List of Genes Involved in *Myxococcus xanthus* Development: Intracellular and Intercellular Signaling

Gene	Proposed Function of Product or Developmental Phenotype of Mutants
<i>Intracellular Signaling</i>	
<i>relA</i>	(p)ppGpp synthetase I; defective in myxospores and fruiting body formation
<i>socE</i>	Essential gene; negative regulator of development and <i>relA</i> -dependent accumulation of (p)ppGpp
<i>csgA</i>	C-signal gene encodes 17 and 25 kDa proteins; 25 kDa protein is a short-chain alcohol dehydrogenase
<i>Intercellular Signaling</i>	
<i>asgA</i>	A-signal gene; putative response regulator-kinase
<i>asgB</i>	A-signal gene; putative transcription factor (helix-turn-helix protein)
<i>asgC</i>	A-signal gene; σ^{70} subunit of RNA polymerase
<i>bsgA</i>	B-signal gene; 90.4 kDa putative ATP-dependent protease (Lon homologue)
<i>csgA</i>	C-signal gene encodes 17 and 25 kDa proteins; 25 kDa protein is a short-chain alcohol dehydrogenase
<i>dsgA</i>	D-signal gene; homologue of <i>E. coli</i> translation initiation factor (IF)-3
<i>esgA</i>	E-signal gene; E1 decarboxylase of branched-chain keto-acid dehydrogenase (with <i>esgB</i> product)
<i>esgB</i>	E-signal gene; E1 decarboxylase of branched-chain keto-acid dehydrogenase (with <i>esgA</i> product)

TABLE 19-4. A Partial List of Genes Involved in *Myxococcus xanthus* Development: Aggregation and/or S-motility

Gene	Proposed Function of Product or Developmental Phenotype of Mutants
<i>dsp</i>	Function unknown; mutants lack fibrils
<i>ifp-20</i>	IFP-20; fibril protein; mutants defective in side-to-side cohesion
<i>pilA</i>	Major pilus subunit, prepilin
<i>pilB</i>	Type II secretion with PilC
<i>pilC</i>	Type II secretion with PilB
<i>pilG</i>	ABC transporter system with PilH and PilI
<i>pilH</i>	ABC transporter system with PilG and PilI
<i>pilI</i>	ABC transporter system with PilG and PilH
<i>pilR</i>	Transcriptional regulator of <i>pil</i> operon; part of two-component sensor-kinase-regulator system with PilS
<i>pilS</i>	Histidine-protein kinase; part of two-component sensor-kinase-regulator system with PilR; negatively regulates PilR activity
<i>pilT</i>	Pilus retraction; mutants defective in S-motility but form pili
<i>sglA(pilQ)</i>	Secretin involved in macromolecule transport
<i>sglK</i>	DnaK (HSP70) homologue
<i>tgl</i>	Mutants defective in pilus biogenesis, S-motility, and development
<i>wzm</i>	Lipopolysaccharide (LPS) biosynthesis
<i>wzt</i>	Lipopolysaccharide (LPS) biosynthesis
<i>wbgA</i>	Lipopolysaccharide (LPS) biosynthesis
<i>mrpA</i>	Histidine-protein kinase; mutants defective in sporulation
<i>mrpB</i>	NtrC-like response regulator
<i>mrpC</i>	cAMP receptor protein (CRP)-like transcriptional regulator
<i>rpoN</i>	σ^{54} subunit of RNA polymerase

databases. Studies indicate that SocE acts as a negative regulator of development and myxospore formation, since loss of SocE leads to growth arrest and myxospore formation. SocE depletion in the cell promotes the *relA*-dependent accumulation of (p)ppGpp. The *csgA* gene is discussed below, but it also is involved in the early signaling events required for growth stoppage and entry into the developmental cycle (Table 19-3).

Five intercellular signals, designated A, B, C, D, and E, have been identified as mediators of the aggregation and fruiting body formation (Table 19-3). The earliest acting signal involved in fruiting body and myxospore formation is the **A-signal**, a mixture of both amino acids and peptides produced by proteases secreted by the myxobacteria. Three genes, *asgA*, *asgB*, and *asgC*, are required for A-signal generation. The *asgA* gene product is proposed to be a response regulator-kinase, since analysis of its amino acid sequence demonstrates homology to the transmitter domains of histidine protein kinases and the receiver domain of response regulator. The *asgB* gene encodes a putative transcription factor based on the presence of a helix-turn-helix domain. The third gene, *asgC*, is a *rpoD* homologue encoding the σ^{70} subunit of RNA polymerase.

The B-signal acts early in development but has yet to be identified. It is dependent on the *bsgA* gene product, a 90.4 kDa protein that appears to be an ATP-dependent protease homologous to Lon protease.

The most extensively characterized of the intercellular signals is the C-signal. It is encoded by the *csgA* gene, which encodes for both a 17 and a 25 kDa protein.

TABLE 19-5. A Partial List of Genes Involved in *Myxococcus xanthus* Development: Chemotaxis Signal Transduction Systems

Gene	Proposed Function of Product or Developmental Phenotype of Mutants
<i>Dif System</i>	
<i>DifA</i>	Methyl-accepting chemotaxis protein (MCP) Tar homologue; receptor for Dif system
<i>difB</i>	Unknown function; no homologue
<i>difC</i>	CheW homologue; signal transduction from DifA to DifE
<i>difD</i>	CheY homologue; response regulator
<i>difE</i>	CheA homologue; histidine-protein kinase that phosphorylates DifD
<i>Frz System</i>	
<i>frzA</i>	CheW homologue; signal transduction from FrzCD to FrzE
<i>frzB</i>	Unknown function; no homologue; interacts with FrzCD
<i>frzCD</i>	Methyl-accepting chemotaxis protein (MCP) Tar homologue; receptor for Frz system
<i>frzE</i>	CheA homologue (N-terminus); histidine-protein kinase CheY homologue (C-terminus); response regulator
<i>frzF</i>	CheR homologue (C-terminus); methyltransferase
<i>frzG</i>	CheB homologue; methylesterase
<i>frzS</i>	No homologue; putative regulator of S-motility
<i>frzZ</i>	CheY homologue (N-terminus); CheY homologue (C-terminus); putative response regulator not central part of Frz system
<i>abcA</i>	ABC transporter/exporter; may interact with FrzZ; mutants exhibit "frizzy" phenotype in <i>pilQ1</i> background
<i>rpoE1</i>	Extracytoplasmic function (ECF) σ factor; may interact with FrzZ; mutants are defective in vegetative swarming and developmental aggregation

Both function as the C-signal and are cell surface associated. The larger 25 kDa protein is proposed to be a short-chain alcohol dehydrogenase. *csgA* mutants are unable to aggregate or sporulate and fail to express several developmental genes. At low concentrations, C-signal functions in the growth arrest required for initiating the developmental cycle. It is positively regulated by (p)ppGpp and positively regulates *relA*. At higher concentrations and at high cell densities, C-signal is essential for directed motility during aggregation and fruiting body formation as well as differentiation into myxospores.

The D-signal is one of the most puzzling of the extracellular signals. The D-signal itself is believed to be a mixture of fatty acids, but it is dependent on the ***dsgA* gene** whose product is a homologue of the *E. coli* initiation factor (IF)-3. Hence the confusing nature of this signal, since it is difficult to hypothesize a reasonable relationship between the effect of a *dsgA* mutation on development and the ability to rescue the *dsgA* mutant phenotype with a mixture of fatty acids.

The E-signal is dependent on two genes: ***esgA*** and ***esgB***. The E-signal itself is proposed to be long branched-chain fatty acids. The *esgA* and *esgB* gene products appear to be an E1 decarboxylase of the branched-chain keto-acid dehydrogenase; thus, they may play a role in generating the E-signal from fatty acids in the environment.

Genes Involved in Aggregation and/or S-motility. In addition to the intracellular and intercellular signals, a number of other gene functions and structures are required for aggregation and S-motility (Table 19-4). These include **fibrils** that are essential to cell-to-cell interactions and many social activities. The production of fibrils is dependent on at least two genes: *dsp* and *ifp-20*. The function of the *dsp* gene product is unclear, but *dsp* mutants lack fibrils. The *ifp-20* gene product is one of the fibril proteins. Mutants unable to produce IFP-20 are defective in the side-to-side cohesion important for maximal signal exchange and thus are defective in cell-to-cell interactions required for development.

As mentioned, the production of type IV pili is required for gliding motility and thus social aggregation and fruiting body formation. The genes essential for pilus biosynthesis are organized into an operon, *pilB^{TC}SRAGHID*. The *pilA* gene encodes the major pilus subunit, prepilin. The *pilB* and *pilC* genes are proposed to encode proteins involved in type II secretion. The *pilG*, *pilH*, and *pilI* genes appear to make up an ABC transporter system. The *pilR* and *pilS* genes are proposed to make up a two-component sensor kinase-regulator signal transduction system, with PilS functioning as a histidine-protein kinase that negatively regulates PilR, a transcriptional regulator of the *pil* operon. The *pilT* gene encodes a protein that functions in pilus retraction that is essential in S-motility. With the exception of *pilT* mutants, mutants in any of the *pil* genes do not produce pili; *pilT* mutants produce pili but are defective in S-motility.

Mutations in the *sgl* and *tgl* genes also result in defects in S-motility and fruiting body formation. The *sglA* gene encodes a secretin involved in macromolecular transport—in particular pilus production—and was renamed *pilQ*. A spontaneous mutation, *sglAI* (*pilQ1*), is found to be in a number of laboratory *M. xanthus* strains, which allows for dispersed growth in liquid cultures but does not affect development. Another gene, *sglK*, has been found to encode a DnaK (HSP70) homologue required for S-motility and cell-to-cell cohesion but is not essential for growth. The *tgl* gene encodes a product needed for pilus biogenesis, which accounts for its defect in S-motility and fruiting body formation.

Genes Involved in Chemotaxis Signal Transduction Systems. As mentioned, gliding motility is directed by and dependent on at least two signal transduction systems, Dif and Frz, which function at different stages in the developmental process.

The *dif* genes encode a putative signal transduction system that functions early in the aggregation process (Table 19-5). The *dif* genes are organized into an operon, *difABCDE*. *difA* or *difE* mutants are defective in fruiting body formation. The deficiency appears to lie in the regulation of S-motility-associated gliding behavior, supporting a role for the Dif system as a signal transduction system. Further evidence for this role is the finding that several of the *dif* gene products encode homologues of the Che system of enteric bacteria. The **DifA protein** is a homologue of the methyl-accepting chemotaxis protein (MCP) Tar and thus is proposed to be a receptor for the Dif system. The **difB gene product** does not exhibit homology to any protein in the databases and thus far has not been assigned a function. The **DifC protein** is a homologue of the CheW protein and thus likely interacts with both the DifA protein and the CheA histidine-kinase protein homologue, the **DifE protein**. The **difD gene product, DifD protein**, is a CheY homologue and thus is proposed to function as a response regulator in the Dif system that is regulated by the DifE histidine-kinase. Thus far, no CheB or CheR homologues in the Dif system have been identified.

The Frz system is encoded by genes (Table 19-5) located in the same region on the *M. xanthus* chromosome and organized into an operon, *frzABCDEGF*, and as separate genes, *frzZ* and *frzS*. The *frz* genes are essential for vegetative swarming, aggregation, and formation of the fruiting body. The designation *frz* comes from the frizzy appearance of the aborted mounds seen in *frz* mutants. Similar to the Dif system, many of the *frz* genes encode for Che system homologues. The *frzA* gene product is a CheW homologue that likely acts as an adapter protein, interacting with the MCP homologue **FrzCD protein** and the CheA histidine kinase homologue **FrzE protein**. Both the FrzCD and FrzE proteins exhibit unique features relative to their Dif and Che homologues. The FrzCD protein does not possess a membrane-spanning region and therefore probably remains cytoplasmic, suggesting that signal input must come from an as-yet unidentified membrane receptor transducer or is generated within the cytoplasm.

The FrzE protein appears to have two functions: its N-terminal end is homologous to the CheA protein, but its C terminus exhibits homology to the CheY response regulator. The FrzCD protein interacts with the **FrzB protein**; although it is not known why, since FrzB has no homologue and its function is unknown. The C terminus of the **FrzF protein** is a CheR homologue and is proposed to possess methyltransferase activity. The **FrzG protein** is a homologue of the CheB protein and possesses methylesterase activity. Thus, FrzF is proposed to methylate FrzCD while FrzG demethylates FrzCD.

The *frzS* gene encodes a new component of the Frz system. Mutants in *frzS* exhibit the frizzy phenotype and have only a slight increase in reversal frequency compared with wild-type strains but interestingly exhibit a major defect in vegetative swarming. This suggests that FrzS plays a role in social motility. The **FrzZ protein** possesses two distinct domains, both of which exhibit homology to the CheY protein, and is therefore proposed to be a response regulator. However, phenotypic data suggest that it is not a central part of the Frz system but rather a regulator of directed motility.

The FrzZ protein was found to interact with two other proteins encoded by the *abcA* and *rpoE1* genes, respectively. AbcA protein shows homology to members of the ABC transporter family. Analysis indicates that it may be an exporter important in the secretion of a molecule involved in developmental aggregation. *abcA* mutants exhibit a frizzy phenotype but only in a *sglAI(pilQ1)* background, making their role in development less clear. The RpoE1 protein is a member of the extracytoplasmic function (ECF) σ -factor family. The *rpoE1* gene is located downstream of the *frzZ* gene and upstream of the *frzS* gene. The *rpoE1* and *frzS* genes are apparently part of an operon with two other genes. Mutants of *rpoE1* do not exhibit the frizzy phenotype but show defects in vegetative swarming and aspects of developmental aggregation.

Other Genes. Mutations in three genes, *wzm*, *wzt*, and *wbgA*, involved in *M. xanthus* lipopolysaccharide (LPS) O-antigen biosynthesis, are also defective in S-motility and aggregation. Since these mutants produce normal type IV pili and fibrils, a role is indicated for LPS O-antigen in S-motility.

Recently, three new genes involved in aggregation and sporulation have been described: *mrpA*, *mrpB*, and *mrpC* (for *Myxococcus* regulatory protein). The *mrpA* gene appears to encode a histidine-kinase homologue and is upstream of, and forms an operon with, the *mrpB* gene, which encodes a NtrC-like response regulator. *mrpC* is transcribed separately and encodes for a cAMP receptor protein (CRP)-like transcriptional regulator. Mutants in both *mrpB* and *mrpC* are defective in aggregation

and sporulation, while *mrpA* mutants are defective in sporulation only. All three genes are developmentally regulated, with MrpB being required for *mrpC* expression and MrpC autoregulating itself.

Another gene that is important for *M. xanthus* development and vegetative growth is *rpoN*. The *rpoN* gene encodes a σ^{54} homologue. In *M. xanthus*, σ^{54} appears to be essential for growth but is also required for the expression of several developmental genes. In fact, the MrpB gene is proposed to be a σ^{54} activator involved in the expression of several σ^{54} -dependent developmental genes.

CAULOBACTER DIFFERENTIATION

Stalked bacteria are a broadly diverse collection of Gram-negative bacteria that are members of the α *Proteobacteria*. They are referred to as stalked bacteria because at some stage in their life cycle they all possess at least one extension from the cell surface called a stalk—also known as **prostheca**, appendages, or hypha.

Caulobacter crescentus is the best studied of the so-called stalked bacteria. It is a crescent-shaped bacillus populating aquatic environments. It is interesting and unique for a variety of reasons. For example, unlike most other differentiating bacteria, *C. crescentus* does not differentiate in response to nutritional stress or environmental cues. Instead, differentiation is a central part of the growth cycle of this bacterium, occurring during each cell division. As a part of this growth cycle, two very different cells are formed: one that is motile and incapable of replicating its DNA or dividing (swarmer cells) and the other that is nonmotile but capable of DNA replication and cell division (stalked cells). The generation of different cell forms during the life cycle of *C. crescentus* brings up intriguing questions: How does the dividing cell differentially produce structures at opposite poles prior to cell division? Why is one cell form able to reproduce whereas the other is not?

Life Cycle of *Caulobacter crescentus*

A schematic representation of the *C. crescentus* life cycle is presented in Figure 19-5. As for most of the stalked bacteria, each division cycle yields two very different cells: a swarmer cell and a stalked cell. The swarmer cell has a single polar flagellum and pili (at the flagellar pole) and is capable of chemotaxis. The main purpose of the swarmer cell is dissemination of the bacteria to new and different environs. This presumably enhances the probability of finding niches with sufficient nutrient supply and less competition for available nutrients. In contrast, the stalked cell is sessile—that is, typically attached to some surface via its stalk/holdfast structure. It is capable of replicating its DNA and dividing to form a swarmer cell and another stalked cell. Thus, unlike cell division in most cells, cell division in *Caulobacter* and most other stalked bacteria yields only one daughter cell capable of further reproduction.

The swarmer cell, although metabolically active and capable of chemotaxis, is unable to replicate its DNA and divide. After some time and in response to an as-yet unknown intracellular signal, the swarmer cell sheds its flagellum and pili. A stalk is produced at the same pole, replacing the flagellum while DNA replication is initiated. As stalk synthesis progresses, the holdfast organelle is formed at the tip. The holdfast functions directly in attachment of the stalked cell to surfaces. This newly formed stalked cell

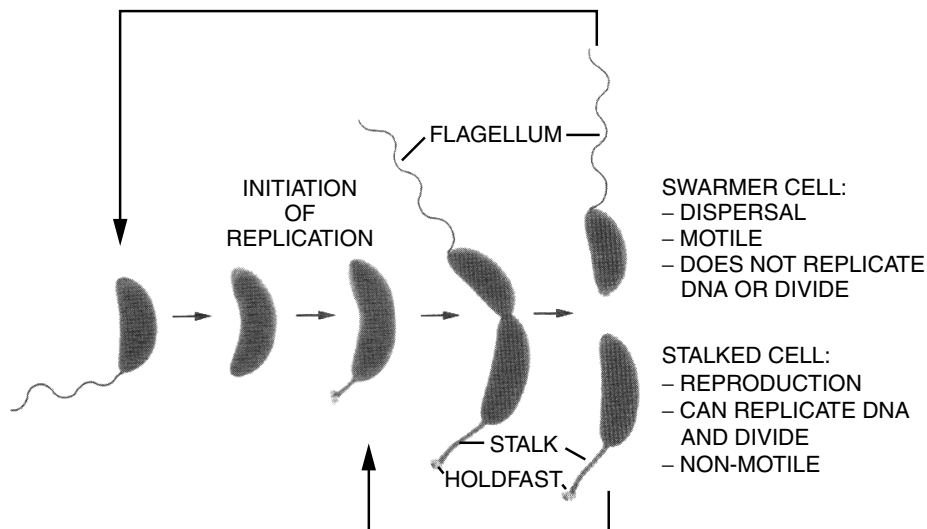


Fig. 19-5. Schematic representation of *Caulobacter crescentus* life cycle. Figure shows electron micrographs of *C. crescentus* cells at different stages of its cell cycle. The flagella are enhanced for illustration purposes. Also at the flagellated poles of swarmer cells but not visible would be pili. (Source: Modified from Brun and Janakiraman, 2000. In Y. V. Brun and L. J. Shimkets (eds.), *Prokaryotic Development*, ASM Press: Washington, DC.)

elongates and begins to synthesize a flagellum at the opposite pole, eventually yielding a **predivisional cell** with a flagellum at one pole and a stalk, with holdfast, at the other. The predivisional cell ultimately divides to yield a swarmer cell and a stalked cell. The swarmer cell is released into the environmental milieu to allow the search for more nutrients and less competition. The stalked cell, on the other hand, is capable of initiating a new round of cell division almost immediately.

The Stalk, the Holdfast, and the Flagellum: Structure, Genetics, and Regulation

The Stalk. The stalk of *C. crescentus* can be described as an extension of the cell body. The center of the stalk is cytoplasmic in origin and appears to be contiguous with the cytoplasm of the cell but devoid of ribosomes and DNA. The surface of the stalk is also continuous with the cell surface structure of the cell body. Along the stalk are **crossbands** of peptidoglycan and perhaps other material. These crossbands link the inner and outer membranes to provide rigidity to the stalk. Interestingly, the numbers of crossbands are believed to reflect the age of the cell. One crossband is formed during each cell cycle, and recent evidence indicates that the stalk grows linearly with each generation.

Stalk biosynthesis is localized to its base. The enzyme apparatus necessary for stalk biogenesis—at least the stalk peptidoglycan—is located there. Components of this machinery include the enzymes that catalyze peptidoglycan biosynthesis, collectively known as penicillin-binding proteins (PBPs). PBPs involved in stalk peptidoglycan biosynthesis appear to be somewhat different from those involved in peptidoglycan production in the cell proper. PBPX, PBPY, and PBP2 appear to be the major PBPs involved in stalk peptidoglycan biosynthesis.

Stalk biogenesis is an impressive undertaking for the cell. It not only requires the redirecting of cell surface biosynthesis to a specific site in the cell and in a perpendicular direction but, it also requires that it be initiated at the correct time in the cell cycle. In addition, the cell must control stalk elongation in response to extracellular phosphate levels.

Although *C. crescentus* differentiation is not responsive to extracellular phosphate concentrations, phosphate limitation does have an extraordinary effect on stalk biosynthesis. Phosphate starvation stimulates stalk biosynthesis, leading to stalks 15 to 30 times the length of stalks (1–2 μm) produced in excess phosphate conditions. Phosphate deprivation has been shown to stimulate stalk biosynthesis in other stalked bacteria as well, suggesting that this is a common environmental stress encountered by these typically aquatic bacteria. The increased stalk length during phosphate limitation can have many benefits. For example, elongating the stalk will increase the surface area of the cell, which allows for more efficient uptake and utilization of available phosphate sources as well as other nutrients. Not surprisingly, some aspects of stalk biosynthesis are under the control of the **pho regulon** of *C. crescentus*. Mutants of the high-affinity phosphate transporter system encoded by the **pstSCAB operon** exhibit a constitutive long stalk phenotype independent of phosphate concentration.

PstSCAB has also been shown in *E. coli* to regulate the Pho regulon by complexing with the sensor-kinase **PhoR**, preventing the phosphorylation of its cognate response regulator **PhoB**. Thus, mutants of the *pstSCAB* genes would not form the repression complex, allowing phospho-PhoR to activate PhoB; phospho-PhoB would then go on to activate, directly or indirectly, genes involved in stalk biosynthesis. This model is supported by findings that *phoB* mutants are defective in stalk elongation in response to phosphate limitation. The stalk biosynthetic genes regulated by PhoB have not been identified.

In addition to Pho regulation, stalk biosynthesis exhibits other levels of control. Mutants of the **rpoN gene**, (encodes the σ^{54} subunit of RNA polymerase), do not produce flagella or stalks, and have cell division defects. Mutants of, the histidine-protein kinase encoding, **pleC gene** do not produce stalks or pili and have nonfunctional flagella. Mutants of the **pleD gene**, which encodes the putative cognate response regulator, are defective in stalk elongation but appear to exhibit normal motility. Moreover, mutants of the global response regulator gene, **ctrA**, exhibit defects in stalk biosynthesis as well as flagellum biosynthesis, DNA replication, and cell division. Interestingly, the Pho regulation of stalk biosynthesis appears to be independent of, or can bypass, the cell cycle control of stalk biosynthesis, since phosphate limitation can induce stalk elongation in mutants that lack stalks during phosphate excess.

The Holdfast. As mentioned, the holdfast is located at the tip of the stalk and functions as an attachment organelle. It mediates the attachment of the stalked cell to a surface or other substrate. Chemical analysis of the holdfast indicates that it is a complex polysaccharide containing a number of acidic residues such as uronic acids. The synthesis of the holdfast is restricted to during differentiation of the swarmer cell to the stalked cell. It is first detected at the tip of the stalk of a predivisional cell and is not found in swarmer cells or at the flagellar pole of the predivisional cell.

The genes involved in holdfast synthesis are still not well characterized. However, a four-gene operon, **hfaABDC operon**, has been identified as being important in holdfast attachment to the cell. The roles of each of these gene are still unclear, but sequence

analysis has revealed some clues. The *hfaA* gene product was proposed to exhibit homology to the C termini of pilus tip proteins of *E. coli* and *Serratia marcescens*. However, this was not confirmed by BLAST or PSI-BLAST (see Chapter 4) analysis of known proteins in the databases. The *hfaB* gene product exhibits partial homology to the CsgA protein of *E. coli*, which is involved in *curli* fiber (temperature-controlled thin, coiled fibrillar pili) formation. The *hfaC* gene product shows similarity to members of the ABC transporter family. The *hfaD* gene product possesses three potential membrane-spanning regions and is proposed to function in anchoring the holdfast to the membrane at the tip of the stalk. Mutants of *hfaA*, *hfaB*, and *hfaD* all show defects in holdfast attachment, exhibiting a high frequency of holdfast shedding. Interestingly, *hfaC* mutants do not show any defects in holdfast production or attachment.

The transcription of *hfaABD* shows cell cycle control. It is greatest in the swarmer compartment of the predivisional cell. However, the holdfast does not appear until after swarmer cell differentiation in the next cell cycle, so it is unclear why *hfaABD* transcription occurs preferentially in the swarmer cell compartment of the predivisional cell. Some researchers have suggested that it provides for the preloading of the swarmer cell with *hfa* mRNA and/or proteins so that it is ready for holdfast production during swarmer cell differentiation into the stalked predivisional cell.

The Flagellum. During each cell cycle, *C. crescentus* divides to generate a nonmotile stalked cell (see above) and a motile swarmer cell. This swarmer cell possesses a **single polar flagellum** formed at the pole opposite to the stalked pole in the predivisional cell. The flagellum of *C. crescentus* is very similar to the flagellum of Gram-negative bacteria described in Chapter 7. Like the flagellum of these bacteria, the biosynthesis of the *C. crescentus* flagellum involves some 50 genes encoding structural, export, and regulatory proteins. In addition, the flagellum is assembled in stages and as three subassemblies.

The first subassembly is known as the basal body, which transverses the inner and outer membranes. The basal body is composed of the C-ring/switch, MS-ring, E-ring, P-ring, L-ring, and distal and proximal rods. The basal body anchors the flagellum to the cell (MS-ring) and also acts as a rotor to turn the flagellum (C-ring/switch) during chemotaxis. Also, associated with the basal body is the flagellar (type III-like) export system. The second subassembly is the hook, which is extracellular and attached to the basal body via the distal rod. The third subassembly is the flagellar filament attached to the hook and acting as the propeller to move the swarmer cell through the environmental milieu.

The flagellum is put together in stages. The genes involved in flagellar regulation and synthesis in each stage are placed into four classes, I through IV. Flagellum synthesis is controlled by cues from the cell cycle. As a result of these cues, the *ctrA* gene (**class I gene**) gets turned on and activates the σ^{70} -RNA polymerase holoenzyme to transcribe the **class II genes**, which include genes that encode for the MS-ring (*fliF* gene) and the flagellar C-ring/switch (*fliG*, *fliM*, and *fliN* genes). The *fliF*, *fliG*, and *fliN* genes are found in an operon with two other genes: *flbD* and *flbE*. FlbD and FlbE are *trans*-acting regulators that negatively control class II gene expression and positively regulate class III gene expression. The *fliM* gene is in an unlinked operon with the *fliL* gene. The *fliL* gene product is needed for flagellar function but not for flagellum assembly. Other class II genes include those that encode the components needed for the export of the extracytoplasmic/external components of the flagellum. These are the

flhA, *flhB*, *flhE*, *fliQ*, *fliR*, *fliI*, *fliJ*, *fliP*, and *fliO* genes. The flagellar export system is homologous to the inner membrane portion of the type III secretion systems of several bacterial pathogens. MS-ring, C-ring/switch, and export system assembly are required not only for the assembly of the remaining flagellum components but also for the expression of class III genes.

FlbD, FlbE, and the integration host factor (IHF) protein positively regulate the expression of **class III genes**. The class III genes are transcribed by an RNA polymerase holoenzyme containing the σ^{54} (the *rpoN* gene product). The next assembled structures are the proximal and distal rods along with the E-, P-, and L-rings of the basal body. The *flgB*, *flgC*, and *flgF* genes encode the proximal rod while the *flgG* gene encodes the distal rod. These genes compose an operon: *flgBCFG*. The formation of the rod is followed by the assembly of the P-ring (encoded by the *flgI* gene) and the L-ring (encoded by the *flgH* gene). The flagellum of *C. crescentus* possesses an additional outer ring called the E-ring, which is encoded by the *flaD* gene. The E-ring lies between the P-ring and the MS-ring within the periplasm.

Upon completion of the basal body, the next subassembly to be put together is the hook, which is also encoded by class III genes. The hook and hook-associated proteins are encoded by the *flgE* gene and the *flgK* and *flgL* genes, respectively. The FlgE, FlgK, and FlgL proteins are secreted through the flagellar export system. The completion of the basal body and hook is required for production of the flagellar filament by suppressing the activity of the *flbT* gene product, which is proposed to be a negative posttranscriptional regulator of flagellin synthesis. The flagellin genes are class IV genes.

Like the class III genes, FlbD, FlbE, and the integration host factor (IHF) protein positively regulate the expression of **class IV genes**, and the class IV genes are also transcribed by a σ^{54} -RNA polymerase holoenzyme. The flagellin protein secretion via the flagellar export system and assembly into the flagellar filament are the last stages in flagellum synthesis. In *C. crescentus*, six different flagellin genes are present in two separate gene clusters. The α cluster is composed of *fljL*, *fljK*, and *fljJ* genes, which encode a 27.5 kDa, 25 kDa, and 28.5 kDa flagellin protein, respectively. The unlinked β cluster possesses three more genes, *fljM*, *fljN*, and *fljO*, each of which encodes a 25 kDa flagellin protein. The flagellin proteins are assembled in specific order. The 28.5 kDa flagellin is added proximal to the hook followed by addition of the 27.5 kDa flagellin. The 28.5 kDa and 27.5 kDa flagellins, although relatively minor components, are required for flagellum biosynthesis. The 25 kDa flagellins are added next and extend the flagellar filament; thus, they represent the most abundant flagellar component.

Several other genes have been identified that are required for the synthesis and/or assembly of a normal hook and/or filament. The *fliK* gene product controls assembly of FlgE monomers into the hook structure. A number of genes are required for correct flagellin synthesis including *flmA* (*flaA*), *flmB*, *flmC*, *flmD* (*flaR*), *flmE* (*flaZ*), *flmG* (*flhA*), and *flmH* (*flaG*). The *flmABCD* gene products are homologues of polysaccharide biosynthetic genes. The *flmEFGH* gene products appear to be similar to tryptophan monooxygenase, an O-linked acetylglucosamine transferase, and an acetyl transferase of other organisms. Based on these observations, the *flmEFGH* operon is proposed to encode a methyltransferase. Mutants of these genes produce flagellin proteins of anomalous molecular masses, indicating that they are involved in posttranslational modification of the flagellin proteins.

Regulation and Checkpoints of the Cell Cycle of *C. crescentus*

The successive stages of *C. crescentus* development all require the completion of the previous stage. The differentiation of the swarmer cell into the stalked predivisional cell must occur prior to the formation of the stalked and flagellated predivisional cell, which in turn must precede separation into the individual swarmer and stalked cells. In each successive stage, certain events must occur in order for the cycle to proceed (Figure 19-5).

The formation of polar structures (flagellum and stalk/holdfast) is connected to the cell division cycle. Initially the flagellum of the swarmer cell is lost and replaced by the stalk and holdfast. The opposite pole of this predivisional cell then yields the flagellum for what will become the swarmer cell of that developmental cycle. The next developmental cycle initiated from that swarmer cell replaces the flagellum with the stalk/holdfast at that pole. If DNA replication is inhibited, this results in the blocking of class II early flagellar biosynthetic gene transcription. Mutants able to replicate their DNA but blocked in some later stage of cell division exhibit defects in their progression through the developmental cycle. The initiation of cell division is especially key to beginning the programmed differentiation of the two daughter cells in each cell cycle.

As mentioned, the *pleC* gene product encodes a histidine-protein kinase important in the regulation of flagellar rotation and stalk formation. An additional phenotype of *pleC* mutants is that they are blocked at early stages of cell division and the developmental cycle. In this role, PleC is believed to phosphorylate the response regulator homologue, the *divK* gene product, DivK. Thus, PleC and DivK are proposed to make up a signal transduction system that couples flagellum rotation activation and stalk formation to cell division.

As in *E. coli* and other bacteria, the control of cell division is a very complex and highly regulated process requiring the functions of several proteins. In many ways, cell division in *C. crescentus* is very similar to the process of cell division described for *E. coli* in Chapter 17. However, in *C. crescentus* it must occur in such a way that it yields two dissimilar daughter cells.

The accumulation and location of the tubulin-like GTPase FtsZ is key to the initiation of cell division. The structure and function of FtsZ is highly conserved among bacteria. As it accumulates, FtsZ, polymerizes, forming a ring structure at the future site of cell division, which is associated with the inner membrane. It is the first protein to localize to the site of cell division and is necessary for the localization of other cell division proteins at this site. As a result of its critical importance, FtsZ is tightly controlled with the cell cycle. FtsZ is only found in the stalked cell following cell division; it is absent in the swarmer cell. As the swarmer cell differentiates, FtsZ begins to accumulate at about the same time as DNA replication initiates. It increases rapidly and peaks around the time that cell constrictions can be seen in the predivisional cell. At this point, FtsZ levels rapidly decrease as a result of proteolytic and transcriptional control.

Unlike in *E. coli*, *ftsZ* transcription in *C. crescentus* is uncoupled from *ftsQ* and *ftsA* transcription due to the presence of a strong transcriptional terminator between *ftsA* and *ftsZ*. The *ftsZ* gene is transcribed from a single promoter just downstream from the terminator. The *ftsZ* gene transcription is turned off in swarmer cells but increased following the initiation of DNA replication during swarmer cell differentiation. Following DNA replication and the initiation of cell division, *ftsZ* transcription declines. After cell division, it increases again in the stalked cell. The CtrA global regulator controls FtsZ levels during the cell cycle by repressing *ftsZ* transcription.

Thus, CtrA expression during the cell cycle is opposite that of FtsZ; it is high in the swarmer cell, declines (due to proteolysis) at the time of DNA replication during swarmer cell differentiation, and then reappears as cell division begins. CtrA also negatively regulates DNA replication by binding to the origin of replication. Thus, the control of CtrA levels coordinates the initiation of DNA replication and cell division during the cell cycle.

In addition to cell cycle transcriptional control, FtsZ also undergoes cell cycle-dependent proteolytic control. The rapid decline in FtsZ levels after the beginning of cell division can only result from the degradation of FtsZ. Experiments with *ftsZ* constitutive mutants indicate that the variation in and localization of FtsZ levels during the cell cycle is primarily controlled by differential FtsZ degradation. The half-life of FtsZ is about 80 minutes during DNA replication to the beginning of cell division, at which point the half-life decreases to about 10 to 20 minutes. This change in stability may be linked to its polymerization into the FtsZ ring at the site of cell division.

Interestingly, in *C. crescentus*, transcription of the *ftsQA* operon occurs for the most part at the same time that *ftsZ* transcription is repressed. This is because in contrast to its role as a negative regulator of *ftsZ*, CtrA is a positive regulator of *ftsQA* transcription. Thus, most of *ftsQA* transcription occurs later than *ftsZ* transcription, peaking at the end of DNA replication and the beginning of the cell division process. Evidence indicates that FtsA is necessary for the final stages of cytokinesis in *C. crescentus*. The sequential transcription of *ftsZ* and *ftsA* may help to coordinate the end of DNA replication with the initiation of cytokinesis. It should be noted, however, that additional regulators besides CtrA play a role in controlling *ftsZ* and *ftsQA* transcription during the cell cycle.

CtrA controls flagellum biosynthesis, stalk biosynthesis, DNA replication, and cell division. CtrA also regulates DNA methylation. It is therefore not surprising that CtrA, itself, is meticulously controlled throughout the cell cycle. It is controlled at the level of stability, exhibiting increases in degradation at key points. The protease involved in CtrA degradation is the ClpXP homologue of *C. crescentus*. However, ClpXP is present throughout the cell cycle, suggesting that another protein/factor is required to promote or prevent CtrA degradation by ClpXP.

In addition to control by degradation, CtrA also exhibits regulation by phosphorylation. The phosphorylation of CtrA activates it. The signals that control the phosphorylation of CtrA with the cell cycle are not known. Several histidine-protein kinases have been proposed to play a role in CtrA control. One is encoded by the *cckA* gene. The CckA kinase is a membrane-bound enzyme that is expressed throughout the cell cycle but exhibits polar localization early in the predivisional cell. Mutants of this gene exhibit phenotypes similar to *ctrA* mutants. Two other kinases have been implicated in CtrA phosphorylation: *divJ* and *divL* gene products. Both these enzymes have been found to phosphorylate CtrA in vitro. Their role in CtrA regulation in vivo is not as clear-cut. For example, DivJ preferentially phosphorylates DivK in the presence of CtrA. However, both *divJ* and *divL* mutants exhibit cell division defects, indicating some role in cell division.

Critical to the cell division process is the delivery of the bacterial chromosome to each of the daughter cells. In *C. crescentus* a single round chromosomal replication occurs per cell cycle; during cell division a copy of the chromosome is delivered to each daughter cell by chromosomal segregation. For this bacterium, chromosomal segregation is proposed to involve homologues of the **ParA** and **ParB** proteins. The ParB protein is known to bind to DNA sequences located within 80 kb of the *oriC*

region, downstream of the *parAB* operon. ParA and ParB localize at the poles of the late predivisional cell, and this localization is dependent on DNA replication, at least for ParB. Based on these findings and results from overexpression and mutants, it is hypothesized that ParA and ParB function in chromosomal partitioning or segregation during cell division.

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CHAPTER 20

HOST–PARASITE INTERACTIONS

For many microbes, we must consider growth and physiology from the point of view of host–parasite interactions as well as natural microcosms. Indeed, for many microbes growth upon or within plant or animal hosts is a significant portion of their life cycle. Not surprisingly, many of the physiologic processes described in previous chapters have been adapted to growth within the host microenvironments. However, many novel mechanisms have also evolved that are unique to the host–parasite relationship. Akin to growth under nonhost conditions, a major goal of the microbe is to obtain nutrients to support its growth. This chapter highlights several physiologic mechanisms and strategies employed by microbes during their interactions with host organisms. Unfortunately for the host, many of the strategies that pathogens employ to obtain nutrients and avoid host defenses result in damage to host cells and tissues, leading to a disease state.

OVERVIEW OF HOST–PARASITE RELATIONSHIPS

The first step in host–parasite interactions is the encounter between the microbe and the host. This can occur through a variety of methods ranging from direct contact with an infected animal or contaminated fomite, inhalation of contaminated aerosolized droplets or particles, ingestion of a contaminated food or beverage, intimate sexual contact, arthropod vectors, or contaminated needles and syringes. Following contact with the microbe, one of two possible situations develops: the microbe will be lost/shed without any significant interaction with the host or the microbe may establish a relationship with the host. This relationship can be a coexistence that is harmless to the microbe and host (**commensalism**), a mutual benefit to both the host and the microbe (**symbiosis**), or a benefit to the microbe at the expense of the host (**parasitism**). There are other types of relationships that can be established, but we will not consider them at this time. The microbe (**parasite**) in commensalism is referred to as a **commensal** while the benefiting

parasite in parasitism is referred to as a **pathogen**. The **host** in or on which the parasite lives (plants, animals, or humans) is unharmed/benefits in commensalism/symbiosis while in parasitism the host is harmed or damaged in some manner. In either scenario, the microbe must establish a relationship with the host or infect the host.

Infection of the host involves the production of structures and/or products that allow the microbe to adhere to and colonize a host surface. These include structures and products described in Chapter 7 such as **pili** or **fimbriae**. In parasitism, damage to the host may result from pathogen-encoded structures or products and/or the host's response to the infection. This damage results in the production of a set of signs and symptoms (e.g., inflammation, fever, sneezing, coughing, aches and pains) characteristic of a **disease state**.

Commensal organisms typically inhabit or infect host body surfaces, and by the nature of their relationship they have evolved so as to occupy and effectively compete within a specific host niche without causing damage to the host under normal circumstances. As such, commensal organisms have become an important asset to the host by supplying important nutrients, breaking down material making nutrients available to the host, and/or occupying space, making it difficult for potential pathogenic microbes to establish themselves within the host. This population of commensal microbes can produce a biofilm that covers many of the host surfaces and is generally referred to as **normal** or **resident flora** for the host.

What makes a commensal organism different from a pathogen under normal conditions? It may be related to the recent finding that commensal organisms can actively block the host's immune response (e.g., inflammation) to their presence. Commensals are proposed to block the degradation of I κ B, which normally inhibits the function of NF- κ B. NF- κ B is a key transcription factor in host cells; in particular, it activates the transcription of tumor necrosis factor (TNF) and interleukin (IL)-8, which in turn promote an inflammatory response. Thus, commensals are proposed to actively suppress the host inflammatory response.

For a pathogen, the next step after colonization is the entry of the microbe or a microbial product(s) across an epidermal or epithelial cell surface to enter deeper well-vascularized (both lymph and blood circulation) tissue. With the exception of microbes transmitted by blood-sucking arthropod vectors, or contaminated needles or syringes, entry into the host is for many microbes the difference between life as a commensal organism, confined to the host's body surfaces, and life as a pathogen, able to infect and elicit damage to host tissue.

Entry into the host may or may not involve the physical entry of the microbe into deeper tissue but instead may be limited to the entry of a microbe-derived product such as an exotoxin. The physical entry of the microbe involves microbial products, which mediate the direct invasion of epithelial cells or specialized epithelial cells called M cells, such as invasins. Microbial exotoxins and factors mediating host cell entry are discussed later in this chapter.

The next phases of host–pathogen interactions generally involve multiplication to increase cell number and/or the level of one or more cell products along with local and/or systemic spread/dissemination. In order to multiply and disseminate, the pathogen must be able to acquire nutrients to support its growth. However, the host, and the normal microbial flora, goes to great lengths to limit the nutrients available to incoming pathogens in order to limit their growth and spread. Thus, many pathogens have evolved mechanisms that allow them to successfully compete with the host for

nutrients. For example, many bacterial pathogens possess systems designed to bind and take up iron from the host environments. But since the host and normal flora also compete actively for available iron, some pathogens have evolved ways to release iron from the host cells. These include the production of exotoxins known as **hemolysins**, which lyse red blood cells, releasing the iron-rich hemoglobin. This is a good example of the strategies that some pathogens employ to obtain nutrients.

The pathogen's goal is to acquire nutrients to support its growth, but the consequence is often damage to the host. This damage and/or that caused by the immune response leads to the development of the signs and symptoms associated with infectious disease.

STRUCTURES AND FUNCTIONS INVOLVED IN HOST-PARASITE INTERACTIONS

Adherence/Colonization

The first step in host-parasite interactions is adherence to host surfaces. This is necessary for the colonization of the host and the ability to cause disease. Because it is a critical step in host-parasite interactions, microbial pathogens have evolved a variety of structures and mechanisms to promote their adherence to host cells. Many express multiple adherence factors that allow them to colonize several hosts and/or a variety of tissues. Adherence factors generally fall into one of two broad groups: **fimbriae** or **pili** and **afimbrial adhesins**.

Fimbriae (Pili). Some of the best-characterized adherence structures are the fimbriae or pili of Gram-negative bacteria such as *E. coli* and *Salmonella enterica*. As described in Chapter 7, fimbriae (meaning threads or fibers) or pili (meaning hair) are terms used to describe similar proteinaceous adherence appendages. Recently, researchers have proposed that the term *fimbriae* be used to describe structures involved in bacterial-host cell adherence and that the term *pili* be used to describe structures involved in bacteria-to-bacteria adherence such as sex pili involved in conjugation. However, since no consensus has been reached on this nomenclature, the two terms are used interchangeably in this discussion.

A complex multiprotein apparatus carries out pilus biogenesis. A common scheme for pilus formation (except type IV pili; see below) involves a **chaperone/usher pathway**. In this scheme, pilus subunits are exported to the periplasm where they interact with **chaperone** proteins that stabilize the pilus subunits, preventing them from prematurely interacting with each other during their time in the periplasm. The pilus subunit is then transferred to the **usher** protein complex, which forms an oligomeric channel in the outer membrane and translocates the subunits across the outer membrane. In this way, the subunits are assembled to form a relatively long thread-like structure possessing a **tip protein** complex that is adhesive and directly interacts with the target receptor. The pilus is anchored to the outer membrane by another protein complex. Type IV pili are formed using a pathway similar to type II secretion systems described below and in Chapter 2.

One classification format groups pili into types, I, II, III, and IV. However, not all fimbriae will fit into these groupings. **Type I pili** were first identified for their ability to agglutinate red blood cells. Interestingly, this binding is blocked by the addition of mannose, so it is referred to as **mannose-sensitive hemagglutination**.

(MSHA). Mannose-sensitive type I pili are produced by a large variety of Gram-negative bacteria—for example, *E. coli*, *Salmonella enterica*, *Klebsiella* spp., and *Vibrio* spp. It was later found that some type I–like pili are not inhibited from binding by mannose and therefore mediate **mannose-resistant hemagglutination (MRHA)**. Type I pili, however, are synthesized through the chaperone/usher pathway.

Type II pili are thought to be mutants of type I pili, since they are similar to type I pili but are nonadhesive. **Type III pili** mediate MRHA but only when red blood cells are pretreated with tannic acid. **Type IV pili** were originally described for their ability to mediate MRHA and as being immunologically different from type I pili. However, it was later shown that they are assembled through a different pathway than other pili.

The classification based on ability to mediate MSHA or MRHA is becoming outdated in light of the ability to characterize pili at the molecular level. For example, many of the pili genes have been sequenced and therefore they can be compared/aligned to one another and grouped based on similarities at the DNA or deduced amino acid sequence level. More recently, pili have been named based on specific characteristics owned by them—for example, pyelonephritis-associated pili (Pap or P pili), bundle-forming pili (BFP), or toxin coregulated pili (TCP). Both BFP and TCP are also type IV pili, while P pili are type I–like (see below).

The **Pap** or **P pili** of **uropathogenic *E. coli* (UPEC)** are among the best-studied pili, particularly in terms of their biogenesis. Thus, they have become a paradigm for the process of pilus formation. P pili are essential adhesin factors for UPEC strains to colonize the urinary tract and cause pyelonephritis. P pili are similar to type I pili in terms of gene organization and analogous sequences, but type I pili are flexible structures whereas P pili are rigid rod-like structures. The P pili are encoded for by the *pap* operon. The major subunit of P pili is encoded by the *papA* gene. The initial PapA subunit is anchored into the outer membrane by the PapH protein.

At the tip of the pilus is the tip fibrillum, PapE protein, and the tip adhesin, PapG. The PapD protein acts as a chaperonin required for transport of the pilus subunits from the inner membrane to the outer membrane, where they are accepted by the PapC protein, which is proposed to function as an usher protein. Two additional proteins, PapF and PapK, play a role in tip fibrillum synthesis. The P pili attach to specific host cell receptors in the upper urinary tract via the tip adhesin protein PapG. PapG binds to α -D-galactopyranosyl (1-4)- β -D-galactopyranoside [Gal α -(1-4)-Gal] moieties of glycolipids on cells lining the upper urinary tract.

Afimbrial Adhesins. A pathogen of the respiratory mucosa *Bordetella pertussis*, the causative agent of whooping cough in children, expresses a number of adhesin factors including pili and several afimbrial adhesins. All these adhesin factors contribute to the difficult task of adherence to the ciliated respiratory epithelium and colonization of the upper respiratory mucosa. Examples of the latter include **filamentous hemagglutination (FHA)**, **pertactin**, **pertussis toxin** (see below), and the **BrkA protein**. FHA is a high-molecular-weight protein sharing homology with the high-molecular-weight (HMW) afimbrial adhesins of *Haemophilus influenzae* and many eukaryotic cell-to-cell adhesins.

FHA possesses the **RGD (arginine-glycine-aspartate) motif** characteristic of the recognition site of adhesion molecules that bind to eukaryotic surface integrin proteins. FHA binds to receptors on the ciliated respiratory epithelium and the leukocyte integrin CR3; the latter triggers bacterial uptake into macrophages without activating NADPH

oxidase and the respiratory burst. Furthermore, the S2 and S3 subunits of pertussis toxin share similarities with eukaryotic selectin proteins—that is, cell adhesion molecules present on leukocytes (L-selectins) and endothelium (E- and P-selectins) that bind to mucin-like cell adhesion molecules (CAMs). Both pertactin and BrkA also possess RGD motifs and appear to be involved in host cell adherence. Thus, these four adhesins provide good examples of how bacteria have evolved adherence structures that may mimic host adherence molecules to promote their interactions with host cells.

Another example of an afimbrial adhesin molecule is the **YadA protein** of *Yersinia enterocolitica*, which binds to cell-associated fibronectin. Similarly, *Streptococcus pyogenes* also produces at least two afimbrial adhesin structures that bind to cellular fibronectin: **Protein F** and **M protein/lipoteichoic acid (LTA) complexes**. The LTA of *Staphylococcus aureus* also binds to fibronectin on cells, mediating adherence for these bacteria.

Host Cell Specificity. The specific adhesin–host receptor interactions dictate the host tissues/cells that microbes can adhere to/colonize, thereby causing disease. For example, in enteric bacteria the production of different pili or fimbriae (or more precisely the tip complex of the pili) determines the regions of the intestinal tract that the pathogen will adhere to and colonize.

The extreme acidic pH achieved in the stomach normally kills most ingested bacteria. However, *Helicobacter pylori* is able to colonize the stomach. It binds preferentially to the gastric mucosa by binding to the Lewis^b blood group antigen expressed on gastric mucosal epithelium; however, the adhesin has not been definitively identified. This close interaction with the gastric mucosal epithelium is beneficial, since it is believed that the immediate environment around the gastric mucosa is neutralized to prevent damage to the epithelial cells.

Passage out of the stomach into the intestine provides several new challenges for the pathogen. Strains of *Vibrio cholerae*, enteropathogenic *E. coli* (EPEC), and enterotoxigenic *E. coli* (ETEC) adhere to and colonize the mucosal epithelium of the duodenum and proximal jejunum. EPEC strains produce damage to the jejunal epithelium (**attaching and effacing or A/E lesion**) by forming microcolonies referred to as localized adherence (LA). LA requires the production of **bundle-forming pili (BFP)** encoded by genes located on the EAF plasmid. In addition, the adherence of *V. cholerae* to epithelium of the proximal small intestine requires the chromosomally encoded **toxin coregulated pilus (TCP)**. Interestingly, the colonization of ETEC strains of the proximal small intestine in humans requires plasmid-encoded fimbriae referred to as **colonization factor antigens (CFAs)**; and, in piglets and calves, pili called **K88 pili**. This is a good example of pathogenic microbes expressing different adhesin factors to colonize different hosts.

Progression into the distal jejunum and ileum also requires the expression of certain adhesin factors for colonization by pathogens. *Salmonella enterica* serovar Typhimurium adherence to epithelial cells in the distal small intestine involves at least two pili systems: **long polar fimbriae (LPF)** and **plasmid-encoded fimbriae (PEF)**. LPF appear to bind to specialized epithelial cells, associated with Peyer's patches, called M cells, and PEF appear to bind to receptors on intestinal epithelium. LPF appear to be necessary for adherence associated with *Salmonella* invasion while PEF appear to be involved in colonization of the mucosal surface without invasion. Thus, a microbe can express different adhesin factors in the same host that result in different types of host–parasite interactions.

Colonization of the large intestine or colon requires that the incoming pathogen bind to colonic epithelium while competing with the normal resident flora for available space. Strains of enterohemorrhagic *E. coli* (EHEC) colonize the large intestine, causing A/E lesions via fimbriae encoded on a 60 MDa plasmid. Strains of enteroaggregative *E. coli* (EAEC) adhere to colonic mucosa in a distinct arrangement called aggregative adherence mediated by pili known as **aggregative adherence fimbriae (AAF)**.

Similar roles for afimbrial adhesins in binding to different host cells/tissues have been observed. FHA, pertactin, pertussis toxin, and BrkA of *B. pertussis* function in binding to receptors on ciliated respiratory epithelium, allowing this bacterium to cause disease in the upper respiratory tract; and, at least, FHA binds to the $\alpha\beta_2$ integrin CR3 on macrophages, which is believed to stimulate uptake of the bacterium into the macrophage without inducing the respiratory burst. This allows *B. pertussis* to evade an important host antimicrobial mechanism. Furthermore, strains of *S. pyogenes* produce Protein F and M protein/LTA that bind to cellular fibronectin, but protein F appears to be critical to the colonization of the nasopharyngeal epithelium, whereas M protein is key to binding to cells in the skin, suggesting additional factors may be involved.

Virulence Factor Secretion Systems

The majority of virulence factors are either surface-associated or secreted factors. Thus, systems used to export these factors are critical to the pathogenesis of many microbes. Interestingly, many of the secretion systems are conserved and highly homologous between species. In addition, many unrelated virulence factors can share basic transport systems. Furthermore, protein secretion in Gram-negative bacteria is more complex than that of Gram-positive bacteria due to the presence of an outer membrane.

Gram-Negative Bacterial Secretion Systems. To address some of these complexities, Gram-negative bacteria have evolved a number of different secretion systems designated types I through IV as well as other systems (described in detail in Chapter 2).

Type I secretion systems represent a family of structurally and functionally related protein complexes involved in the export of proteins, lacking the classic N-terminal signal sequence, through the inner and outer membrane to the exterior without a periplasmic intermediate. Thus, these are *sec*-independent pathways. Type I secretion pathways utilize three components to accomplish their task: an inner membrane ABC transporter protein (provides the energy for protein export), an outer membrane protein, and a periplasmic spanning protein that is anchored into the inner membrane and associated with the ABC transporter component.

The substrates for these secretion systems, although lacking an N-terminal signal peptide, possess a signal sequence within the C-terminal 60 amino acids. The prototype type I secretion system is involved in *E. coli* α -hemolysin secretion. However, several other virulence factors have been shown to exhibit type I secretion including: invasive adenyl cyclase (CyaA) of *Bordetella pertussis*, leukotoxins (LktA) of *Pasteurella haemolytica*, alkaline protease (AprA) of *Pseudomonas aeruginosa*, LipA lipase and PtrA protease of *Serratia marcescens*, and PtrBC protease of *Erwinia chrysanthemi*.

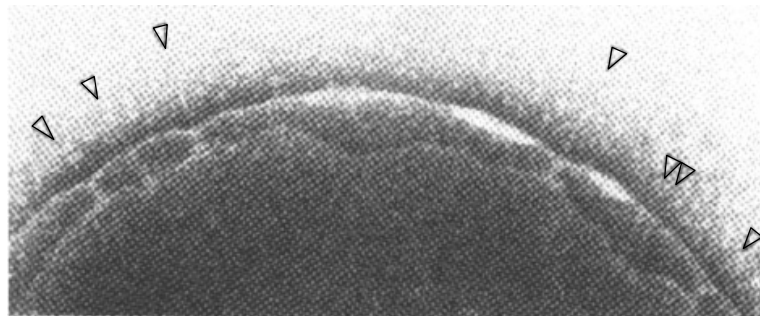
Type II secretion systems are widely distributed among Gram-negative bacteria. This type of secretion involves two genetically and biochemically distinct translocation

steps. The first step is mediated by the Sec pathway and requires an N-terminal signal sequence peptide. This step translocates the protein from the cytoplasm to the periplasm, where the signal peptide is cleaved and the protein folds into its secondary shape and perhaps tertiary assembly prior to being translocated across the outer membrane by the type II secretion system. Type II secretion systems are composed of at least 12 different gene products. Many plant and animal pathogens use this type of secretion for the export of exotoxins and hydrolytic enzymes important for their pathogenesis. The best-studied type II secretion system is the Pul system of *Klebsiella oxytoca* involved in pullulanase secretion; others include the Xcp system of *Pseudomonas aeruginosa*, which secretes elastase, exotoxin A, and phospholipase C; the Esp pathway involved in cholera toxin secretion in *Vibrio cholerae*; the Out system, which secretes pectic enzymes and cellulases in *Erwinia chrysanthemi*; and the Exe pathway implicated in amylase and protease secretion in *Aeromonas hydrophila*.

Type III secretion systems (TTSS) have relatively recently been described as major routes for export of virulence factors in human and animal pathogens. Export via TTSS occurs independently of the Sec system. However, many of the components of the TTSS apparatus appear to require the Sec pathway for their export. The TTSS apparatus is a multiprotein complex that spans the inner-to-outer membrane and extends from the surface forming a needle-like structure (Fig. 20-1).

Unlike other secretion systems, which can export proteins that are active in the extracellular milieu, TTSS appear to have specifically evolved for the direct translocation of proteins from the bacterial cytoplasm into the host cell cytoplasm. Thus, secretion via TTSS is believed to be regulated by bacterial adherence to the surface of target cells. Over the past few years, several plant, animal, and human pathogens have been shown to encode TTSS apparatus. These TTSS and the proteins they secrete, for the most part, are encoded on specific regions of the bacteria's chromosome that have apparently been acquired via horizontal transfer (from other genera/species rather than parental ancestors).

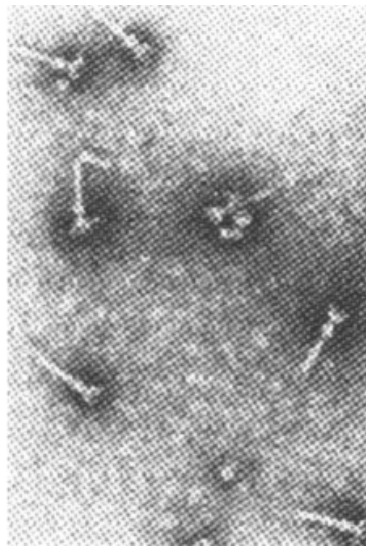
The chromosomal loci that encode the TTSS and its secreted proteins are referred to as **pathogenicity islands**. Examples of pathogens (Table 20-1) that produce TTSS include (1) *Yersinia* species (*Y. pestis*, *Y. enterocolitica*, and *Y. pseudotuberculosis*), which secrete Yops proteins that inhibit host phagocyte activity among other functions; (2) *Salmonella enterica* serovars, which encode at least two different TTSS on two separate *Salmonella* pathogenicity islands (SPI-1 and SPI-2) and secrete proteins that promote invasion of nonphagocytic cells within the small intestine (SPI-1) and systemic infection and disease (SPI-2); (3) enteropathogenic (EPEC) and enterohemorrhagic (EHEC) *Escherichia coli*, rabbit-diarrheagenic *E. coli* (RDEC-1), and the rodent pathogen *Citrobacter rodentium*, which produce related TTSS apparatus encoded on a pathogenicity island referred to as the **locus of enterocyte effacement (LEE)** and secrete proteins required for the generation of the attaching and effacement (A/E) lesion; (4) *Shigella* species, which secrete Ipa proteins required for invasion of nonphagocytic cells and induction of apoptosis in host cells; (5) *Pseudomonas aeruginosa*, which secretes various exoenzymes, such as exoenzyme S, needed for hematogenous dissemination and inhibition of host T cell functions; and (6) the plant pathogens *Pseudomonas syringae*, *P. solanacearum*, *Xanthomonas campestris*, and *Erwinia chrysanthemi*, which secrete proteins called harpins that are involved in generating plant tissue damage.



(a)



(b)



(c)



(d)

Fig. 20-1. Electron micrographs demonstrating the Type III secretion apparatus of *Salmonella enterica* serovar Typhimurium. (a) and (b) EM of osmotically shocked nonflagellated *S. Typhimurium* showing needle complexes spanning the inner-to-outer membranes and extending from the surface (open arrows). (c) and (d) EM of isolated needle complexes from an enriched fraction of a CsCl gradient. Excerpted from T. Kubori, et al. 1998. *Science* **280**:602–605, copyright (1998) American Association for the Advancement of Science.

TABLE 20-1. A Partial List of Components and Effector Proteins, and Their Functions, of Bacterial Type III Secretion Systems (TTSS)

Bacteria	Effector Protein	Function
<i>Yersinia</i> spp.	YopB/D	Pore-forming translocase
	YopK	Control of translocation
	YopN	Regulation of secretion
	YopE	Rho GTPase-activating protein causing F-actin filament disruption
	YopT	Causes F-actin filament disruption
	YopH	Protein tyrosine phosphatases that blocks phagocytosis and respiratory burst
	YpoJ/P	Bind to mitogen-activated protein kinase kinase (MAPKK) and I κ B kinase β , preventing cytokine synthesis and promoting host cell apoptosis
	YopM	Binds to host thrombin
	YpkA	Ser/Thr protein kinase
	YscM	Gene regulation
	Tir	Translocated intimin receptor needed for intimin-mediated adherence
Enteropathogenic <i>Escherichia coli</i> (EPEC)	EspA	Activation of epithelial cell signal transduction
	EspB/D	Tyrosine phosphorylation of Tir; cell signaling
<i>Salmonella enterica</i> (SPI-1)	InvJ	Part of TTSS apparatus
	SspA ^a	Binds to F-actin, stabilizing F-actin
	SspB/C/D ^a	Effector protein translocation; binds to IL-1 converting enzyme (ICE or caspase I), promoting apoptosis
	SopB	Inositol phosphate phosphatase that promotes fluid secretion and inflammation
	SopE	Guanine-nucleotide exchange factor (GEF) and activates Rho GTPases, promotes membrane ruffling
	SptP	Tyrosine protein phosphatase; GTPase, activating protein Rac and Cdc42 (counteracts SopE effect)
	SspH1	Calf virulence
	SseB	Component of TTSS surface translocon
	SpiC	Blocks endosomal trafficking
	SspH1/2	Calf virulence
<i>Shigella flexneri</i>	IpaA	Binds to vinculin and controls cell invasion
	IpaB	Binds ICE protease to promote apoptosis
	IpaB/C	Binds to $\alpha_2\beta_1$ integrins and CD44 to promote membrane ruffling
<i>Pseudomonas syringae</i>	IpaD	Regulates secretion
	AvrB	Promotes plant hypersensitivity response (HR)
	AvrPto	Binds to Pto kinase, resulting in induction of HR
	HopP	Promotes HR
	HrpA	TTSS pilus
	HrpW/Z	Effector protein translocation (?) and HR induction

^aSspA, SspB, SspC and SspD are also known as SipA, SipB, SipC and SipD, respectively.

Type IV secretion systems are another mechanism of macromolecular export in Gram-negative bacteria. Type IV systems are a diverse family of secretion pathways that export an array of substrates from large nucleoprotein conjugation intermediates to both monomeric and polymeric protein complexes. Moreover, the cells targeted for secreted substrate delivery are also diverse, ranging from bacteria to fungi to plants to animals. The three defining type IV secretion systems are the VirB/D4 transfer system of *Agrobacterium tumefaciens* that exports T-DNA into susceptible plant cells, leading to plant tumors; the conjugal transfer system of conjugative IncN plasmid pKM101; and the pertussis toxin (see below) exporter Ptl of *Bordetella pertussis*. Additional type IV secretion systems identified to play a role in virulence include the Cag system of *Helicobacter pylori* required for export of the CagA protein, which causes rearrangements of the host cell cytoskeleton, and the Dot/Icm system of *Legionella pneumophila* which secretes an unidentified effector, allowing survival and growth in macrophages possibly by blocking phagosome-lysosome fusion. Further systems have been proposed to be encoded by *Brucella*, *Bartonella*, and *Rickettsia* spp. that are thought to aid in intracellular survival of these bacteria.

Other types of secretion systems have also been described that do not fit into the type I–IV definitions. **Autotransporters** are represented by a family of proteins that appear to mediate their own export through the outer membranes. This type of secretion utilizes the Sec pathway for export across the inner membrane, but all the information for translocation across the outer membrane is located in the protein itself. Thus, this represents an additional mechanism of export utilized by Gram-negative bacteria. The paradigm for this form of secretion is the IgA protease of *Neisseria gonorrhoeae*. Other members of this family include IgA protease, putative adherence and invasion proteins of *Haemophilus influenzae*; BrkA, pertactin, and other outer membrane proteins in *Bordetella pertussis*; a serine protease of *Serratia marcescens*; the vacuolating cytotoxin of *Helicobacter pylori*; SepA protein secreted by *Shigella flexneri*; and EspC protein secreted by EPEC strains.

An additional secretion mechanism has recently been described in Gram-negative bacteria. The **two-partner secretion (TPS) pathways** are composed of a secretion domain-possessing exoprotein (protein being transported) of the TpsA family and a channel-forming outer membrane transporter protein of the TpsB family. Interestingly, each TpsB transporter appears to be specific for the export of its cognate exoprotein and tends to be encoded by genes organized into an operon. Based on analysis of genome sequences, TPS pathways appear to be widespread among Gram-negative bacteria. Three TPS systems have been relatively well characterized: secretion of the filamentous hemagglutinin (FHA) of *Bordetella pertussis*, the ShlA Ca^{+2} -independent cytolysin (hemolysin) of *Serratia marcescens*, and the high-molecular-weight adhesins (HMW1 and HMW2) of *Haemophilus influenzae*. Other less-well-characterized TPS systems include systems for the secretion of the cytolysins of *Proteus mirabilis*, *Edwardsiella tarda*, and *Haemophilus ducreyi*.

Gram-Positive Bacterial Secretion Systems. Secretion of proteins in Gram-positive bacteria is generally simpler than in Gram-negative bacteria, primarily because of the absence of a second outer membrane. Surface proteins or secreted proteins all possess a classic signal sequence peptide. In addition, such proteins have a region of tandem repeats and a C-terminal region rich in proline and glycine residues and a conserved hexapeptide (LPXTGE). Examples of surface-associated or secreted proteins

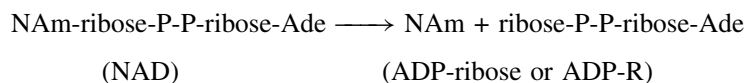
in Gram-positive bacteria include F, M, and G proteins of *Streptococcus pyogenes*, the P1 adhesin of *Streptococcus mutans*, internalin of *Listeria monocytogenes*, and the fibronectin-binding protein and protein A of *Staphylococcus aureus*.

Exotoxins

From the point of view of the microbial pathogen, its host is like a warehouse of food for which you need membership to gain access. The host and its normal resident flora are established and geared to efficiently obtain nutrients. The incoming pathogen must either outcompete the host/resident flora or develop ways to increase the availability of nutrients to support its growth. Thus, many pathogens either secrete, or put on their surface, enzymes that can degrade host constituents and/or lyse cells to release nutrients. For example, pathogens may secrete proteases/peptidases that degrade host proteins (e.g., elastase, collagenase, IgA protease) to smaller peptides or amino acids or secrete lipases that degrade lipids or phospholipids to their fatty acid and sugar moieties. The latter may result in cell lysis, since cell membranes are rich in (phospho)lipids. Therefore, the bacterium attempting to obtain nutrients damages the host through the breakdown of its constituents. It is not surprising then that many pathogens secrete one or more exotoxins, and these exotoxins play a key role in pathogenesis. Although a variety of microbes produce exotoxins, there are relatively few general structural or mechanistic schemes that have evolved for microbial exotoxins.

A-B Toxins. Many exotoxins, although they may differ in their activity or target cell affected, exhibit a common structural scheme. They possess one or more **binding** or **B subunits** that lack enzyme activity but bind to host receptors. These subunits are required for entry of the toxin into the target cell. An **activity** or **A subunit** possesses the enzyme activity or toxic activity that acts on the target cell. A-B toxins are synthesized by both Gram-negative and Gram-positive bacteria and target a variety of different host cell proteins and functions.

One group of A-B toxins possesses a common enzyme activity but differs in the protein or function it targets. This group utilizes the redox cofactor nicotinamide adenine dinucleotide (NAD) as a substrate, cleaving the bond between the nicotinamide (NAm) group and the ribose molecule to yield NAm and **adenosine diphosphate ribose (ADP-ribose or ADP-R)**.



The ADP-R is then transferred to a target molecule (e.g., host protein), altering its structure and/or function and resulting in some physiologic effect. This activity is referred to ADP ribosylation activity, and the toxins possessing this activity are known as **ADP ribosyltransferases** or **ADP-ribosylating toxins**.

Examples of ADP-ribosylating toxins are diverse in terms of the bacteria that produce them and the protein targeted. The classic example of an ADP ribosylating toxin is the **diphtheria toxin (DT)** produced by *Corynebacterium diphtheriae*. DT and the identically acting **exotoxin A (ExoA)**, from *Pseudomonas aeruginosa*, both ADP-ribosylate the eukaryotic elongation factor (EF)-2, inhibiting protein synthesis and ultimately leading to cell death (Fig. 20-2). DT binds to the heparin-binding epidermal

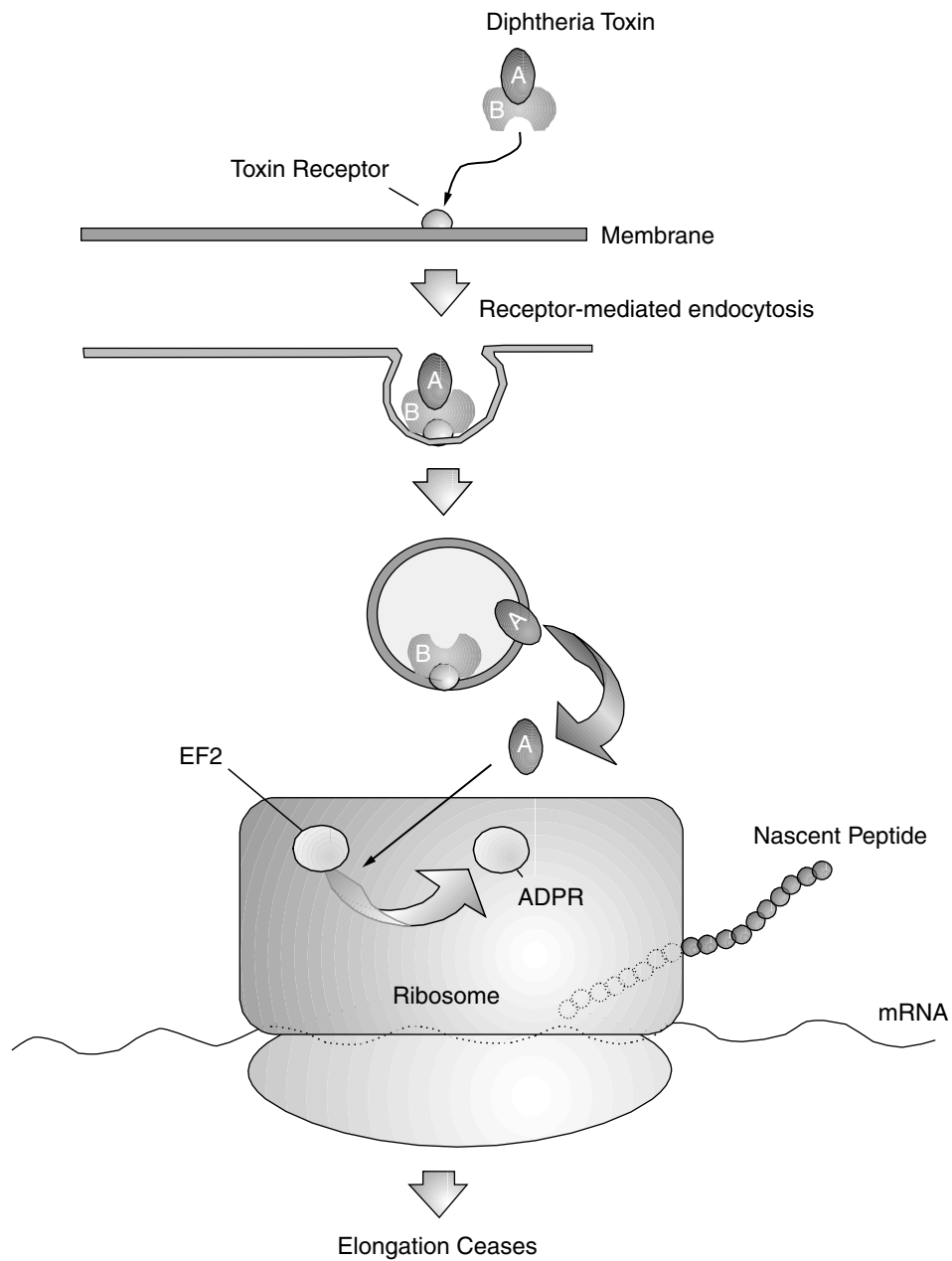


Fig. 20-2. Schematic representation of the proposed mode of action of diphtheria toxin (DTX). DTX binds to EGF receptor on host cell surface, enters the cell via receptor mediated endocytosis. The A subunit is then released from the endosome, targets the ribosome ADP-ribosylating EF-2 inactivating it. Inactivation of EF-2 stops elongation of the nascent polypeptide.

growth factor (EGF), entering the cell through an early endosome while ExoA binds to α_2 -macroglobulin, entering the cell following retrograde transport to the endoplasmic reticulum (ER).

The **cholera toxin (CTX)** of *Vibrio cholerae* and **heat-labile enterotoxin (LT)** of enterotoxigenic *E. coli* (ETEC) are both ADP-ribosylating toxins that target the $G_{s\alpha}$ regulatory protein of adenylyl cyclase (AC), resulting in increased AC activity and elevated levels of cAMP (Fig. 20-3). Elevated cAMP levels in the enterocytes affected by CTX and LT cause increased Cl^- ion secretion and inhibition of Na^+ ion absorption, ultimately resulting in watery diarrhea in the patient. CTX is encoded by a bacteriophage (CTX ϕ) while LT is plasmid-encoded. Both bind to surface gangliosides such as GM1 and enter the cell via retrograde transport to the ER, where the A subunit is activated.

Pertussis toxin (PT) of *Bordetella pertussis* is an ADP-ribosylating toxin that targets the $G_{i,o}$ regulatory protein of AC, preventing its inhibition and resulting in elevated levels of cAMP. In contrast, to CTX and LT, PT targets respiratory epithelium and results in many of the symptoms associated with whooping cough. PT is secreted via a type IV secretion system and binds to a surface receptor on respiratory epithelium via its B subunits, delivering the A subunit into the cell cytoplasm. The **C2 toxin** from *Clostridium botulinum* types C and D, **toxin A (ToxA)** and **toxin B (ToxB)** of *C. difficile*, and **iota toxin** of *C. perfringens* are ADP-ribosylating toxins that also possess binding/translocation activity. They enter the cell through endosomes and ADP-ribosylate G-actin monomers and prevent actin polymerization.

Other A-B toxins do not possess ADP-ribosylating activity but instead act through a variety of different mechanisms. The **Shiga toxin (STX)** of *Shigella* spp. and the **Shiga-like toxins (SLT)** of enterohemorrhagic *E. coli* (EHEC) strains both possess *N*-glycosidase activity that targets the 28S rRNA, damaging the ribosome, blocking protein synthesis, and eventually killing the cell. Following binding via its B subunits, the A subunit enters the cell through retrograde transport to the ER. **Invasive adenylyl cyclase toxin (ACT)** of *Bordetella pertussis* enters the cell directly, where in the cytoplasm it acts as a calmodulin-dependent adenylyl cyclase that raises cAMP levels in the cell, generally disrupting intracellular signaling pathways.

The plasmid-encoded tripartite **anthrax toxin** of *Bacillus anthracis* is composed of an edema factor (EF), lethal factor (LF), and protective antigen (PA). The PA subunit acts as the binding or B subunit for both EF and LF, which act independent of each other. The EF, like ACT, is a calmodulin-dependent adenylyl cyclase that produces similar effects on the cell. The LF is Zn^{+2} -metalloprotease, which cleaves host cell mitogen-activated protein kinase kinases (MAPKK), inhibiting cell proliferation and likely causing cell death.

The **cytotoxic distending toxins (CDT)** of *Haemophilus ducreyi*, *Shigella* spp., *Campylobacter* spp., and several *E. coli* strains are A-B toxins that indirectly inhibit $p34^{cdc2}$ dephosphorylation, arresting the cell cycle in G2 and resulting in cytoplasmic distention and cell enlargement; the CDT of *C. jejuni* possesses deoxyribonuclease I-like activity that appears to be required for its action.

The **tetanus neurotoxin (TeNT)** of *Clostridium tetani* and **botulinum neurotoxin (BoNT)** of *Clostridium botulinum* both are A-B toxins composed of a heavy chain (B subunit) and light chain (A subunit). The light chain possesses metalloprotease activity that cleave synaptosomal-associated proteins. Both TeNT and BoNT inhibit neuroexocytosis. However, TeNT acts on central inhibitory interneurons, preventing

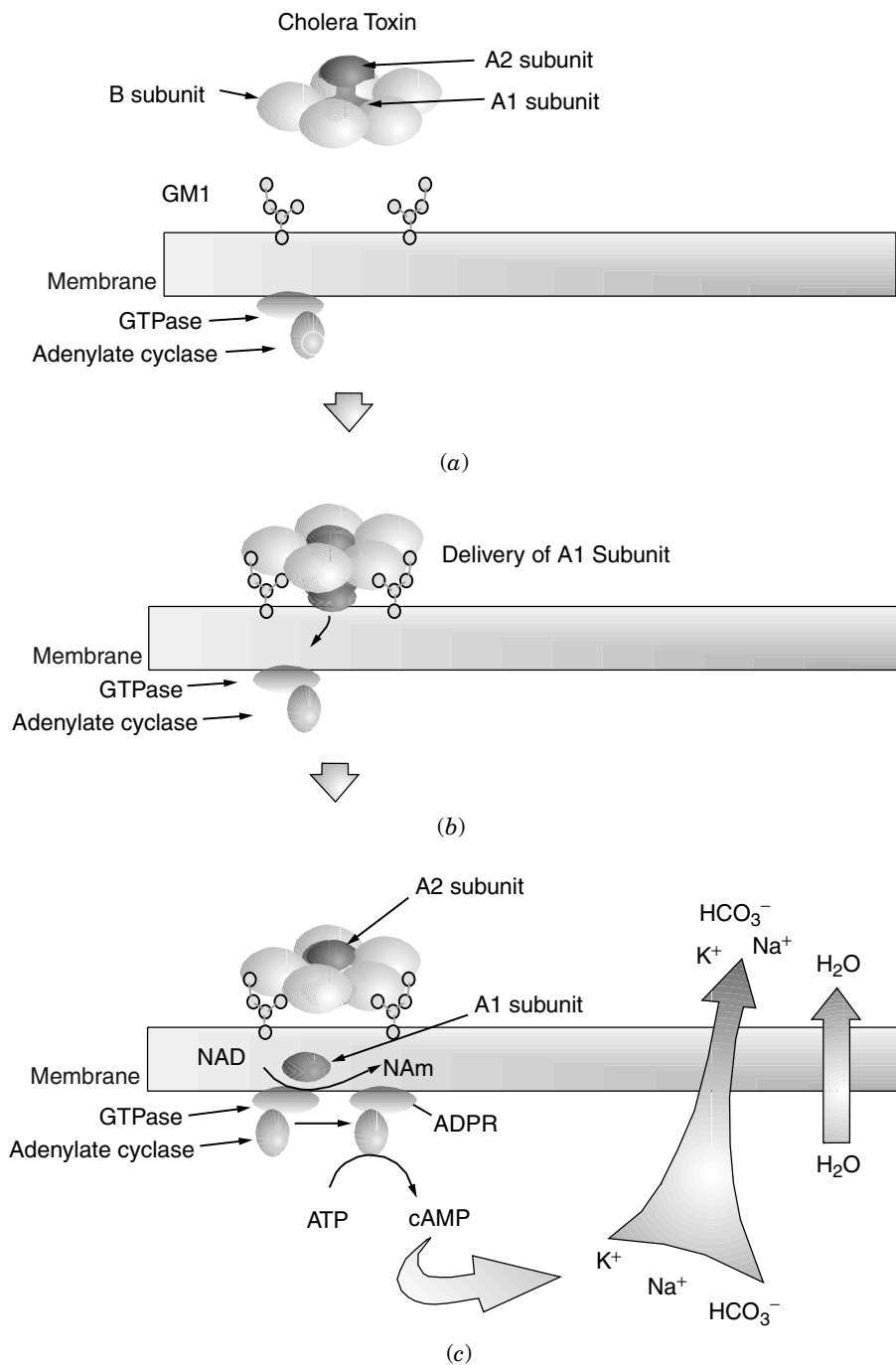


Fig. 20-3. Schematic representation of the proposed mode of action of cholera toxin (CTX). (A) CTX binds via its B subunits to surface receptor, GM1. (B) The A1 subunit is then delivered across the membrane; (C) A1 subunit ADP-ribosylates the $G_{s\alpha}$ (GTPase) subunit, locking Adenylate cyclase in the "on" position leading to elevated levels of cAMP and excretion of electrolytes and loss of water from the cell.

inhibitory neurotransmitter release, causing constant muscle stimulation and a spastic paralysis; BoNT acts on peripheral motor neurons, blocking acetylcholine (ACh) release and muscle stimulation, leading to a flaccid paralysis.

Surface-Acting Toxins. A number of bacteria secrete exotoxins that act at the surface of the target cell to cause some physiologic effect or cell lysis. Strains of both *Staphylococcus aureus* and *Streptococcus pyogenes* produce toxins that act as **superantigens**. They nonspecifically cross-bridge class II MHC molecules on antigen-presenting cells and the T cell (antigen) receptor (TCR), which results in the polyclonal activation of helper T cells and copious cytokine secretion. Superantigen toxins include TSST-1, SPE-A, SPE-C, SSA, SEA-SHE, and SpeB.

The **heat-stable enterotoxins (ST)** produced by strains of ETEC, *Yersinia enterocolitica*, *Vibrio cholerae*, and *Citrobacter freundii* bind to receptors on the cell surface to activate guanylyl cyclase activity, elevating intracellular cGMP levels and activating cGMP kinase II. The net effect is very similar to the effect of LT or CTX on enterocytes, inducing Cl^- ion secretion and blocking Na^+ ion absorption. The result in the patient is watery diarrhea.

Several bacteria produce exotoxins that form pores in the cell membranes of host cells, resulting in cell lysis. Examples of these **lysins** are **streptolysin O (SLO)** of *S. pyogenes*, which binds to cholesterol in the membrane and polymerizes to form large pores, disrupting cell membrane integrity; similarly, **perfringolysin** of *C. perfringens*, **pneumolysin** of *Streptococcus pneumoniae*, **listeriolysin O (LLO)** of *Listeria monocytogenes*, and **tetanolysin** of *C. tetani* also form large pores in the cell membrane, causing enormous electrolyte, metabolite, and protein loss with eventual cell death. Other pore-forming toxins generate smaller pores and include the **α -toxin** and **leukocidin** of *S. aureus* and **aerolysin** of *Aeromonas hydrophila*. Pore formation by these toxins leads to uncontrolled ion and solute flux and stimulates eicosanoids and endonuclease production, cytokine release, cell death, or apoptosis.

Other membrane-acting toxins include the (1) **δ -toxin** of *S. aureus* and *Staphylococcus epidermidis*, which binds nonspecifically to membranes where it self associates to form micelles and leads to osmotic cell lysis; (2) the **vacuolating cytotoxin (VacA)** of *Helicobacter pylori*, which is activated by a cellular protease and forms oligomers in the cell membrane and late endosome membrane, inhibiting endosomal trafficking and causing the formation of large vacuoles; and (3) the secreted phospholipases (PL) of several bacterial pathogens. Examples of PLs include the Zn^{+2} -dependent phospholipase C activity, possessing **α -toxin** of *C. perfringens* and **PLC B** of *L. monocytogenes*, and the phosphatidylinositol-hydrolysing toxin (PLC A) of *L. monocytogenes* that result in cell lysis/hemolysis.

“Injected” Toxins. Several of the effector proteins secreted directly into the cytoplasm of host cells by type III (described previously and in Chapter 2; Table 20-1) or type IV secretion systems alter cell physiology/function or result in cell death, and thus can be considered exotoxins. The **YpkA** and **YopO** of *Y. pseudotuberculosis* and *Y. enterocolitica* both are protein kinases, but their target proteins are unknown. In addition, **YopH** of *Y. pseudotuberculosis* is a protein tyrosine phosphatase that dephosphorylates the Crk-associated tyrosine kinase substrate Cas and inhibits its phagocytosis by host macrophages.

The **YopE** protein [a GTPase-activating protein (GAP) for Rho GTPases] and the **YopT** protein (modifies RhoA in an unknown manner, leading to its redistribution in

the cell) of *Y. pseudotuberculosis* and *Y. enterocolitica* cause disruption of the actin cytoskeleton of the host cell. **Exoenzyme S** or **ExoS** of *Pseudomonas aeruginosa* has two functional domains: the N-terminal domain acts as a GAP on Rho GTPases, causing disruption of the actin cytoskeleton, while the C-terminal domain acts to ADP-ribosylate the Ras protein. The **SopE** protein of *Salmonella enterica* sv. Typhimurium has guanine-nucleotide exchange factor (GEF) activity and activates Rho GTPases, causing actin cytoskeleton reorganization that results in membrane ruffling (Fig. 20-4). The **SptP** protein is a GAP for Rac and Cdc42 and counteracts the effects of SopE.

The **YopP** and **YopJ** proteins of *Y. pseudotuberculosis* and *Y. enterocolitica* bind to I κ B kinase β , ultimately inhibiting the eukaryotic transcription factor NF- κ B, and bind to MAPKK, affecting signaling pathways to inhibit cytokine production. Additionally, YopJ as well as **SipB/C/D** of *Salmonella enterica* and **IpaB** of *Shigella* spp. bind to interleukin-1 (IL-1)-converting enzyme (ICE or caspase I) and stimulate apoptosis of the host cell.

In *Shigella flexneri*, the **IpaA** and **VirA** proteins bind to vinculin, control cell invasion, and later control cell-to-cell spread. IpaB, in addition to stimulating apoptosis, binds to $\alpha_5\beta_1$ integrin along with **IpaC/D** to cause membrane ruffling and to promote cell division. IpaC/D also stimulates phagosomal lysis to release the bacteria into the cytoplasm.

The **SopB** protein secreted by *S. enterica* serovar Dublin and *S. flexneri* is an inositol phosphate phosphatase, which ultimately results in increased Cl⁻ ion secretion by enterocytes and diarrhea in the host. The **ExoY adenyl cyclase** of *Pseudomonas aeruginosa* is a calmodulin-independent enzyme that results in increased cAMP levels in host cells.

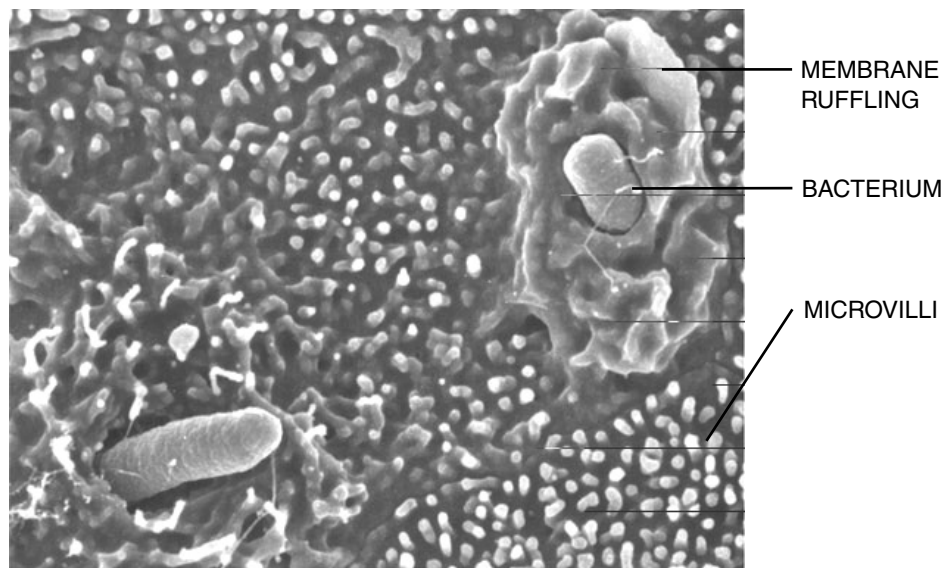


Fig. 20-4. Enhanced electron micrograph showing membrane ruffling and invasion of *Salmonella enterica* serovar Typhimurium. Taken from Galán and Zhou. 2000. *Proc. Natl. Acad. Sci. USA* 97:8754–8761. Copyright (2000) National Academy of Sciences, U.S.A.

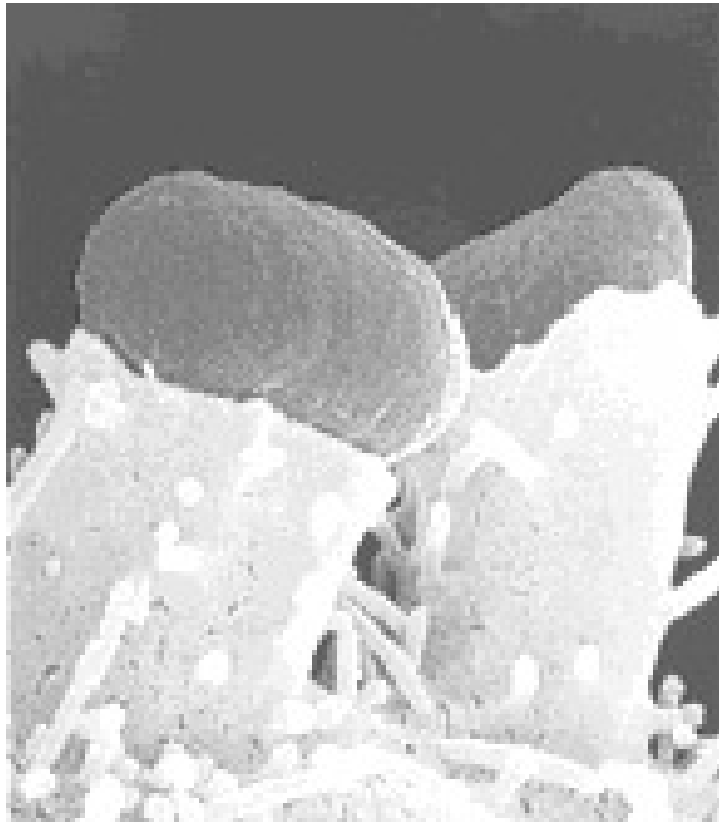


Fig. 20-5. Enhanced electron micrograph showing the formation of the “pedestal” under adhering enteropathogenic *Escherichia coli* (EPEC) cells. Taken from I. Rosenshine et al., 1996. *EMBO J* 15:2613–2624.

EPEC strains secrete **translocated intimin receptor (Tir)**, into the host cell which functions as the receptor for the **intimin** adhesin that allows bacteria to attach, actin to be nucleated, and formation of a “pedestal” under the adhering bacterium (Fig. 20-5).

The **CagA** protein of *Helicobacter pylori* is secreted by a type IV secretion system. It is tyrosine phosphorylated by cellular kinases to stimulate pedestal formation in a similar manner to Tir.

Quorum Sensing

Although bacteria are generally thought of as single-cell organisms living solitary lives, it has become increasingly clear that this is not the case. In fact, many bacteria sense and respond to the presence of, and communicate with, other bacteria in their environment. These interactions are cell density dependent and may occur between individual cells of a single species and/or with other bacterial species. This phenomenon is referred to as quorum sensing (see Chapter 5). In general, each member in the population releases a signal molecule, called an autoinducer or pheromone, which can be sensed by neighboring cells. As the number of cells in the population increases, the concentration of autoinducer/pheromone increases. Once a threshold of autoinducer

is reached, the bacteria respond by activating a transcriptional regulator protein that induces or represses a set of genes. This allows bacteria to gauge the cell density of their own population and/or other bacteria.

It is now apparent that many pathogens link quorum sensing to virulence factor secretion. This provides a mechanism for bacteria to achieve sufficient numbers in, and produce a successful infection of, the host before producing virulence factors that may elicit host defense systems. A possible rationale is that early in the infection sufficient nutrients are likely present to support growth of a small number of cells, but as cell numbers increase, nutrients become more limiting, so the bacteria produce virulence factors to cause the release of nutrients from host tissues/cells. Coincidentally, by that point, they have reached sufficient numbers to withstand the host defenses. Studies of numerous pathogens have revealed that generally the mechanisms used by Gram-negative bacteria and Gram-positive bacteria are fundamentally different, particularly in the nature of the signal molecule employed.

Gram-Negative Bacterial Pathogens. Most quorum-sensing systems identified in Gram-negative bacteria use *N*-acyl homoserine lactones (AHL) as their autoinducer (AI) molecules. However, other signaling molecules have also been described that appear to be chemically different from AHL. These molecules then bind to and activate a transcriptional regulator, or R protein, which regulates the expression of various target genes. A partial list of Gram-negative bacterial autoinducer synthases and regulators along with their cognate autoinducers are presented in Table 20-2.

The paradigm system of quorum sensing is that of the squid parasite, *Vibrio fischeri*, which colonizes the light organ of squids (see Chapter 5). In this bacterium, AI is produced by the **LuxI protein** and binds to the **LuxR protein**; the LuxR–AI complex then transcriptionally activates genes necessary for bioluminescence as well as *luxI* itself. Thus, colonization of the squid light organ resulting in high cell density leads to sufficient AI production to induce the bioluminescence genes. Many Gram-negative bacterial pathogens utilize a similar quorum-sensing scheme.

The best-characterized quorum-sensing systems related to virulence are those of *Pseudomonas aeruginosa*. *P. aeruginosa* infects a wide range of organisms from invertebrates to plants to animals. In humans, it is a frequent opportunistic pathogen of patients who (1) are immunocompromised (e.g., cancer and AIDS patients), (2) are suffering from burns, (3) have indwelling medical devices, (4) are undergoing prolonged aggressive antimicrobial therapies or (5) are suffering from cystic fibrosis (CF). Its virulence reflects an extraordinary arsenal of cell surface–associated and secreted virulence factors. The expression of many of these virulence factors is found to be cell density dependent and under the control of at least two separate quorum sensing systems: *las* and *rhl*.

Virulence genes controlled by quorum sensing include *lasA* (staphylolytic protease), *lasB* (elastase), *apr* (alkaline protease), *toxA* (exotoxin A), and *xcp* (encode a type II secretion system). The *las* system is composed of the **LasR** transcriptional activator and **LasI** autoinducer synthase, which synthesizes the cognate AI for this system. A second quorum-sensing system, *rhl*, is composed of the **RhlR** transcriptional activator and **RhlI** autoinducer synthase, which produces the cognate AI for this system.

LasR and RhlR are LuxR homologues while LasI and RhlI are LuxI homologues. As in the *lux* system, LasR–AI and RhlR–AI complexes induce the expression of their cognate AI synthases as well as other genes. In addition to several virulence

TABLE 20-2. A Partial List of Quorum-Sensing Systems and Functions in Bacterial Pathogens

Bacteria	Regulator/AI Synthase	Signal Molecule ^a	Function
<i>Vibrio fischeri</i>	LuxR/LuxI	3-Oxo-C ₆ -HSL	Bioluminescence
<i>Pseudomonas aeruginosa</i>	LasR/LasI	3-Oxo-C ₁₂ -HSL	Virulence factor production; RhlR expression; biofilm formation
	RhlR/RhlI	C ₄ -HSL	Virulence factor production; rhamnolipid synthesis; RpoS expression; secondary metabolites
<i>Burkholderia cepacia</i>	CepR/CepI	C ₈ -HSL	Virulence factor production
<i>Escherichia coli</i> (EHEC)	SdiA/?	Unknown	Cell division control; virulence factor production
<i>Salmonella enterica</i> serovar Typhimurium	SdiA/?	Unknown	Cell division control; virulence plasmid gene regulation
<i>Erwinia carotovora</i>	ExpR/ExpI	3-Oxo-C ₆ -HSL	Virulence factor production
	CarR/CarI	3-Oxo-C ₆ -HSL	Carbapenem production
<i>Agrobacterium tumefaciens</i>	TraR/TraI	3-Oxo-C ₈ -HSL	Ti plasmid conjugation
<i>Aeromonas hydrophila</i>	AhyR/AhyI	C ₄ -HSL	Exoprotease production
<i>Aeromonas salmonicida</i>	AsaR/AsaI	C ₄ -HSL	Exoprotease production
<i>Enterobacter agglomerans</i>	EagR/EagI	3-Oxo-C ₆ -HSL	Unknown
<i>Yersinia enterocolitica</i>	YenR/YenI	C ₆ -HSL	Unknown
<i>Yersinia pseudotuberculosis</i>	YesR/YesI	C ₈ -HSL	Unknown

^aHSL, homoserine lactone.

genes, RhlR–AI induces *rhlAB* (required for rhamnolipid biosynthesis), *rpoS* (the stress/stationary-phase σ factor, σ^S), and the genes for the production of pyocyanin. *P. aeruginosa* appears to produce a number of minor AHL molecules, but the functions of these are not clear. A recent report indicates that *P. aeruginosa* induces another AI molecule that is chemically different from the AHL molecules. This AI appears to be a quinolone derivative, 2-heptyl-3-hydroxy-4-quinolone, known as PQS. The genes encoding the LuxI/LuxR homologues for this AI have yet to be identified.

However, PQS production is under LasR control, and RhlR is essential for its activity. In addition to controlling virulence gene expression, a direct role for quorum sensing in *P. aeruginosa* pathogenesis has been determined in the nematode, plant, and burned mouse models. Mutants deficient in quorum sensing or virulence factors

controlled by quorum sensing are attenuated in all three models of *P. aeruginosa* virulence. Furthermore, transcripts for *lasR*, *lasA*, *lasB*, and *toxA* accumulate in *P. aeruginosa*, infecting the lungs of CF patients.

Quorum sensing has also been implicated in the related bacterium *Burkholderia cepacia*, which is an emerging pathogen in CF patients usually as a coinfecting agent with *P. aeruginosa*. *B. cepacia* possesses both a LuxR and LuxI homologue called **CepR** and **CepI**, respectively. It is not clear whether *B. cepacia* actually produces its cognate AHL in sufficient amounts to act as an AI in vivo. The same conditions that lead to the accumulation of *P. aeruginosa* AHLs result in a 1000-fold less amount of *B. cepacia* AHL. This, along with the findings that *P. aeruginosa* AHLs can increase the pathogenicity of *B. cepacia*, has led to the proposal that *B. cepacia* uses AIs synthesized by other bacteria to regulate aspects of its pathogenicity and thus profits from the energy expenditure made by other bacteria.

In this same vein, a similar scenario has been proposed for both *E. coli* and *Salmonella enterica* serovar Typhimurium. Both possess a LuxR homologue encoded by the *sdiA* gene. However, neither appears to possess a LuxI AI synthase homologue that produces a cognate AI molecule. Interestingly, they both can produce an AI molecule that induces the *Vibrio harveyi* luminescence genes, but the SdiA protein from both organisms does not respond to this AI. The SdiA protein of both organisms does respond to AI molecules from other bacterial species. Thus, it is proposed that pathogenic strains of *E. coli*, such as EHEC O157:H7 strains, and *S. Typhimurium* can sense and respond to AHL produced by other bacteria in the vicinity via SdiA, which then goes on to regulate various target genes. Among the genes regulated by SdiA in both EHEC and *Salmonella* is the *ftsQAZ* operon. Virulence genes regulated by SdiA in EHEC include the genes encoding EspD and intimin, both of which are negatively regulated by SdiA; thus, SdiA appears to regulate colonization by O157:H7 strains.

In *S. Typhimurium*, the SdiA–AI complex was determined to positively regulate four genes on the *S. Typhimurium* virulence plasmid that appear to compose an operon. These four *sdiA*-regulated genes are *srgA* (ORF8), *srgB* (ORF9), *rck*, and *srgC* (ORF11). The *srgA* gene encodes a *dsbA* homologue, suggesting that it may function as a disulfide bond isomerase involved in protein folding in the periplasm. The *srgB* gene encodes a putative secreted protein (possesses a lipoprotein signal sequence) that shows no significant homology to any gene in the databases. Rck is an outer membrane protein that confers resistance to complement killing (i.e., human serum resistance) and may also play a role in adherence to host cells. The *srgC* gene encodes a putative regulatory protein of the AraC family, although a function for SrgC has not been reported.

The plant pathogen *Erwinia carotovora* also employs quorum sensing to control virulence factors during its pathogenesis in plants. Key to this pathogenesis is the production of various plant tissue-degrading enzymes. The regulation of these exoenzymes is cell density dependent and controlled by the LuxR/LuxI homologues, **ExpR** and **ExpI**. Interestingly, while *expI* mutants are deficient in exoenzyme production, *expR* mutants do not exhibit any defects in exoenzyme production and ExpR overexpression decreases enzyme production, indicating that ExpR acts as a negative regulator. Additionally, *E. carotovora* possess a second quorum-sensing system involved in the production of the antibiotic carbapenem. This second quorum-sensing system involves the **CarR** and **CarI**, LuxR/LuxI homologues. In this system, CarR–AI positively regulates the carbapenem biosynthetic genes. The carbapenem

would then inhibit any competing bacteria from participating in the banquet of nutrients released from the degradation of plant tissues. Thus, this bacterium links the regulation of plant tissue degradation and carbapenem production.

The etiologic agent of crown gall tumors in plants, *Agrobacterium tumefaciens*, uses quorum sensing to mediate conjugal transfer of its Ti plasmid. The Ti plasmid can be transferred to the nucleus of plant cells, where it induces tumor formation. The Ti plasmid is also a conjugative plasmid possessing *tra* genes capable of mediating its transfer via conjugation between bacterial cells. The **TraR** protein, a LuxR homologue, and an autoinducer synthase, **TraI**, positively control conjugal transfer of the Ti plasmid. The TraR–cognate AI complex positively regulates *tra* regulon expression, including *traI*. TraR exhibits regulation by either octopines or agrocinopines A and B through the regulatory proteins OccR (octopine-responsive activator) and AccR (agrocinopines-responsive repressor). *A. tumefaciens* can carry one of two types of Ti plasmids that direct the production of octopines or agrocinopines by the plant and their utilization by the bacteria: octopine-type Ti plasmids or nopaline-type Ti plasmids, respectively. Thus, accumulation of octopine results in OccR activation of *traR* expression, or accumulation of agrocinopines A and B relieves repression by AccR, leading to increased *traR* gene expression; then, once a cell density is achieved for sufficient AI accumulation to occur to activate TraR, the *tra* genes are expressed and conjugative transfer is stimulated.

Gram-Positive Bacterial Pathogens. The quorum-sensing systems of Gram-positive bacteria are fundamentally different from those of Gram-negative bacteria. They do not produce AHL and use sensor-kinase/regulator two-component signal transduction systems. Instead of AHL molecules, they produce peptide signals that interact with the sensor-kinase, which then phosphorylates the cognate regulator protein. Quorum sensing regulates a number of processes in Gram-positive bacteria such as conjugation in *Enterococcus faecalis* and competence in *Bacillus subtilis* and *Streptococcus pneumoniae*.

A good example of how quorum sensing is used to control virulence in Gram-positive bacteria is its role in regulating *Staphylococcus aureus* virulence. Like *P. aeruginosa*, *S. aureus* is the etiologic agent of a variety of diseases ranging from relatively minor skin lesions (folliculitis) to potentially fatal pneumonias or bacteremias. *S. aureus* also produces an impressive collection of virulence factors expressed in stages during infection. Early on, the bacterium synthesizes a number of surface-associated factors involved in adherence to host tissues (collagen and fibronectin-binding proteins) and host defense evasion (Protein A). Presumably these factors allow *S. aureus* to grow and multiply to reach a local high cell density. At this point, surface factor expression is downregulated and expression of a plethora of secreted products is increased, including several lipases and proteases, hyaluronidase, hemolysins, α -toxin, and enterotoxins.

This successive regulation of virulence factors is under the control of two regulatory loci: *agr* (accessory gene regulator) and *sar* (staphylococcal accessory gene regulator). The *agr* locus contains two divergently transcribed operons encoding two polycistronic mRNAs referred to as RNAII and RNAPIII. The *agrBDCA* operon (RNAII) encodes **AgrC**, a signal transducer protein, and **AgrA**, a response regulator. AgrB and AgrD are involved in producing the quorum-sensing signal molecule, an octopeptide cleaved from AgrD and exported extracellularly by the AgrB protein. RNAPIII encodes the δ -hemolysin and also functions as a regulatory RNA molecule. During quorum sensing,

AgrC binds to the peptide signal molecule, phosphorylates itself (autophosphorylation), and then phosphorylates AgrA. Phospho-AgrA (AgrA~P) then induces RNAIII expression, which goes on to stimulate the expression of numerous secreted products and to downregulate specific surface protein expression as well as increase the expression of *agrBDCA*. Additionally, the SarA protein is a positive regulator, inducing the expression of both RNAII and RNAIII.

PARADIGMS OF BACTERIAL PATHOGENESIS

Enteropathogenic *Escherichia coli*

Enteropathogenic *E. coli* (EPEC) strains cause enteritis not by invading the intestinal epithelial cell but by tightly adhering to the enterocyte surface and inducing cytoskeletal reorganization, producing attachment and effacement (A/E) lesions. EPEC strains bind to the apical surface of the enterocyte via EAF (EPEC adherence factor) plasmid-encoded bundle-forming pili (BFP). This initial adherence is referred to as localized adherence (LA) and is the first stage of EPEC pathogenesis.

LA is followed by induction of a number of signal transduction pathways including increases in intracellular inositol triphosphate (IP₃) and Ca⁺² concentrations. This signal transduction induction is dependent on gene products encoded within the locus of enterocyte effacement (LEE) found on a 35 kb pathogenicity island on the EPEC chromosome. LEE encodes a type III secretion system (*sep/esc*; TTSS), the secreted proteins EspA-D, and the adhesin intimin and its secreted receptor Tir. These are all needed for host cell signal transduction that results in intimate adherence, which contributes to A/E lesion formation.

Translocation of Tir, and its subsequent tyrosine phosphorylation, and the EspA, EspB, and EspD proteins, requires the TTSS while the EspC protein is a member of the autotransporter family and mediates its own export. The interaction of intimin with phospho-Tir mediates tight intimate adherence of the bacterial cell to the host cell surface. The effect of intimate adherence and secreted proteins on the cell cytoskeleton (colocalization of cytoskeletal components) results in the formation of a pedestal structure (Fig. 20-5) characteristic of the A/E lesion. Similar A/E lesions are induced by enterohemorrhagic *E. coli* (EHEC) O157:H7, rabbit-diarrheal *E. coli* (RDEC-1), *Citrobacter rodentium*, and *H. pylori* strains.

Salmonella Enterica Serovars

S. Typhimurium, shortly after adherence to the apical surface of enterocytes, stimulates a striking response at the cell surface. It causes an outward extrusion of the surface at the site of adherence referred to as **membrane ruffling** and entry into the cell via a macropinocytosis process (Fig. 20-4). Furthermore, Ca⁺² and IP₃ levels increase activating their cognate signal transduction pathways and actin filaments are reorganized. *Salmonella* invasion is rapid, with the bacteria being internalized in vacuoles within just a few minutes. Interestingly, the host cell surface structure returns to normal soon afterward in a process promoted by bacterial-encoded products. Genes encoding invasion factors are located in one of several pathogenicity islands identified on the *S. enterica* chromosome, referred to as SPI-1. SPI-1 encodes a TTSS and several secreted proteins. Several of these secreted proteins mediate actin reorganization

and membrane ruffling by regulating host cell Rho GTPases or functioning as a IP₃ phosphatase.

Salmonella serovars are found in membrane-bound vacuoles within both non-phagocytic and phagocytic cells. These ***Salmonella*-containing vacuoles (SCV)** possess host cell markers that appear to be determined by some bacterial-encoded function(s). Interestingly, the presence of SCV membrane markers changes over time. Furthermore, the SCV escapes normal endocytic trafficking in both enterocytes and macrophages. Intracellular survival is dependent on genes located in a second pathogenicity island, SPI-2. SPI-2 encodes another TTSS as well as several secreted effector proteins. At least one of these, SpiC, inhibits endosomal trafficking. The SCV is larger than normal and is sometimes referred to as a spacious phagosome. It is proposed that *Salmonella* can actively prevent fusion of the phagosome with lysosomes; however, this appears to be incomplete, since there is evidence that they can and do survive within the phagolysosome. Clearly, though, the phagosome or phagolysosome that they occupy is different from the normal phagosome or phagolysosome. One key event that is critical for virulence of these bacteria is the acidification of the immediate milieu in the SCV. Another interesting aspect of *Salmonella* invasion is that entry into macrophages frequently induces apoptosis while entry into epithelial cells does not. This suggests that *Salmonella* enters epithelial cells and macrophages by two distinct mechanisms.

Essential to *Salmonella* pathogenesis is the entry into and passage through specialized epithelial cells called M cells with subsequent release of the bacteria into the submucosa. In the submucosa they typically enter macrophages where the bacteria can survive and multiply. Survival in macrophages is promoted by functions encoded on SPI2 and regulated by the PhoP/PhoQ two-component regulatory system. PhoP/PhoQ sense and respond to low Mg⁺² ion concentrations, activating and repressing specific genes. Dissemination from the intestinal tract or the associated lymphoid tissue is influenced by the host and the serovar involved. In mice, *S. Typhimurium* can disseminate systemically to the spleen and liver, whereas in humans it remains localized within the intestines and does not typically spread beyond the regional lymph nodes.

Systemic spread is dependent on functions encoded on pathogenicity islands and the virulence plasmid. Specifically, four genes, *spvABCD* (*spv*, meaning *Salmonella* plasmid virulence), composing an operon on the virulence plasmid, are needed for systemic spread of *Salmonella* in appropriate hosts. The expression of this operon is dependent on an upstream regulatory gene, *spvR*, which is divergently transcribed from the *spvABCD* operon. Expression of *spvR*, and thus *spvABCD*, is induced by a specific set of environmental conditions found within macrophages. This regulation assures that the *spv* operon is expressed inside the correct host niche. Serovars causing enteritis in humans do not typically disseminate. However, entry of these organisms into the intestinal epithelium stimulates the epithelial cell to produce a number of chemokines (soluble proteins that act as neutrophil chemotaxins) that cause neutrophils to migrate to the area, transmigrate across the epithelial cell layer into the intestinal lumen, and contribute to inflammation and ulceration of the bowel wall.

Listeria Monocytogenes

L. monocytogenes is a Gram-positive bacterium that can actively invade both nonphagocytic and phagocytic cells. Its invasion is mediated by InlA, a member of

the internalin family of bacterial invasion proteins. Like the Ipa proteins of *Shigella*, InlA mediates epithelial cell invasion at the basolateral rather than apical surface of the cell. Not coincidentally, the InlA receptor is E-cadherin, a cell-to-cell adhesin molecule expressed on the basolateral surface of epithelial cells. A related surface molecule called InlB mediates entry into hepatocytes. InlA functions in epithelial cell invasion but not hepatocytes, while the opposite is true for InlB. Upon internalization, the bacteria escape the vacuole through the action of **listeriolysin O (LLO)** (see this chapter under surface acting toxins) and perhaps additional phospholipase C activities. LLO is activated by low pH, which may be achieved in the initial vacuole in which it resides. Interestingly, only a relatively small percentage of internalized bacteria escape the vacuole into the cytoplasm; those that do not are killed.

Free in the cytoplasm, *L. monocytogenes* directs the condensation of actin at one pole of the bacterial cell and reorganizes it into a **polar tail** or “**comet**” of **polymerized actin** that moves the bacteria through the cytoplasm and into adjacent host cells. The listerial surface protein ActA localizes to one pole of the cell and mediates this actin reorganization and accumulation. Additional host cytoskeletal components and bacterial proteins also appear to be involved. This allows the bacteria to infect and spread within host tissue evading typical host immune responses. *Shigella flexneri* exhibits a similar pathogenic scheme. It also can escape the internalized vacuole, can polymerize actin to form comets, and can spread from cell to cell (Fig. 20-6).

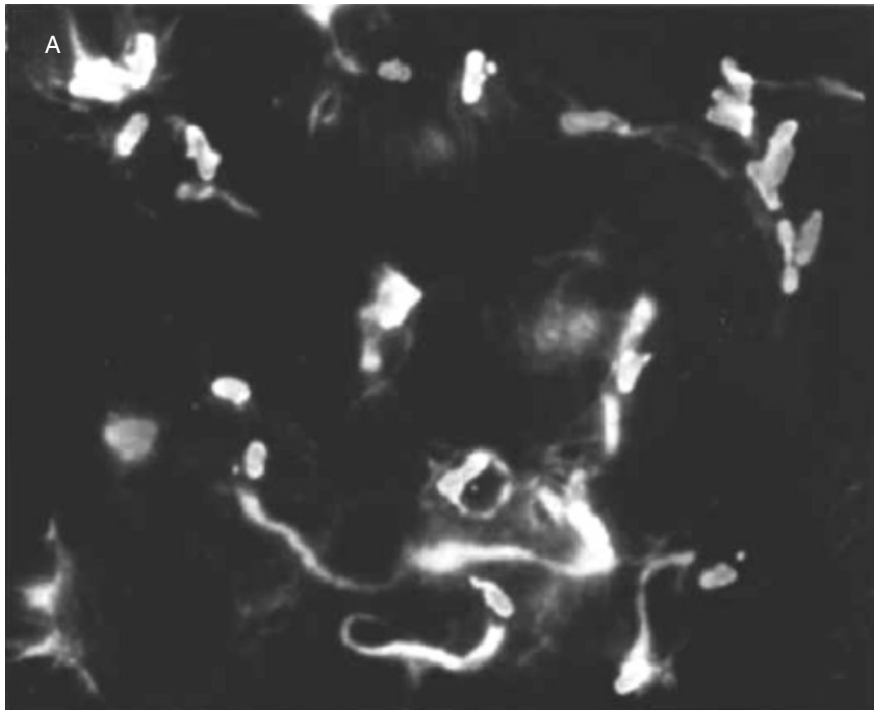


Fig. 20-6. Immunofluorescent photomicrograph showing actin tail or “comets” associated with individual *Shigella* bacterium. Courtesy of ●●●.

***Chlamydia* spp**

Unlike *Salmonella* and *Listeria*, which are **facultative intracellular pathogens**, chlamydiae are **obligate intracellular pathogens** and can only grow within living cells. They can be found in two cell forms: the **elementary body (EB)**, which can adhere to and invade host cells but cannot replicate itself, and the **reticulate body (RB)**, which is the intracellular actively growing cell form but is not infectious. Binding of the EB to the target cell results in the uptake of the cell into a vacuole. This vacuole remains nonacidic throughout the time spent inside the cell. It is atypical in that it lacks significant amounts of host markers in the vacuole membrane. This vacuole also avoids the normal endocytic and exocytic trafficking pathways and does not fuse to lysosomes. Within the vacuoles, the EB converts to the RB and begins to divide. RB multiplication leads to the formation of a cell-packed inclusion in the infected cell. Interestingly, sphingomyelin from the Golgi apparatus of the cell gets incorporated into the inclusion and eventually into the cell surface of the bacteria. The bacteria also appear to secrete proteins that become incorporated into the membrane of the bacterial-laden vacuole, which are proposed to direct vacuolar trafficking so as to promote intracellular growth of the RB. Late in the infection cycle, the RB cell form converts to the EB, which is released from the cell by cell lysis.

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